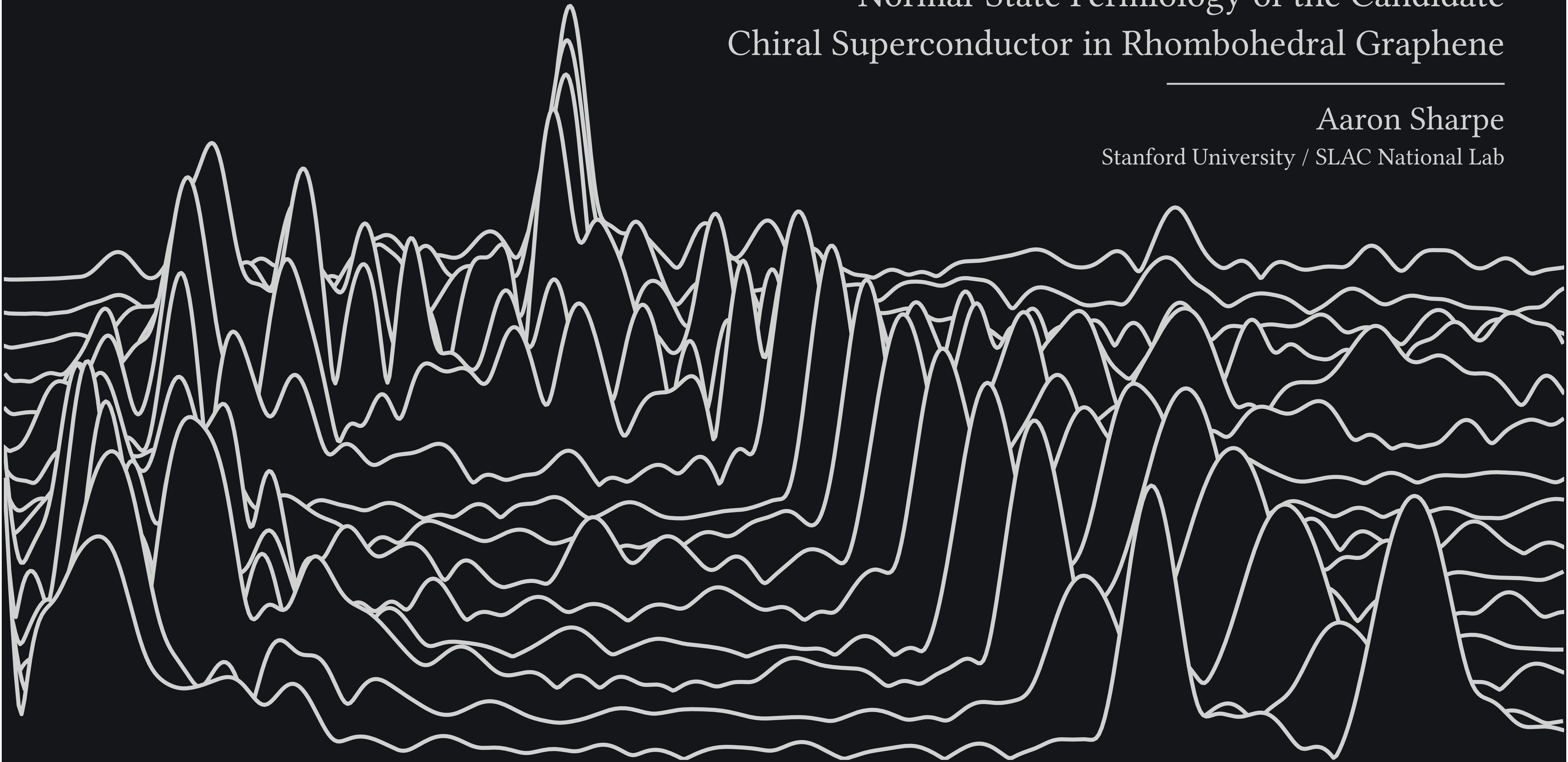
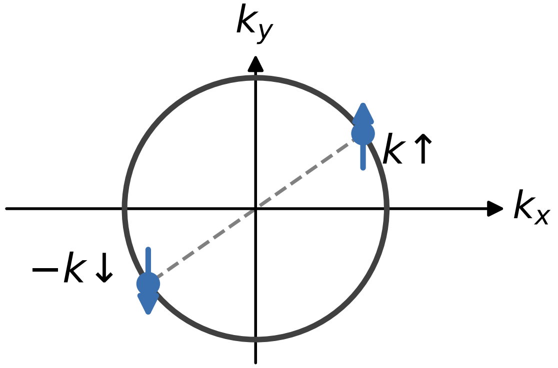
A Multitone Mystery:
Normal-State Fermiology of the Candidate Chiral Superconductor in Rhombohedral Graphene

Aaron Sharpe
Stanford University / SLAC National Lab



Pairing and Time Reversal

- Conventional pairing: electron at k binds its **time-reversed partner**



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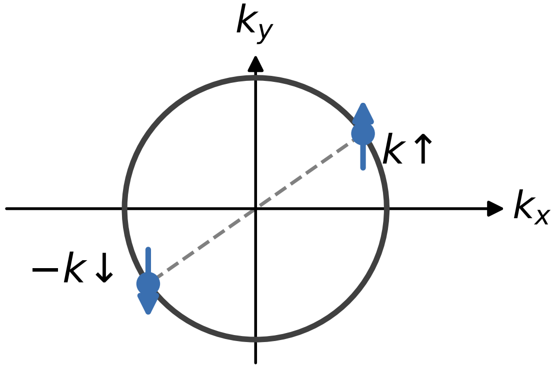

Bardeen, Cooper & Schrieffer Phys. Rev. (1957)
Slides available @ aaronsharpe.science

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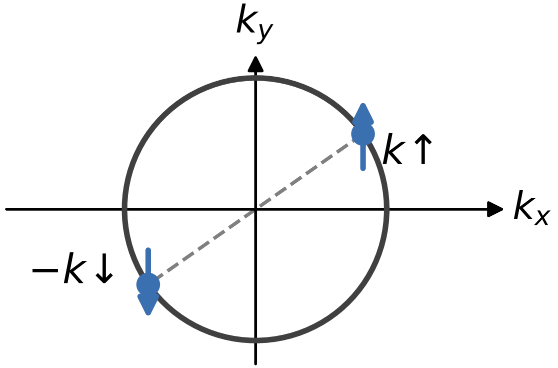

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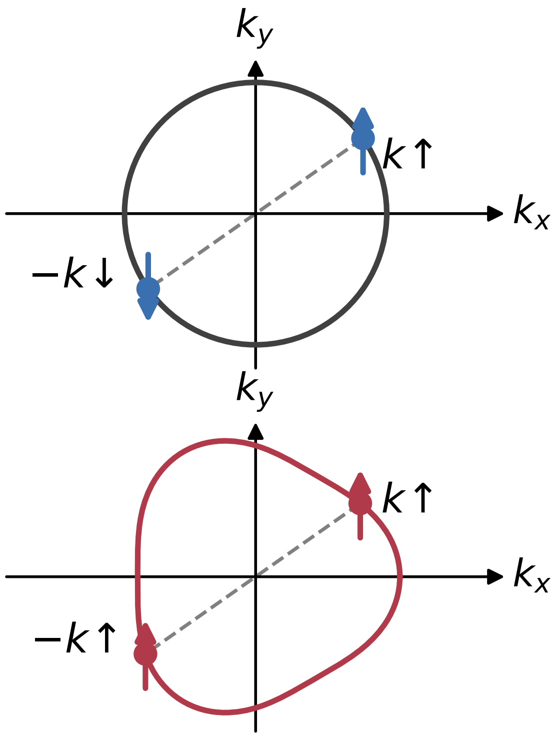

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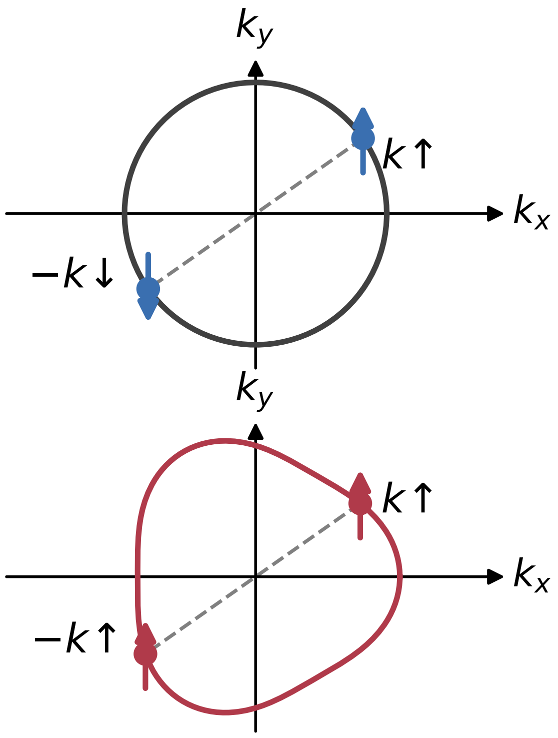

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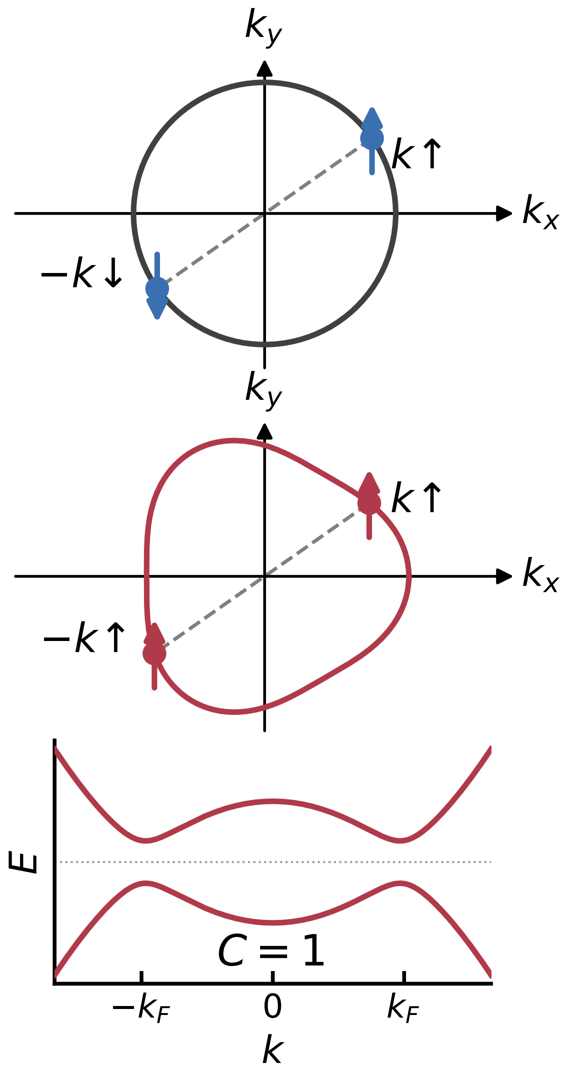

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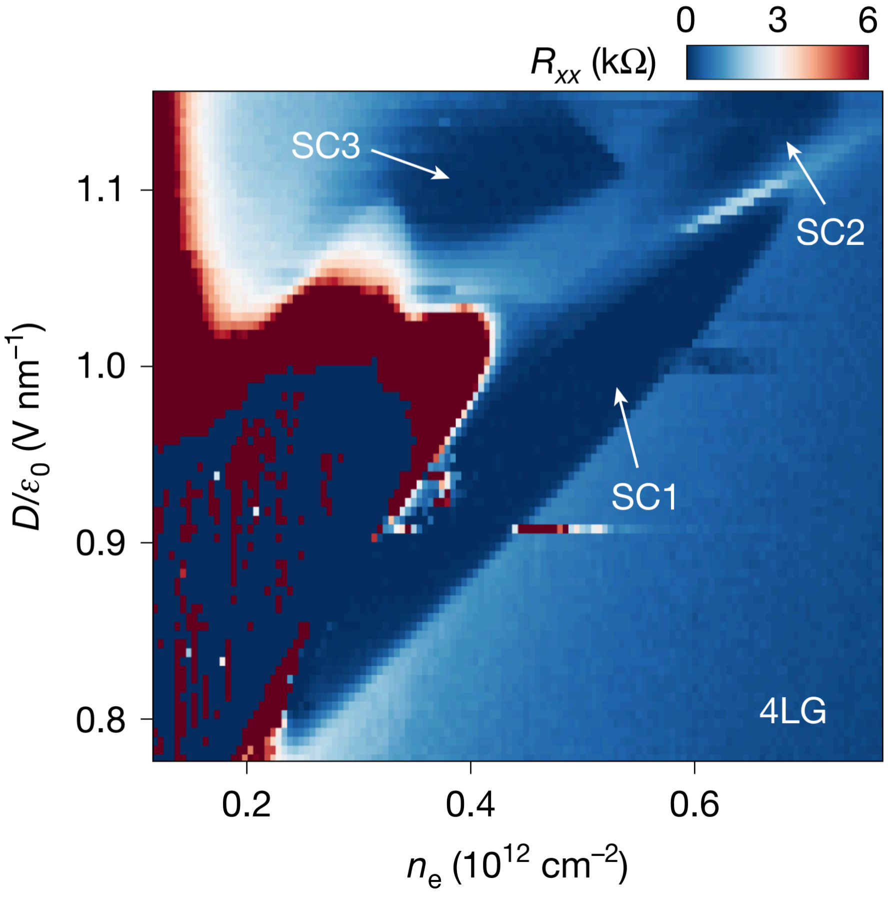

Han et al. Nature (2025)
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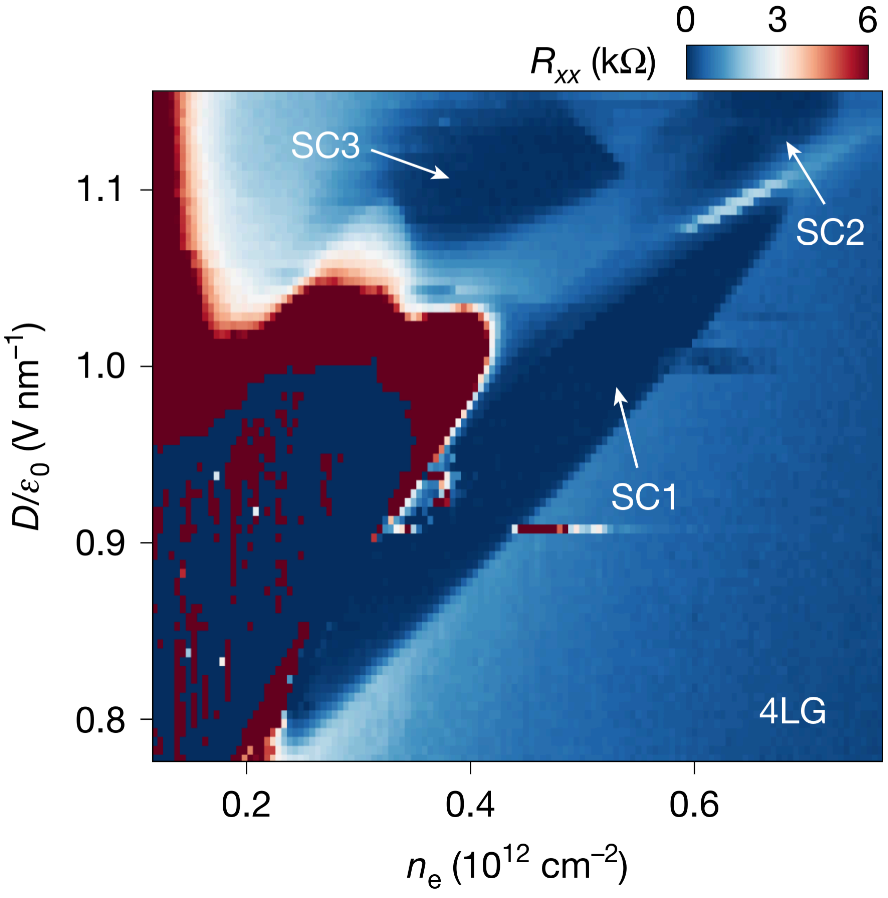

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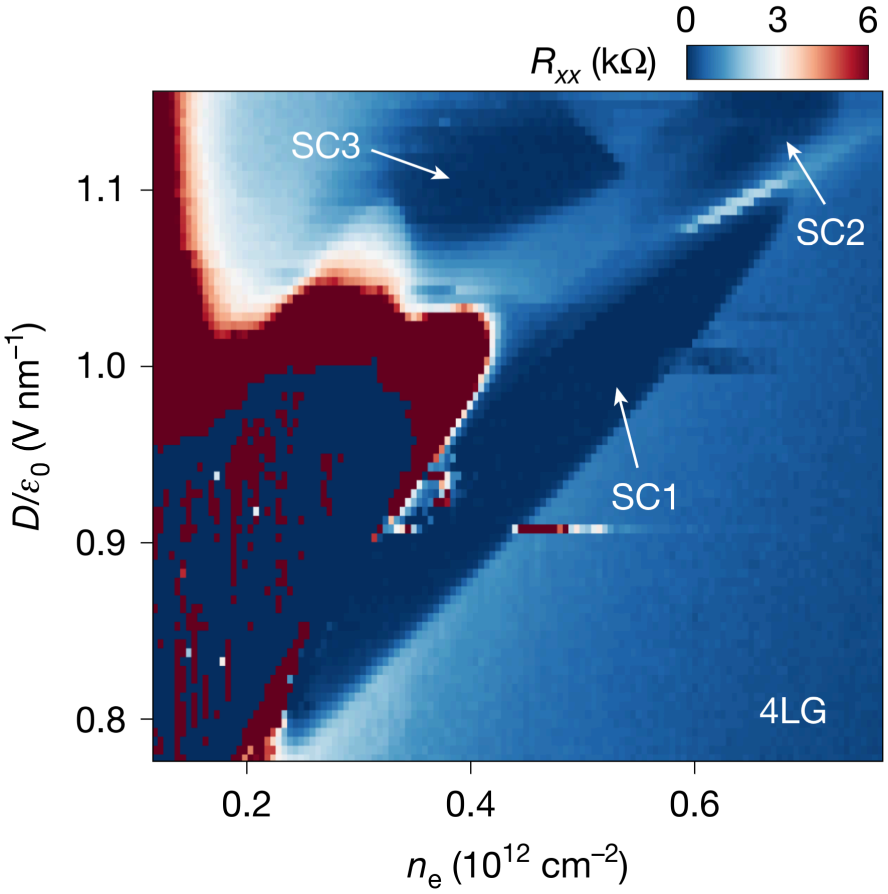

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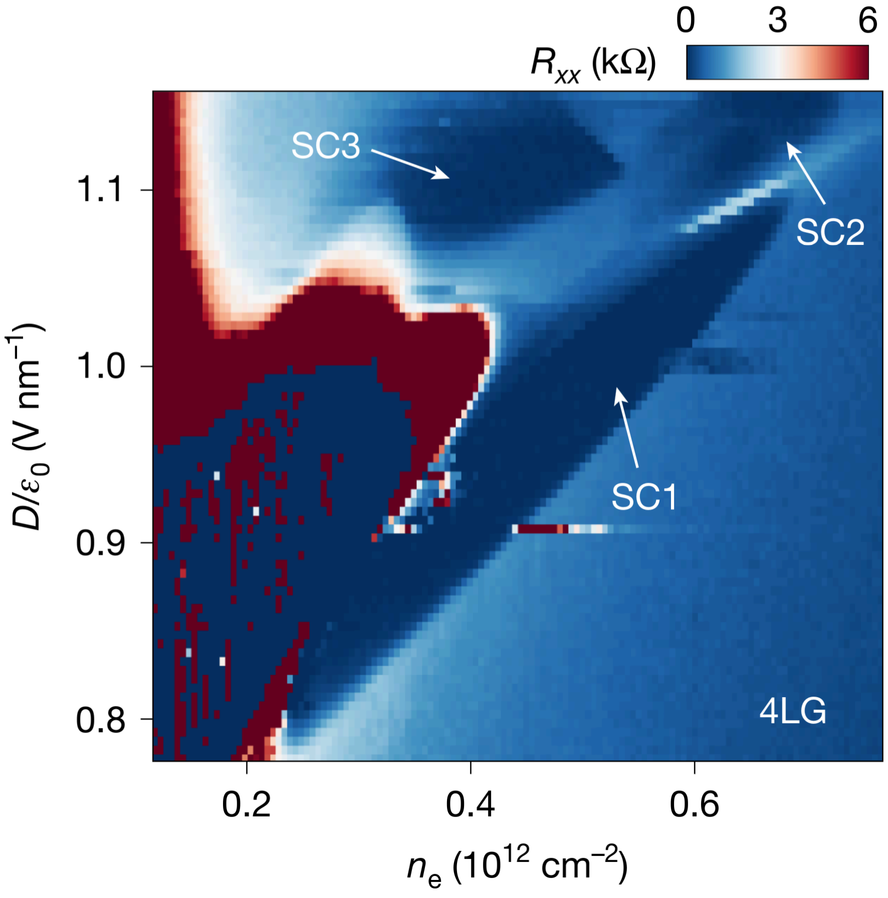

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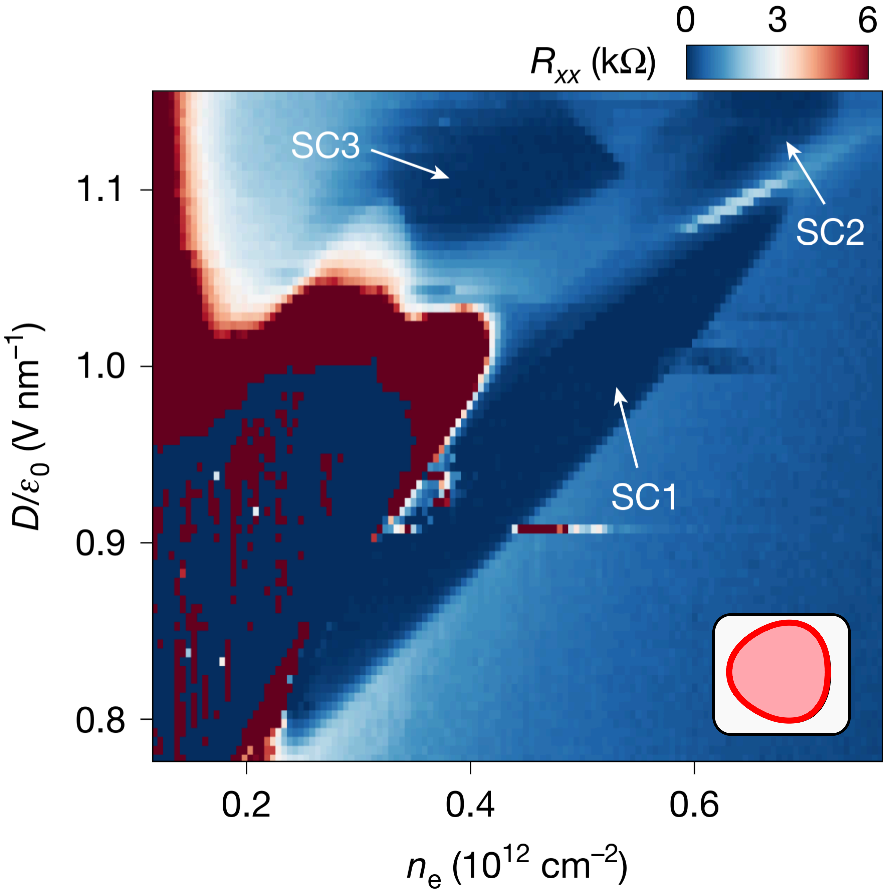

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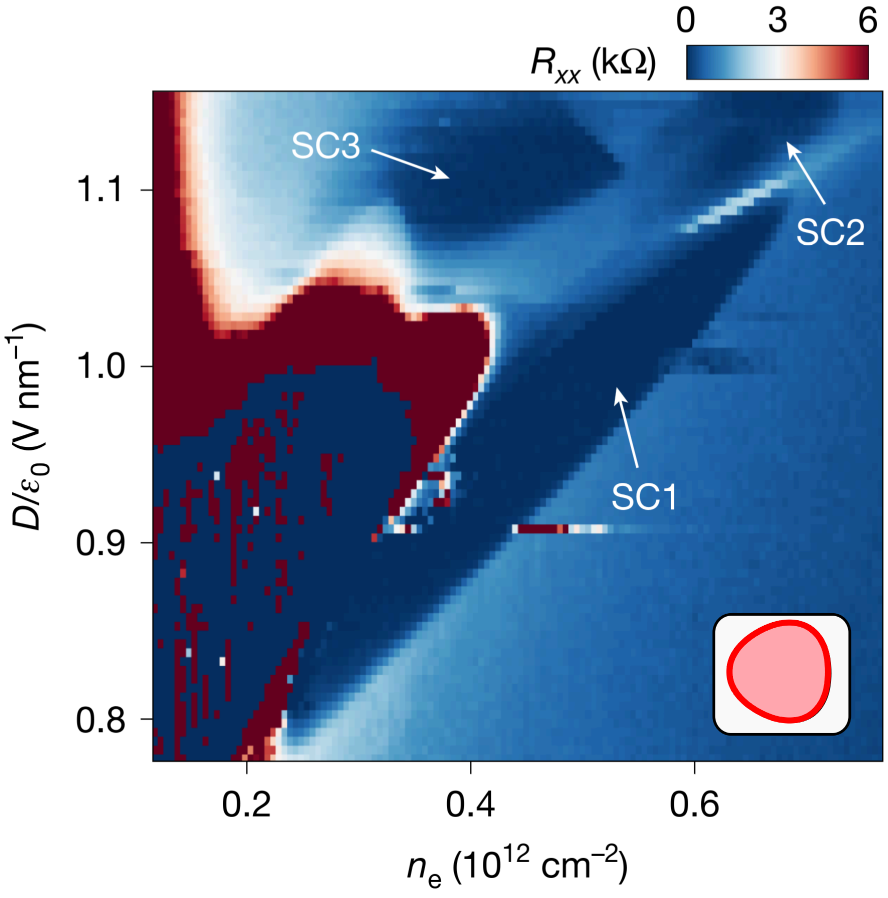
- Normal state?
- Chiral + single Fermi surface $\rightarrow$ topological!



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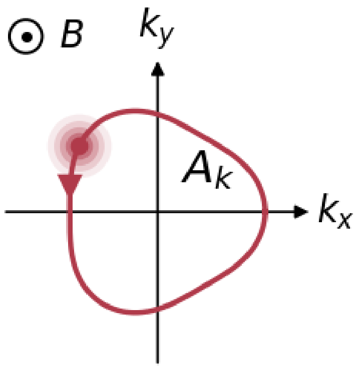

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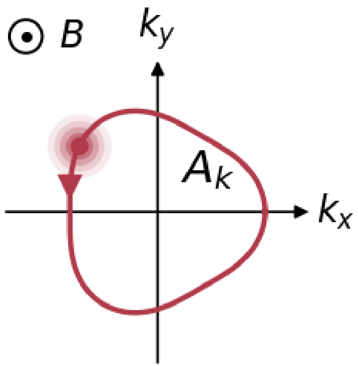

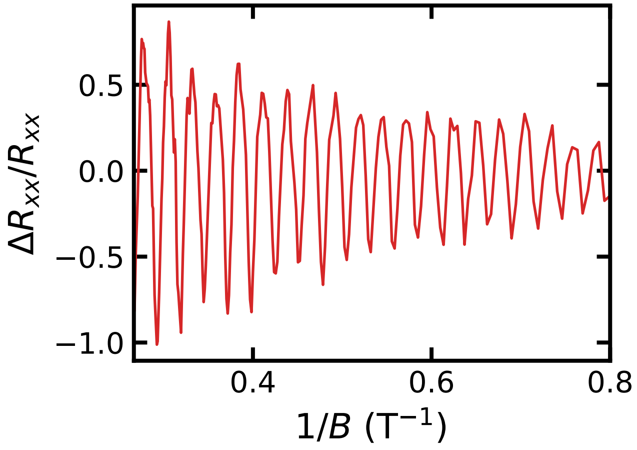

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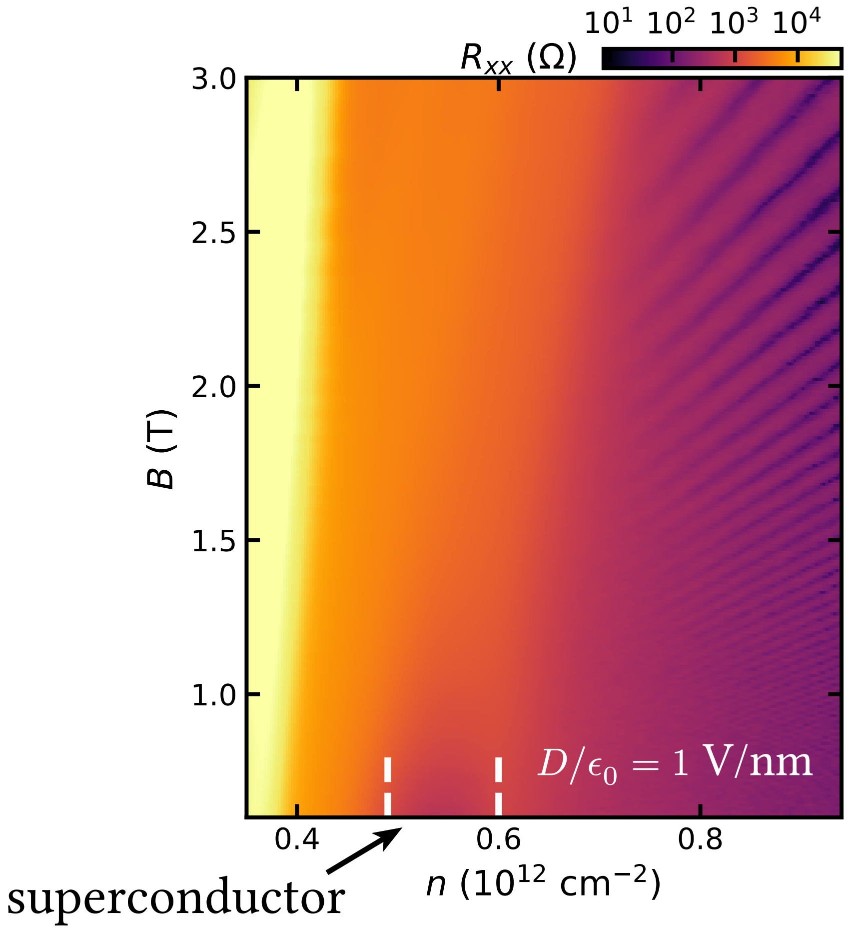

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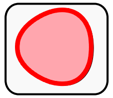
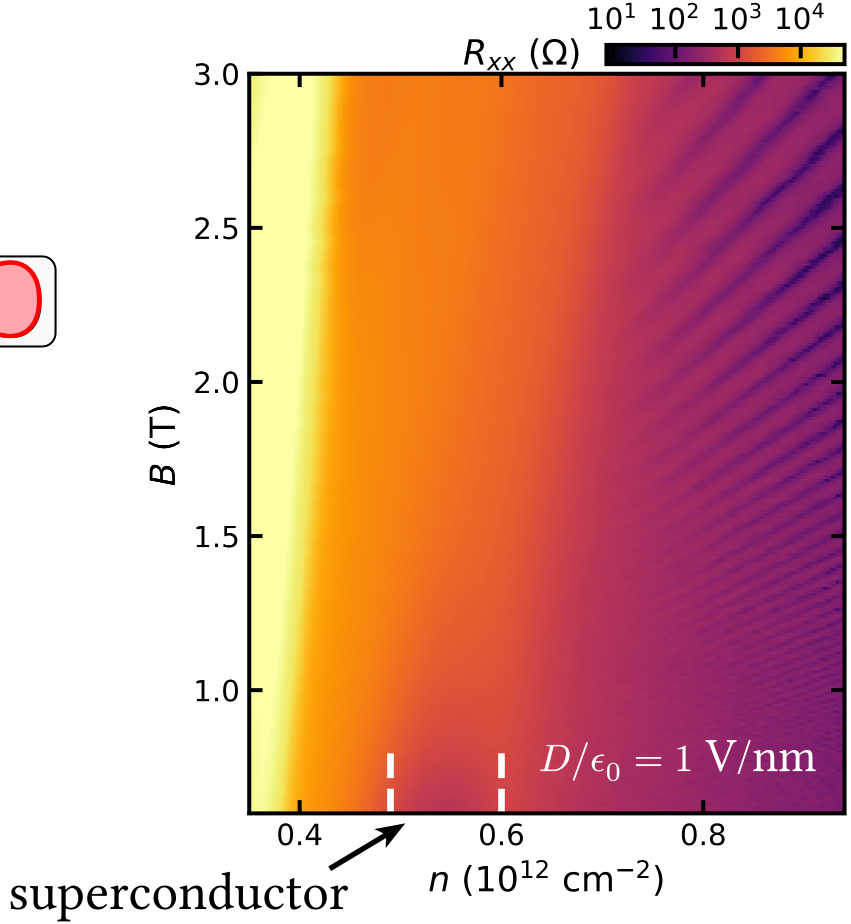
- High n: single simply-connected Fermi surface





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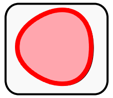

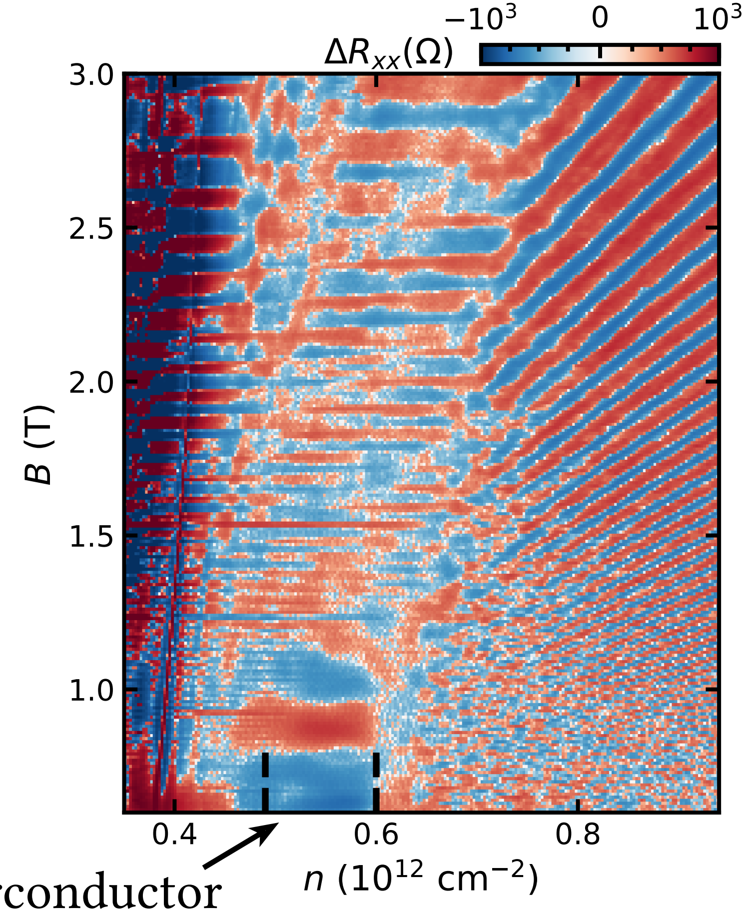

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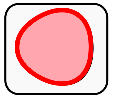

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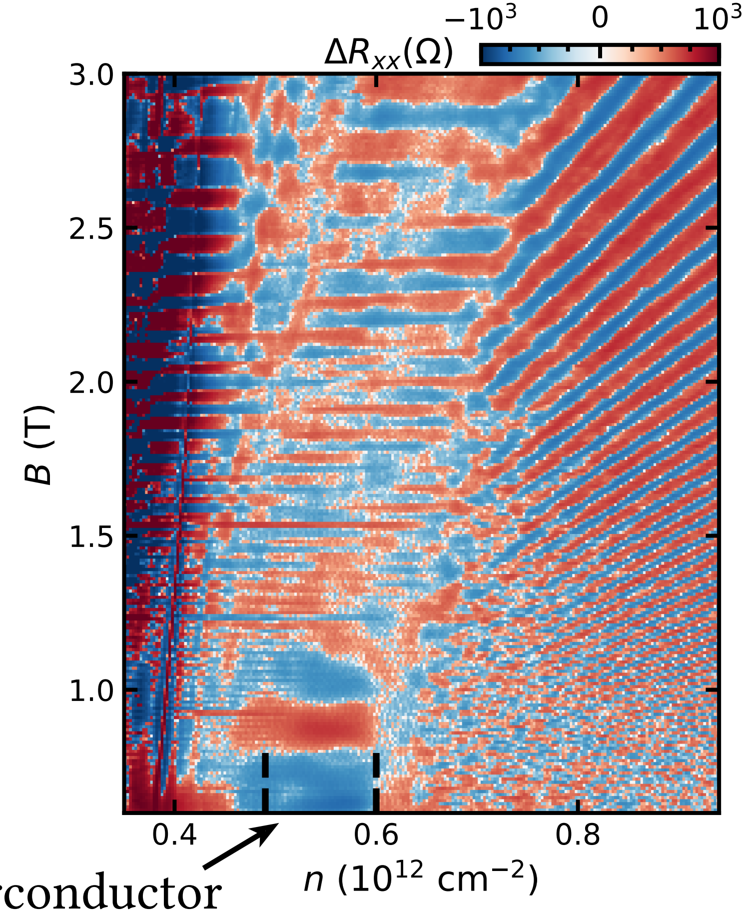
  - Likely the normal state of CSC



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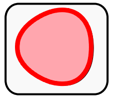
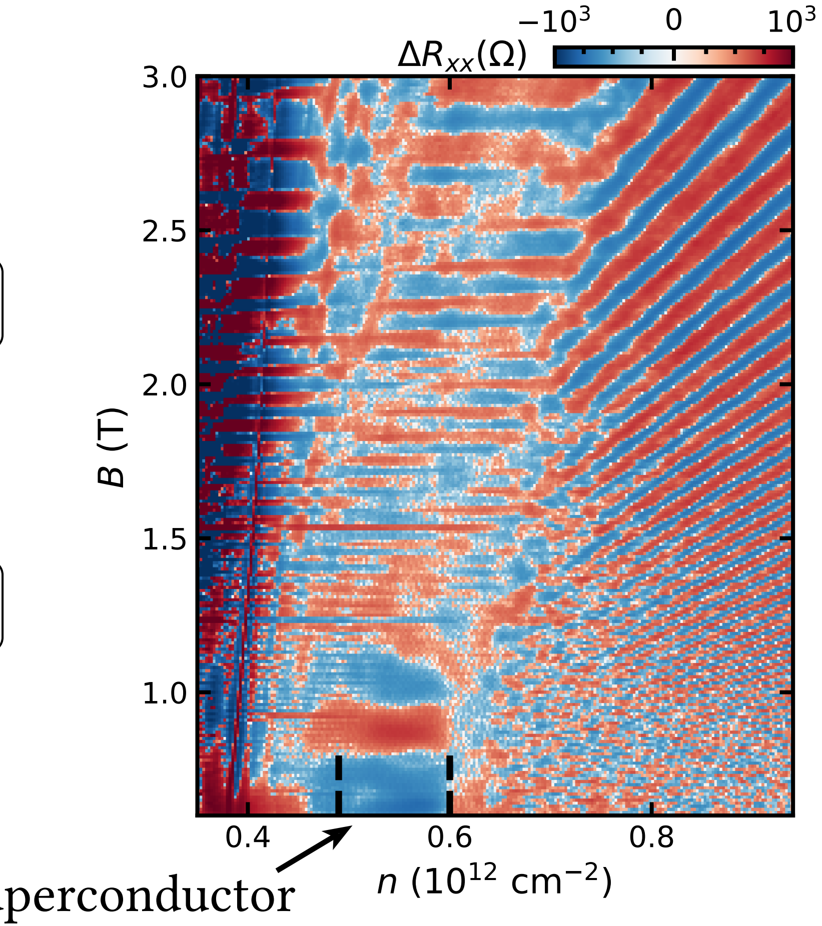


- Lower n: **“multitone”** state
  - Likely the normal state of CSC
  - No obvious Fermi liquid picture for oscillations



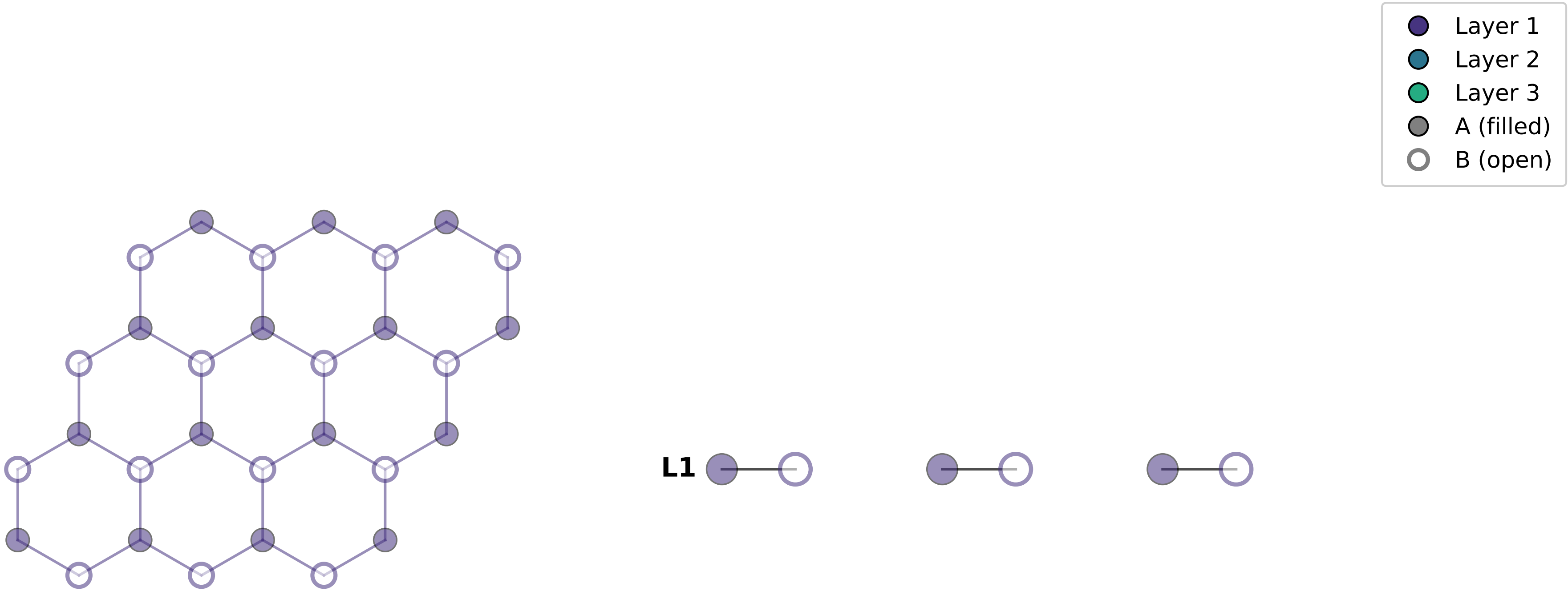
Outline

0. **Introduction: rhombohedral graphene**

1. **Superconductivity in tetralayer rhombohedral graphene**

2. **Fermiology from quantum oscillations**

Bernal (AB) Stacking



Top view

Side view

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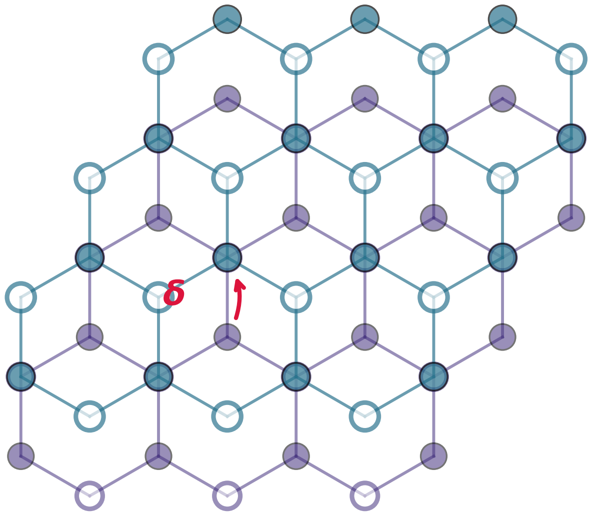

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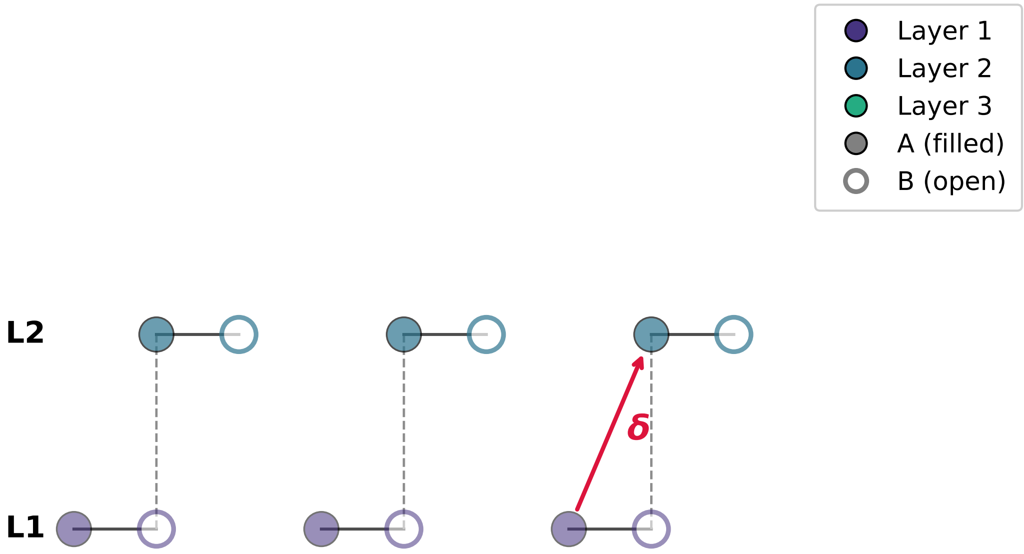

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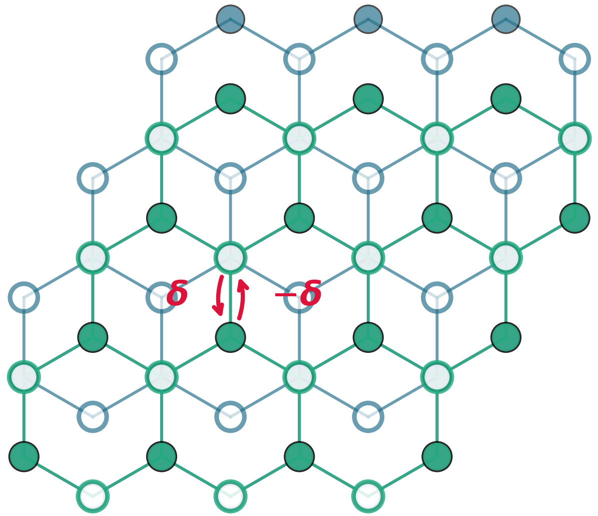

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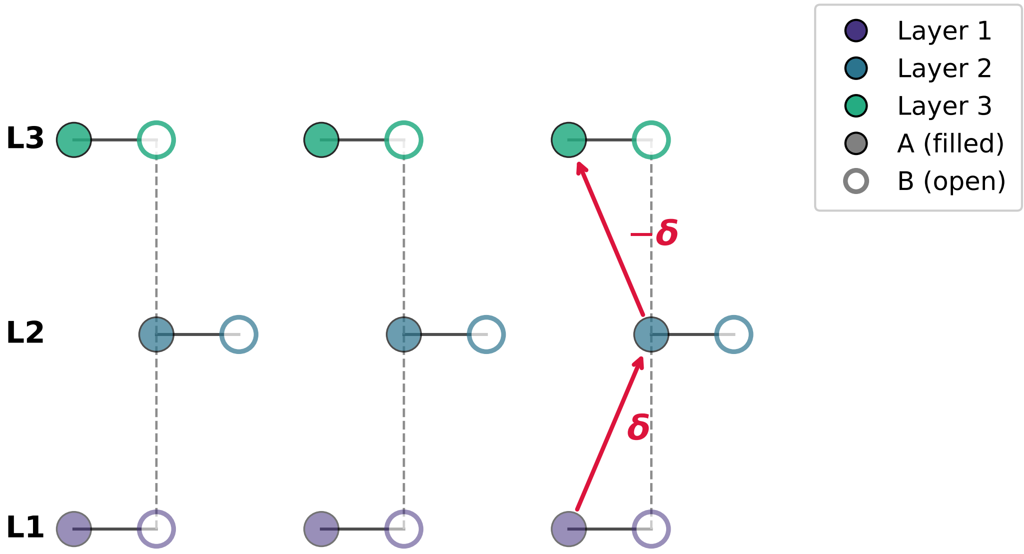

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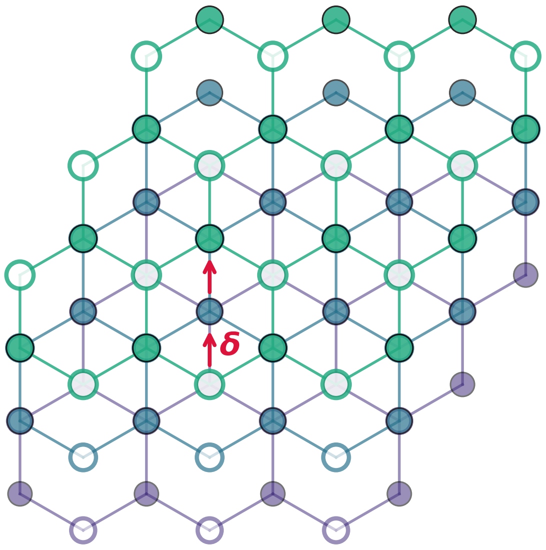
Rhombohedral Multilayer Graphene (RMG)



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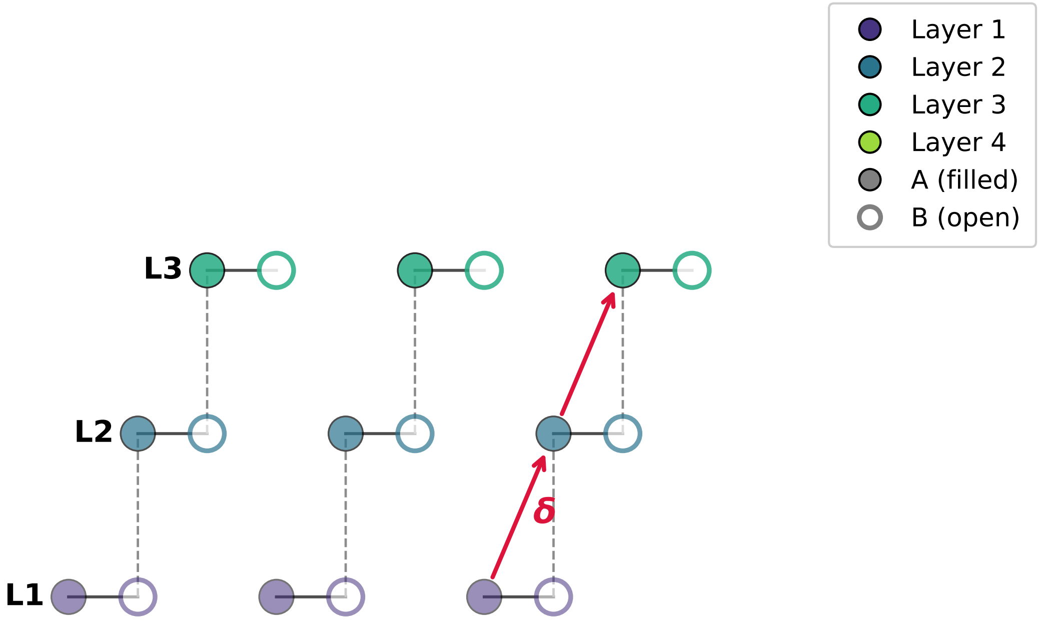

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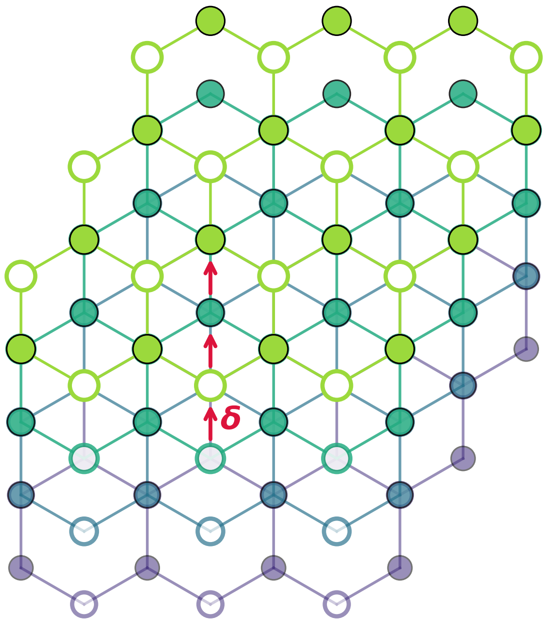

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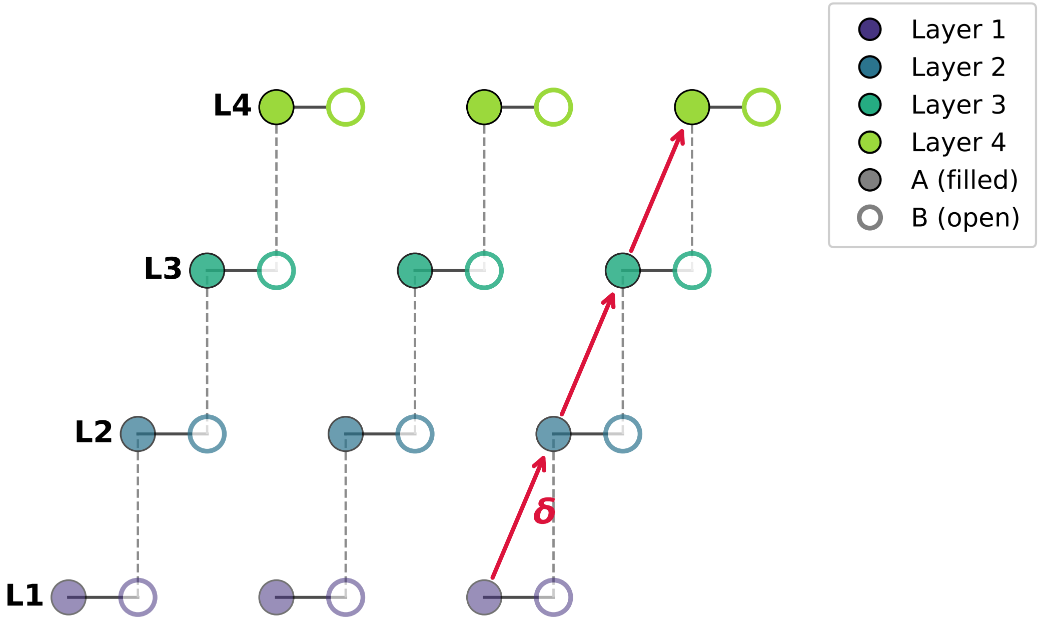

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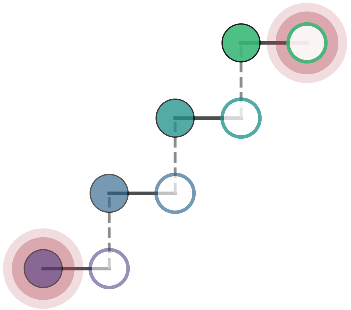

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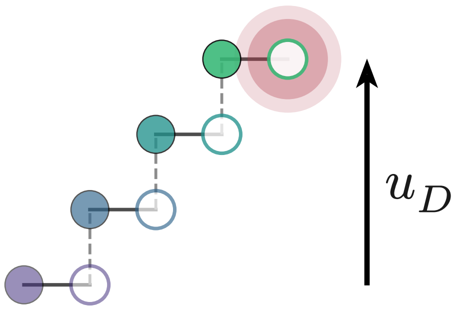

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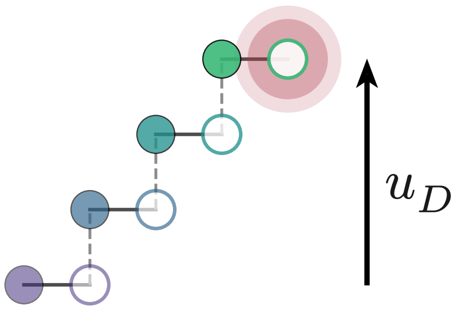

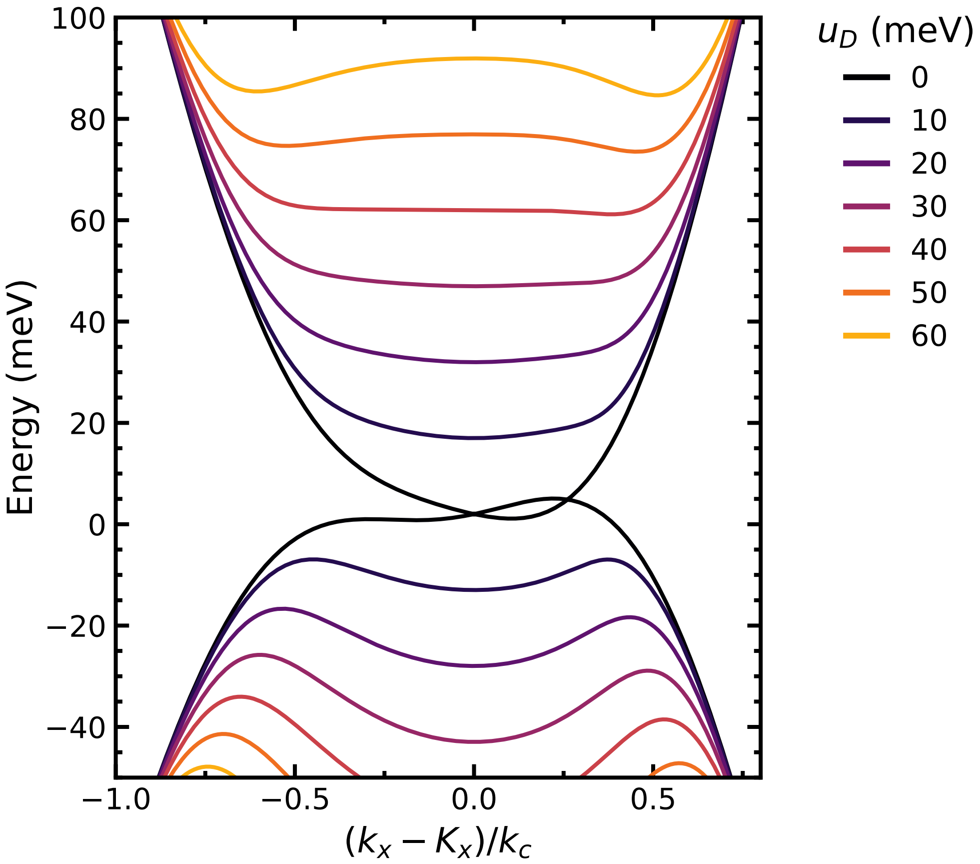

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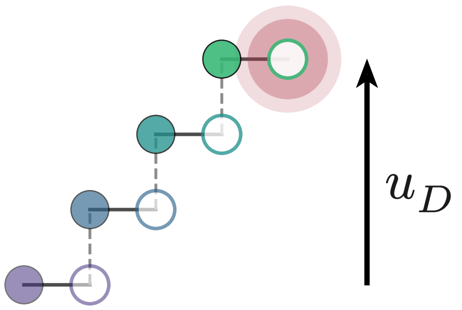

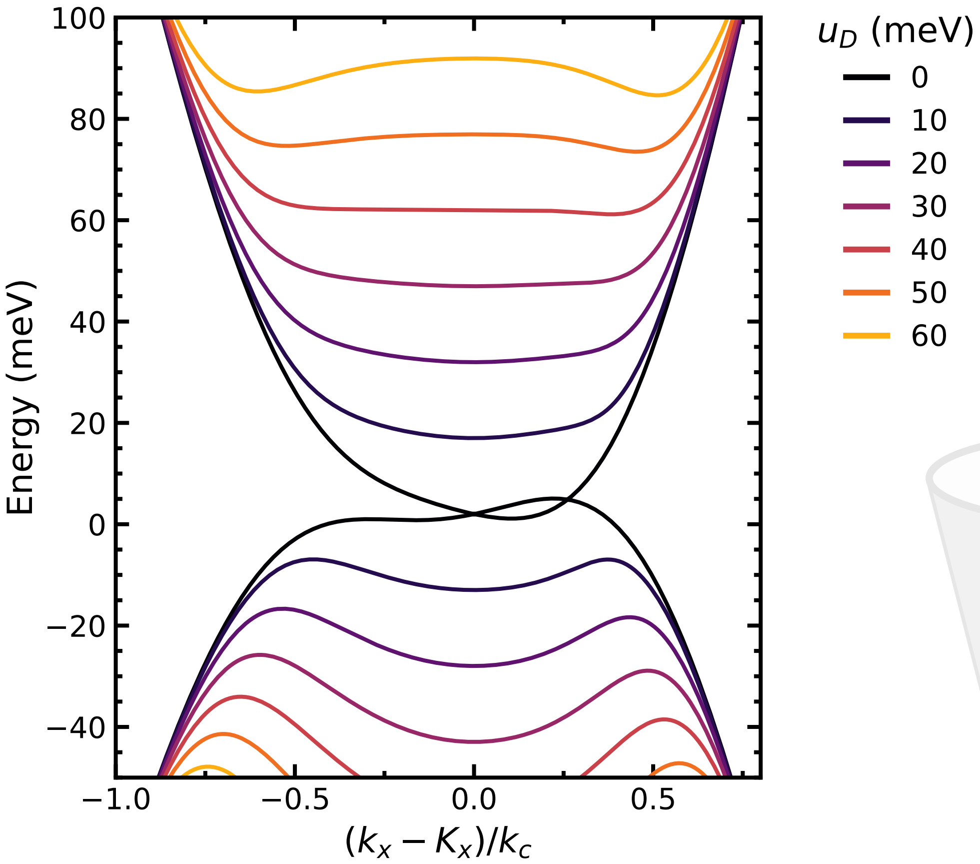

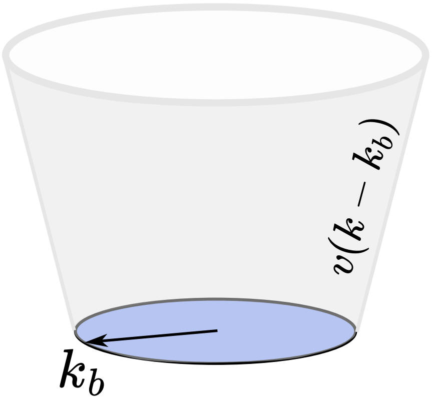

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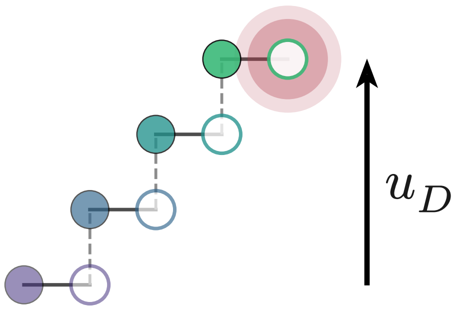

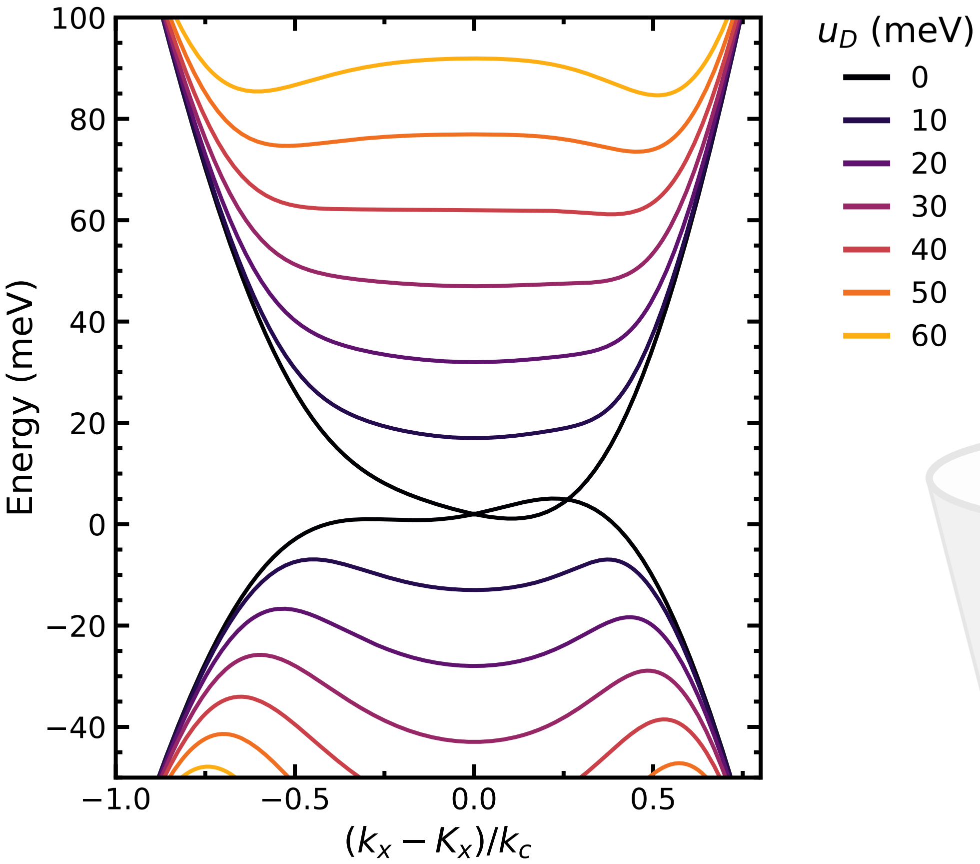

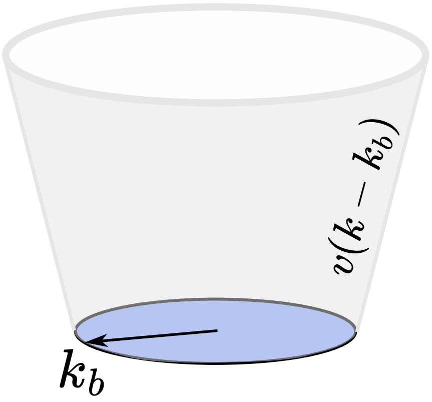

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- $\rightarrow$ flat minima
- $\rightarrow$ to a Mexican hat

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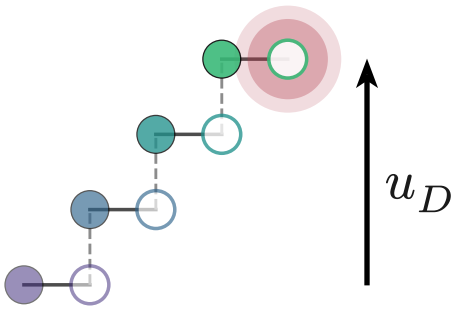

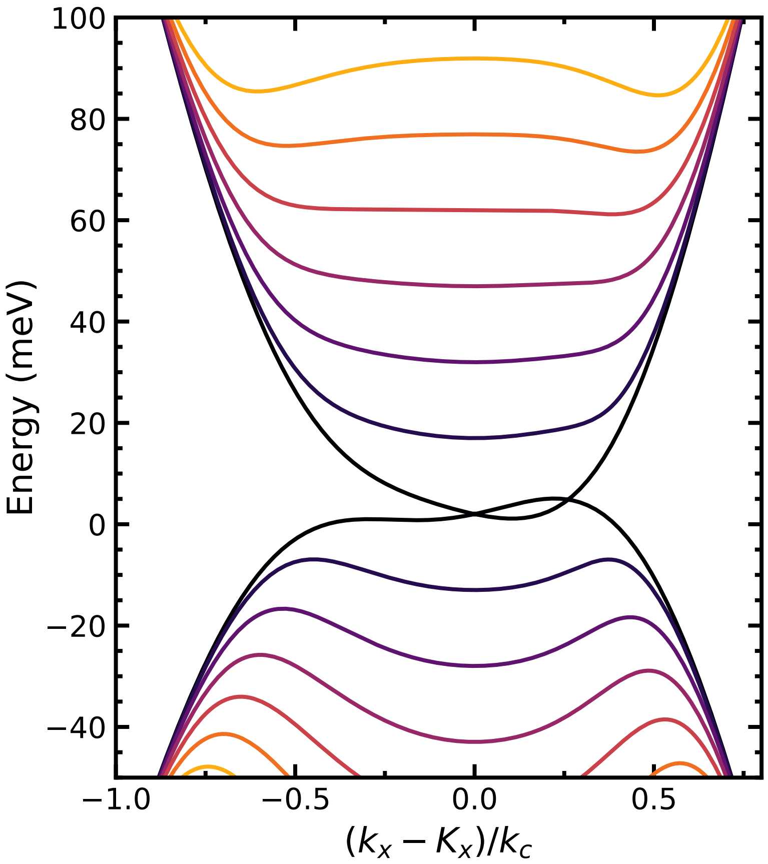

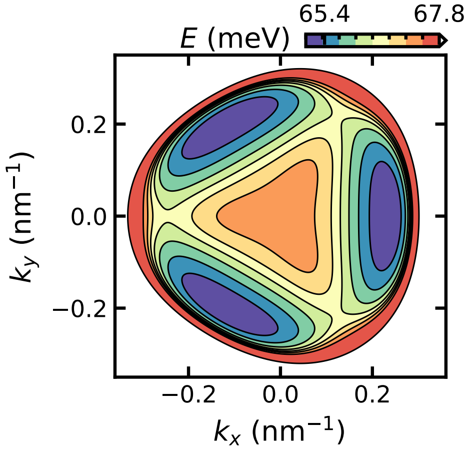

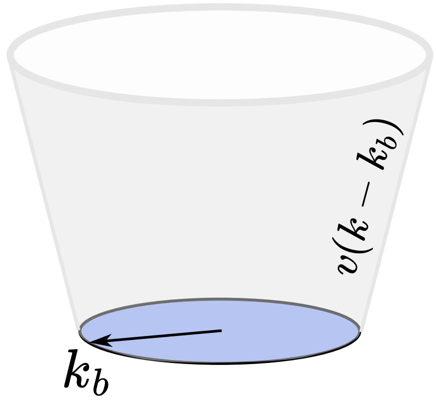

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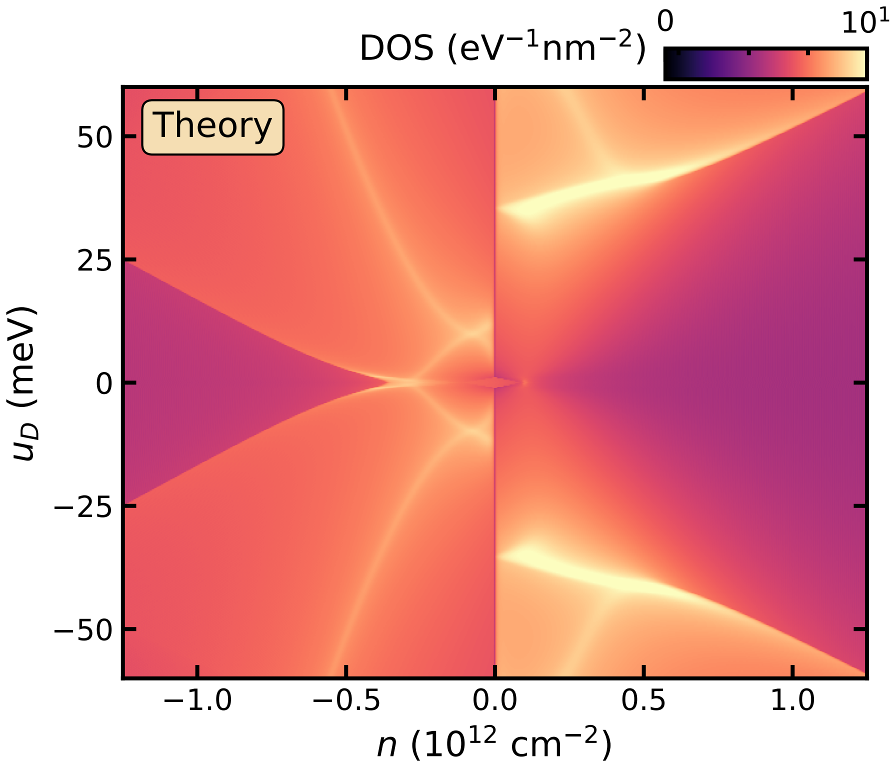
R4G Conduction Band Fermiology



Assume spin- and valley-polarized

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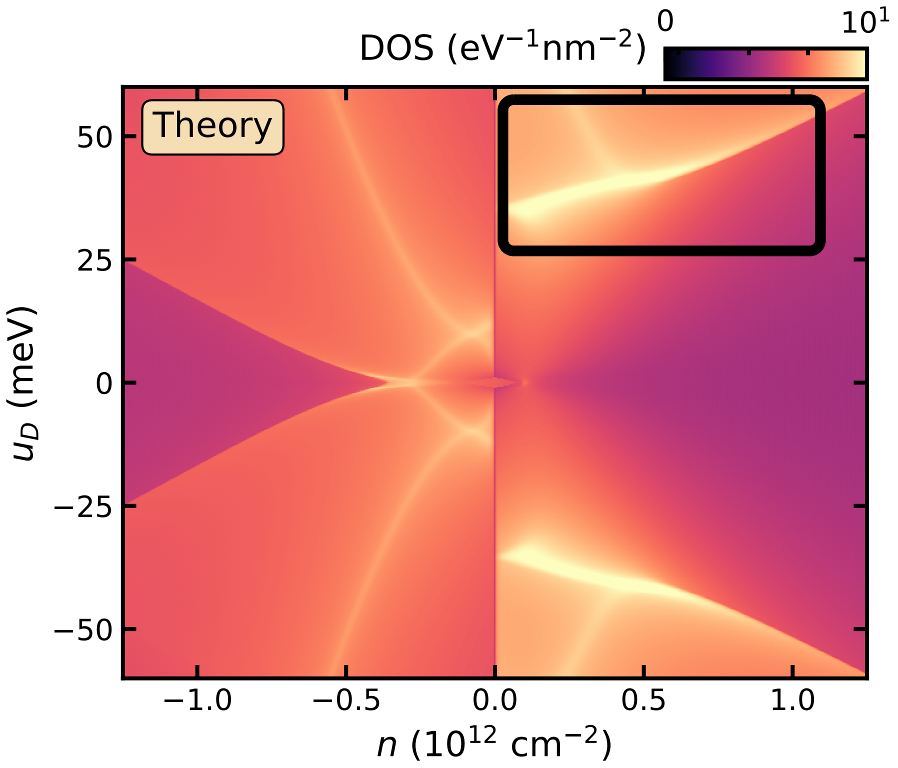

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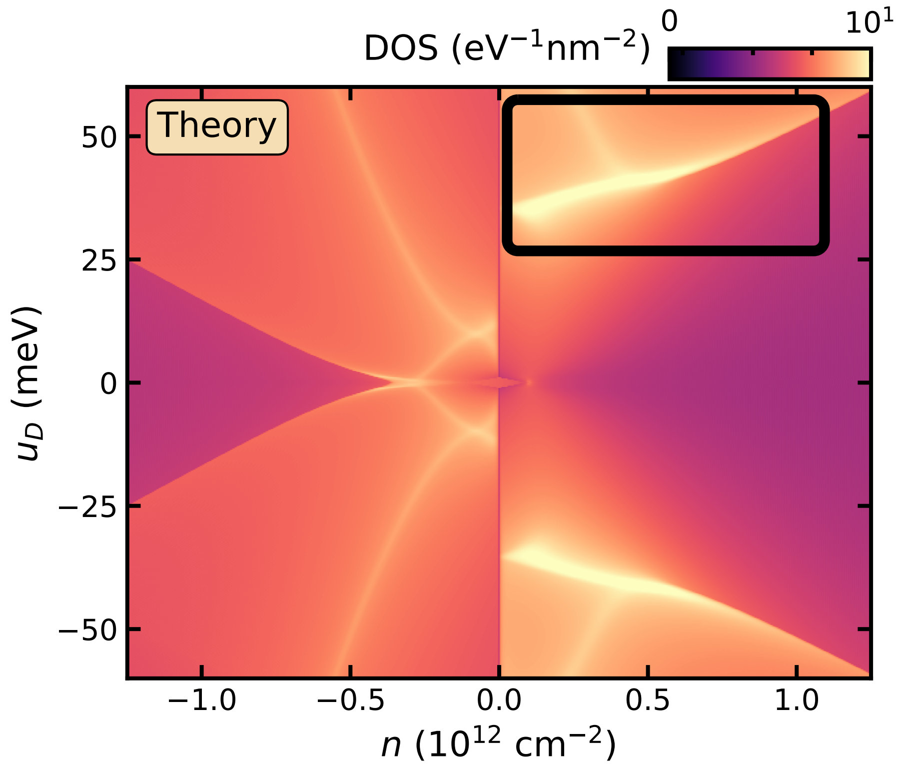

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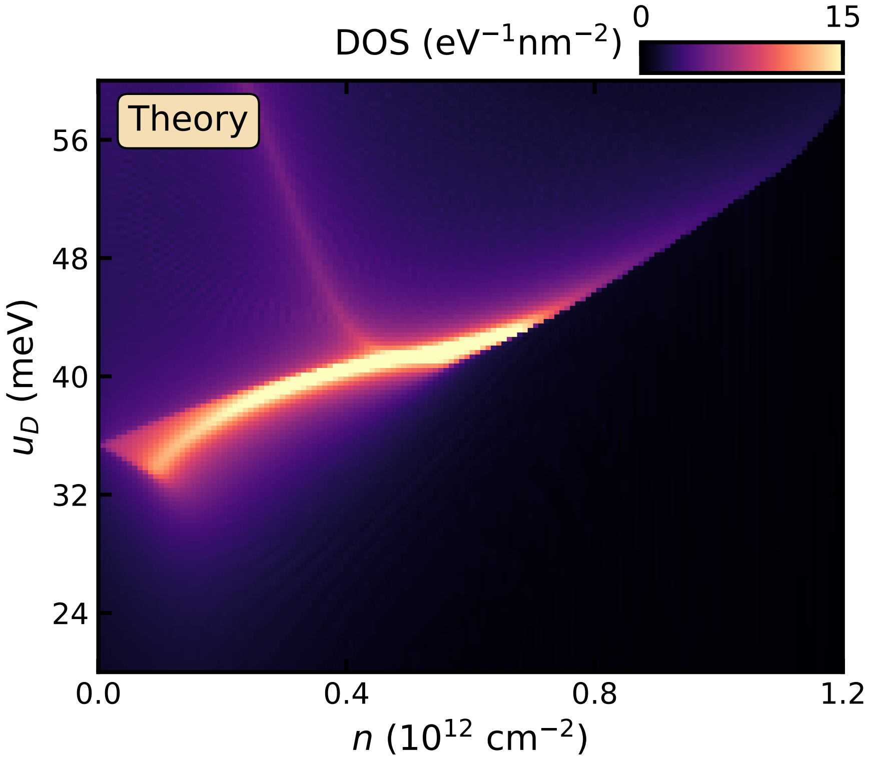

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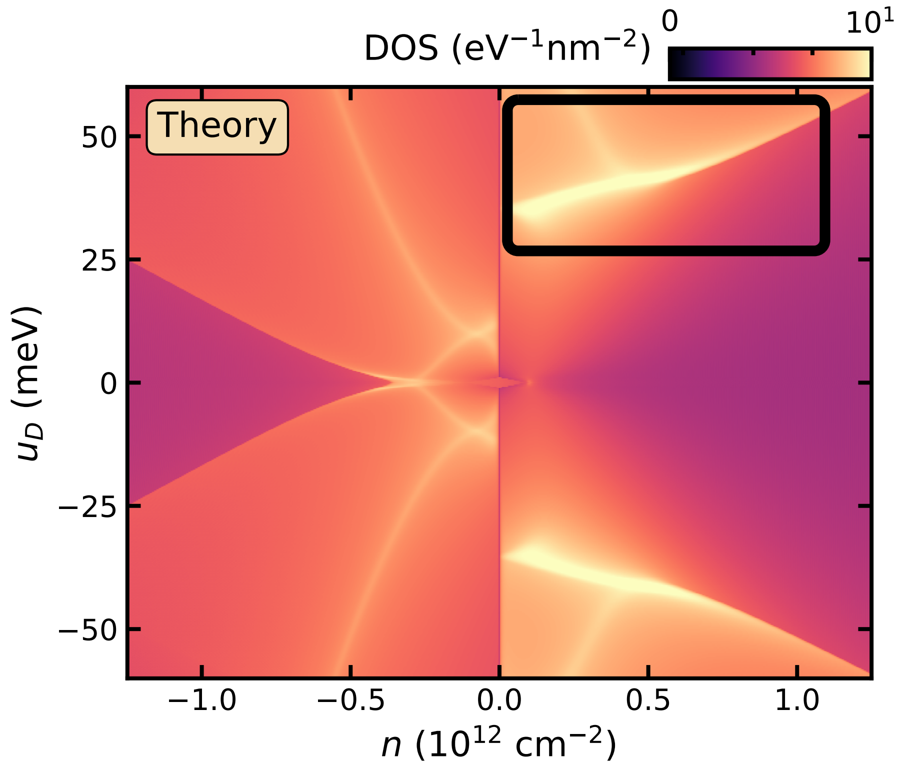

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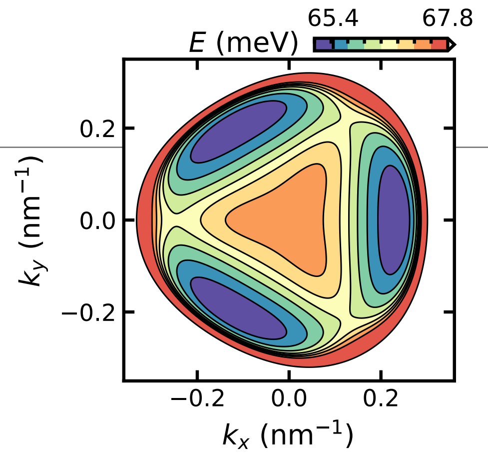

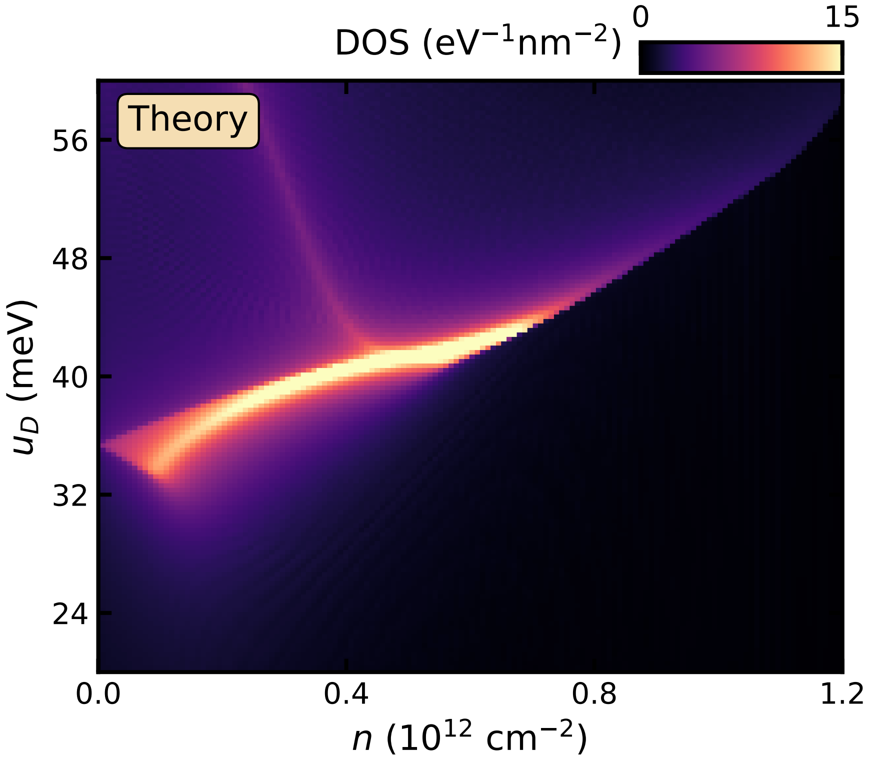

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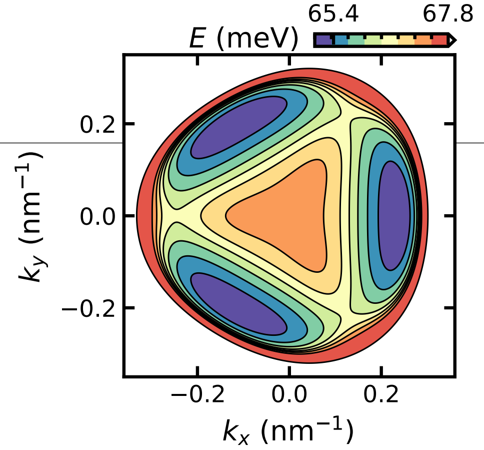

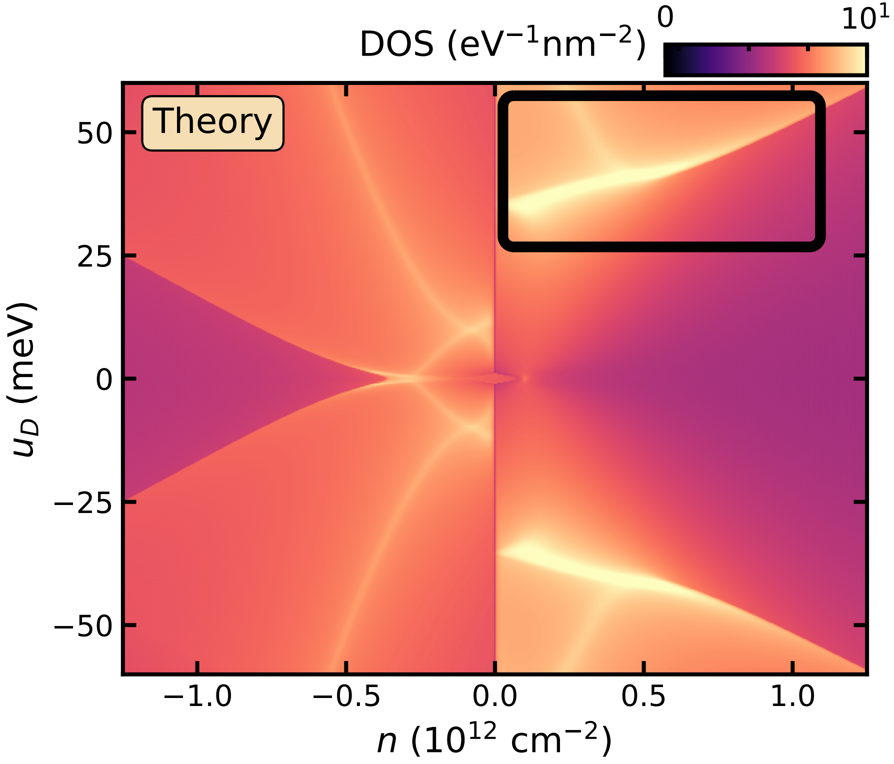

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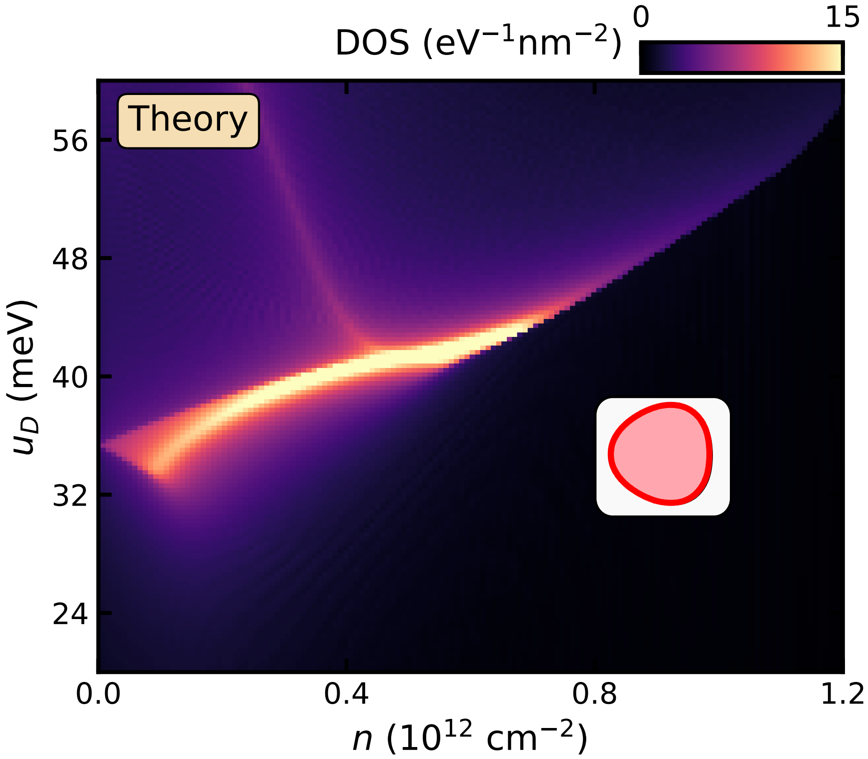
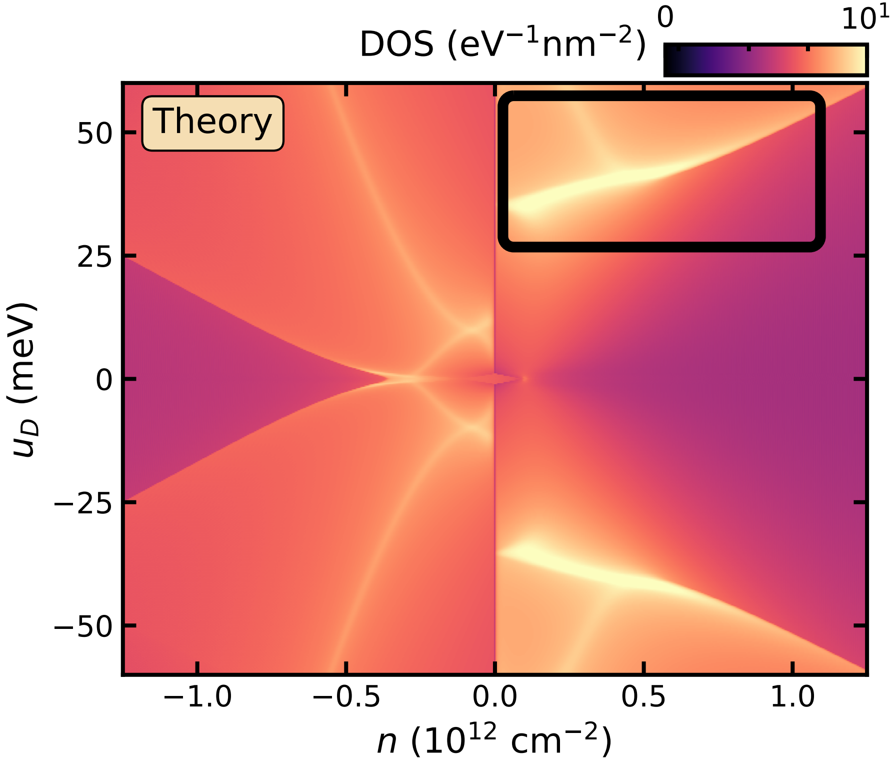


R4G Conduction Band Fermiology



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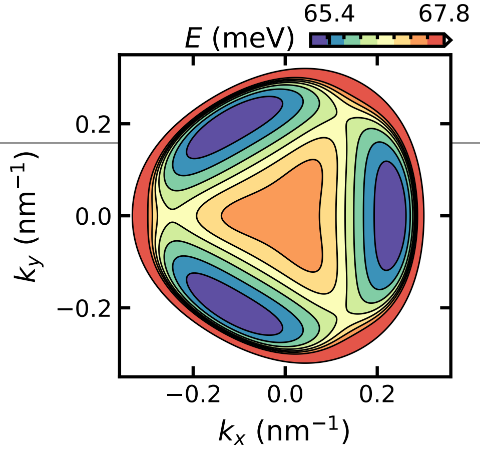

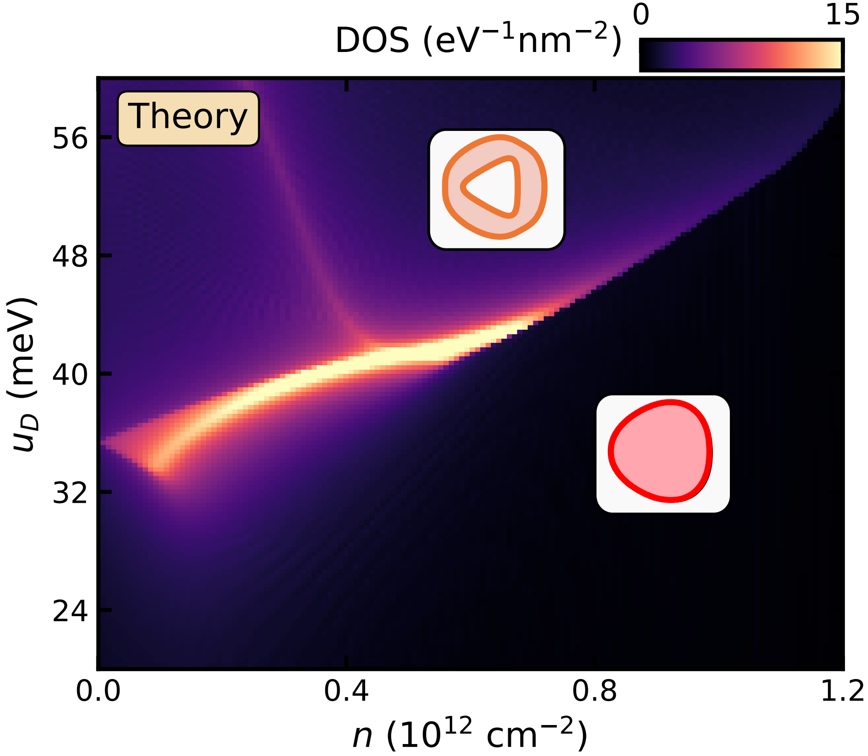

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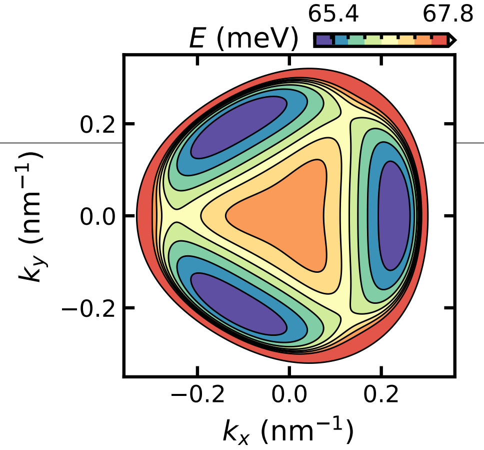

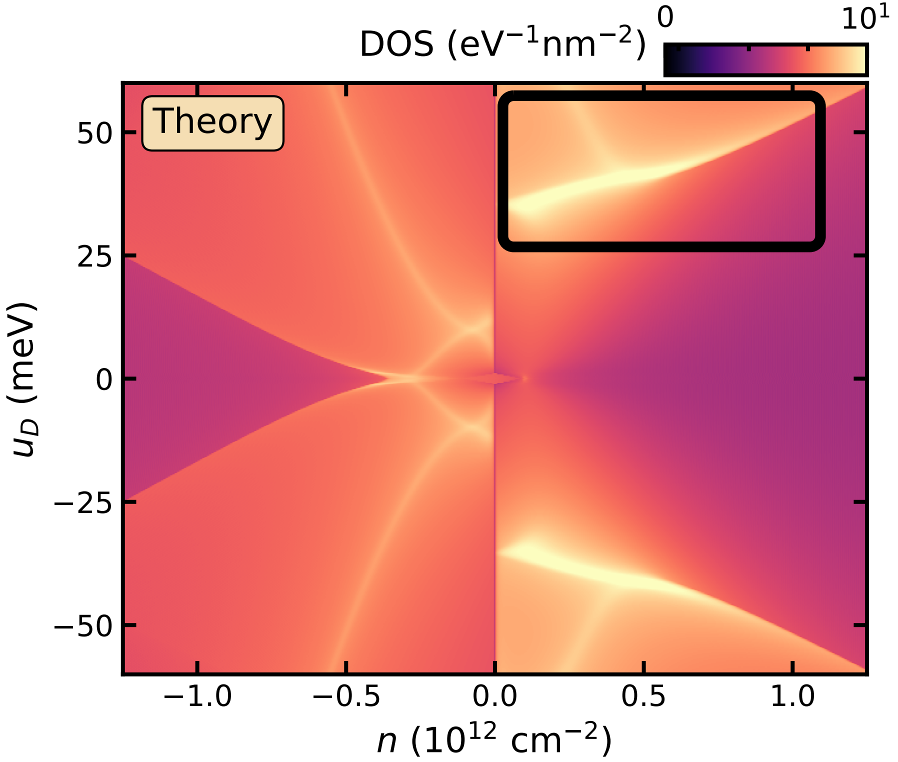

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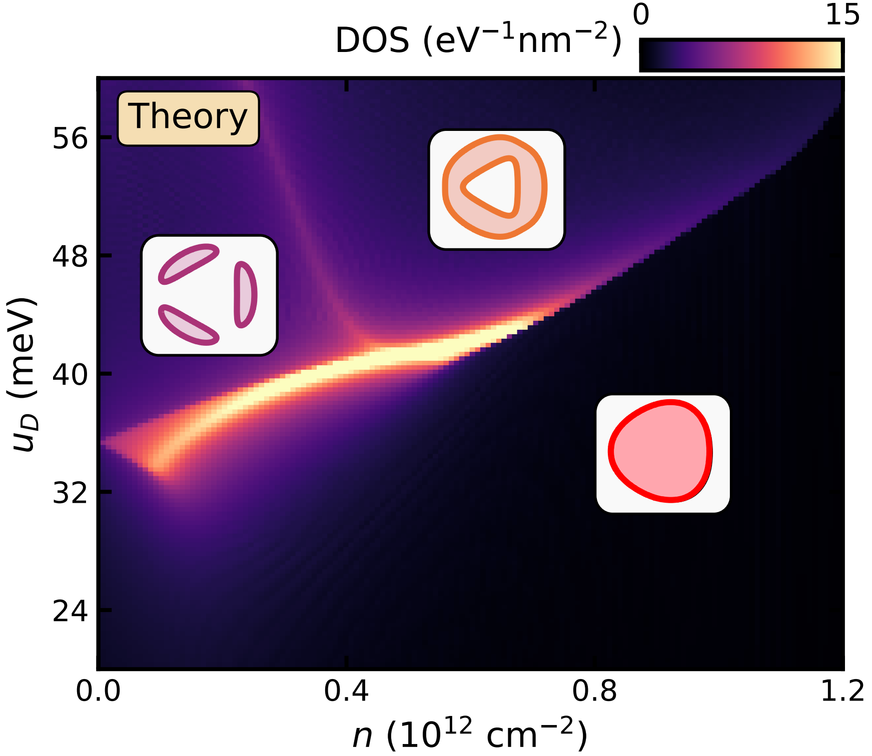

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Experimental Phase Diagram of 4-layer Rhombohedral

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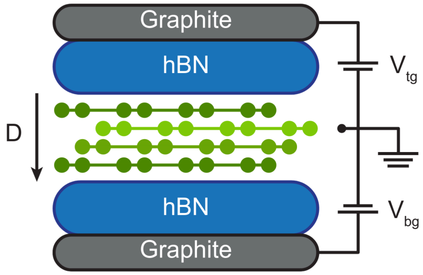

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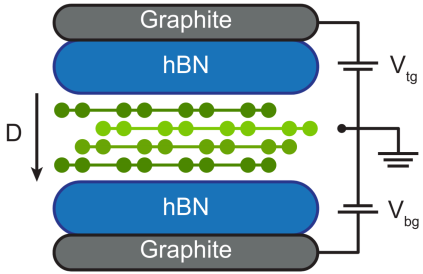

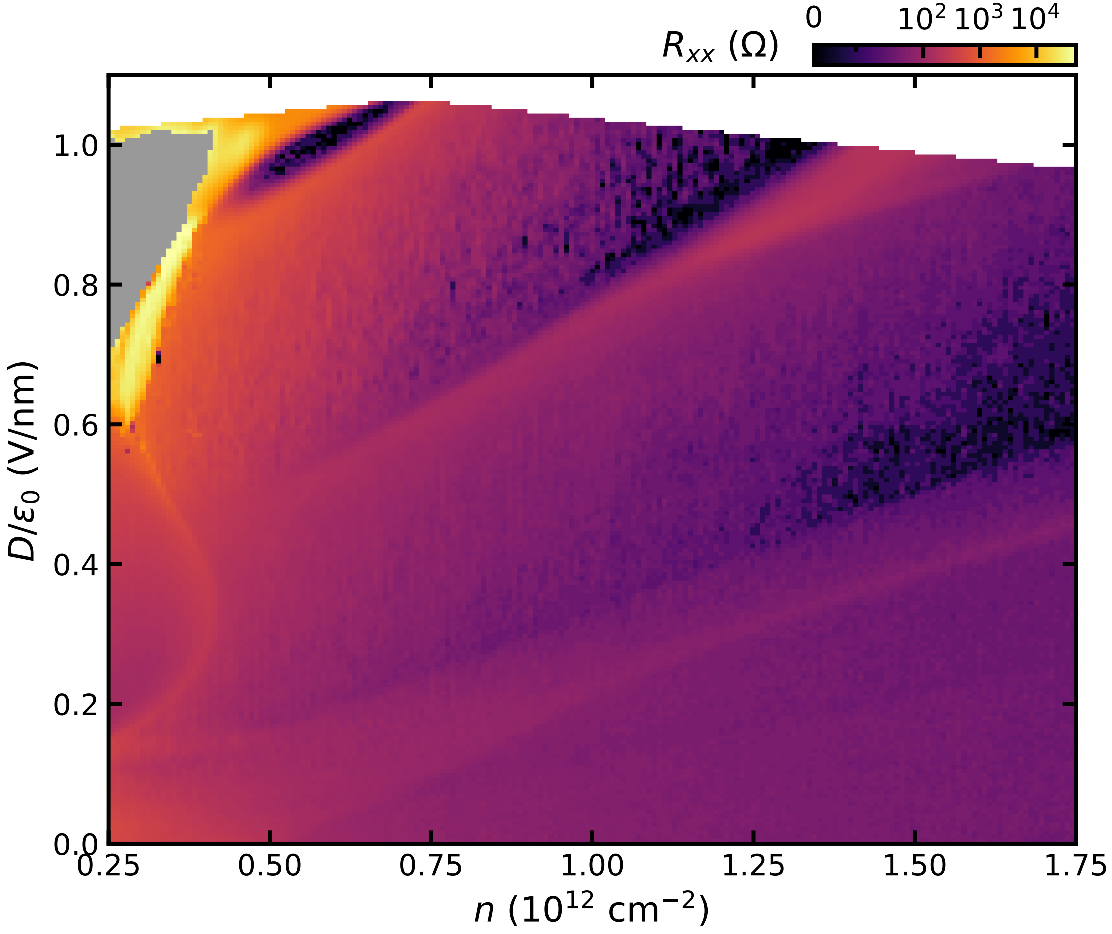

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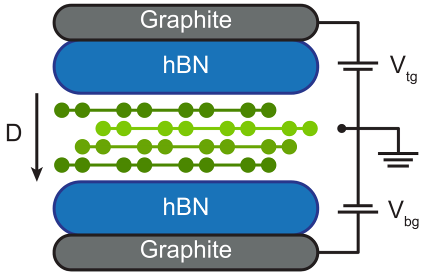

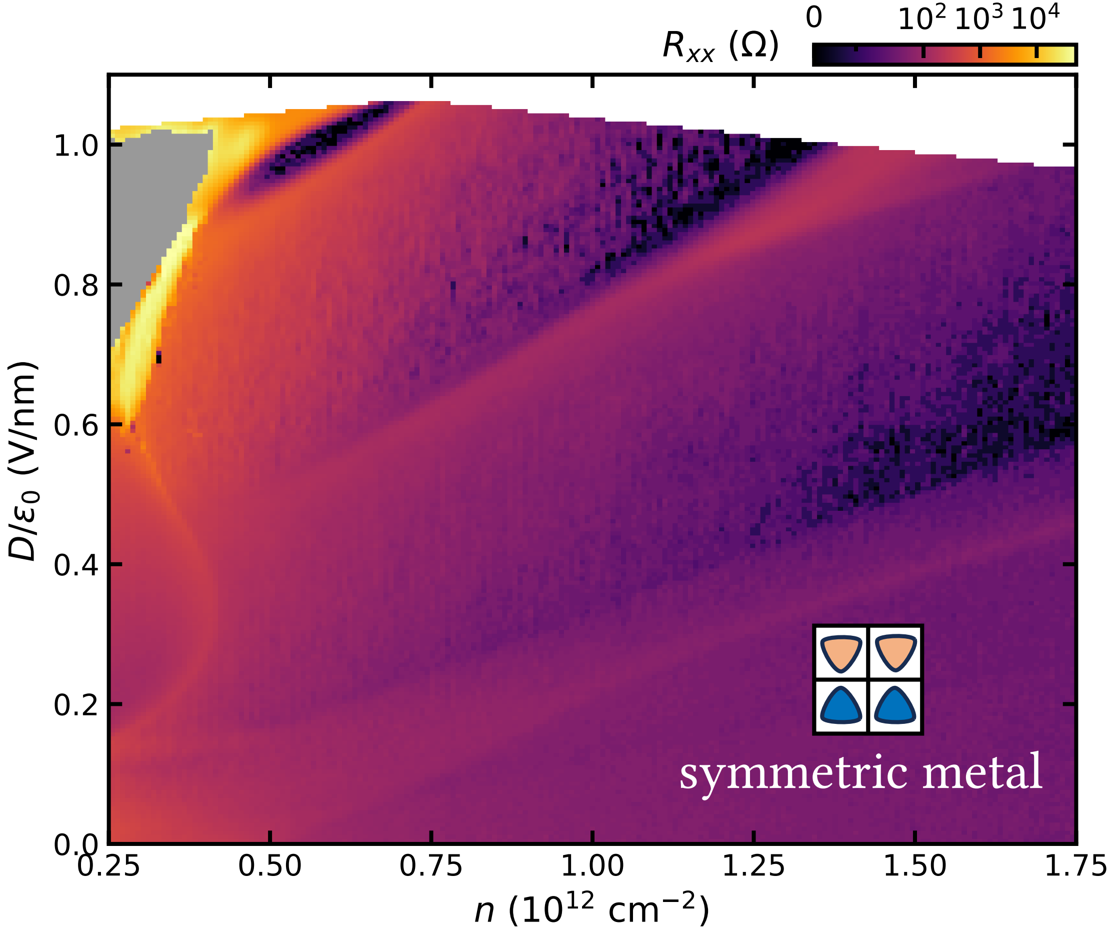

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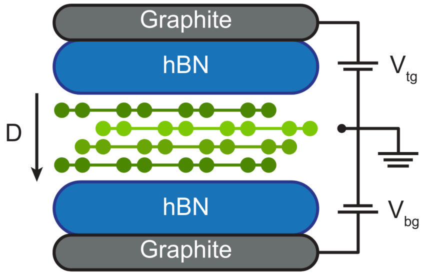

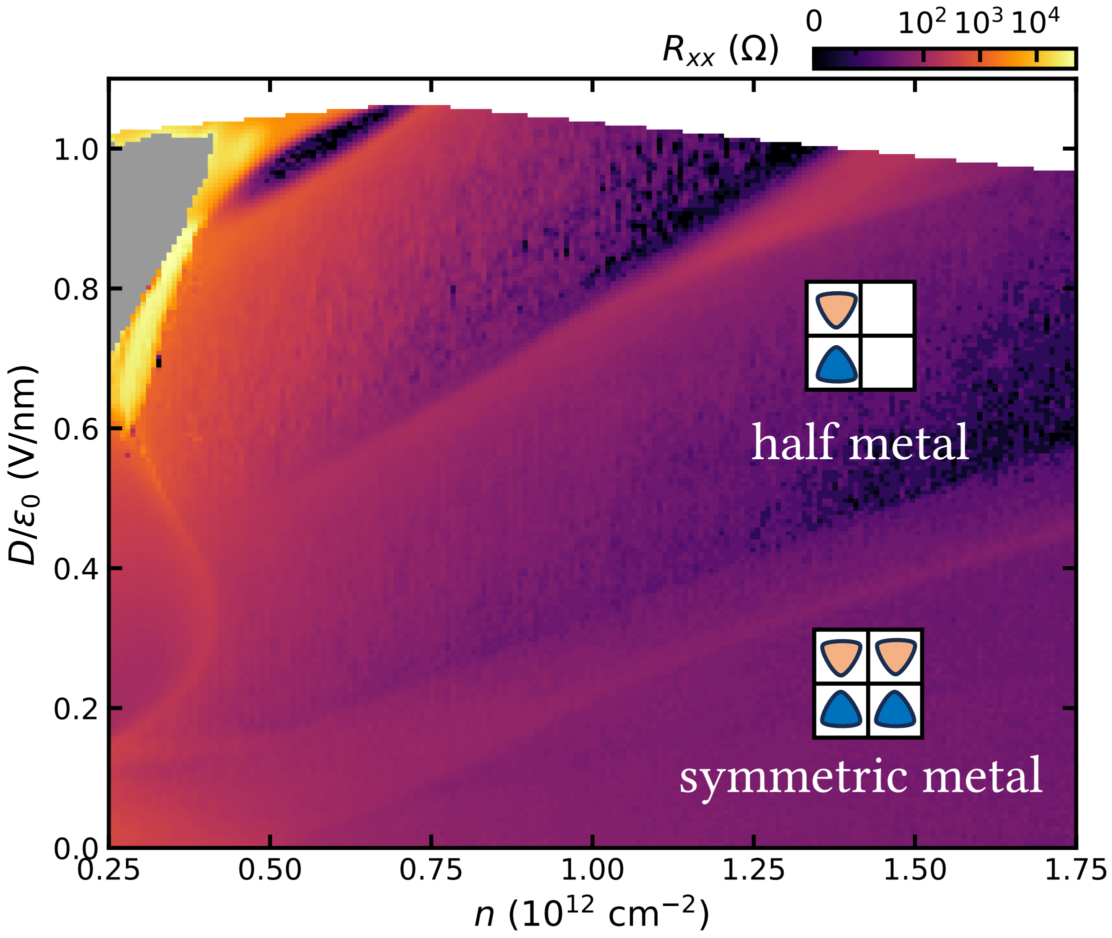

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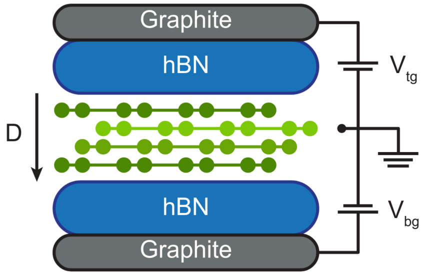

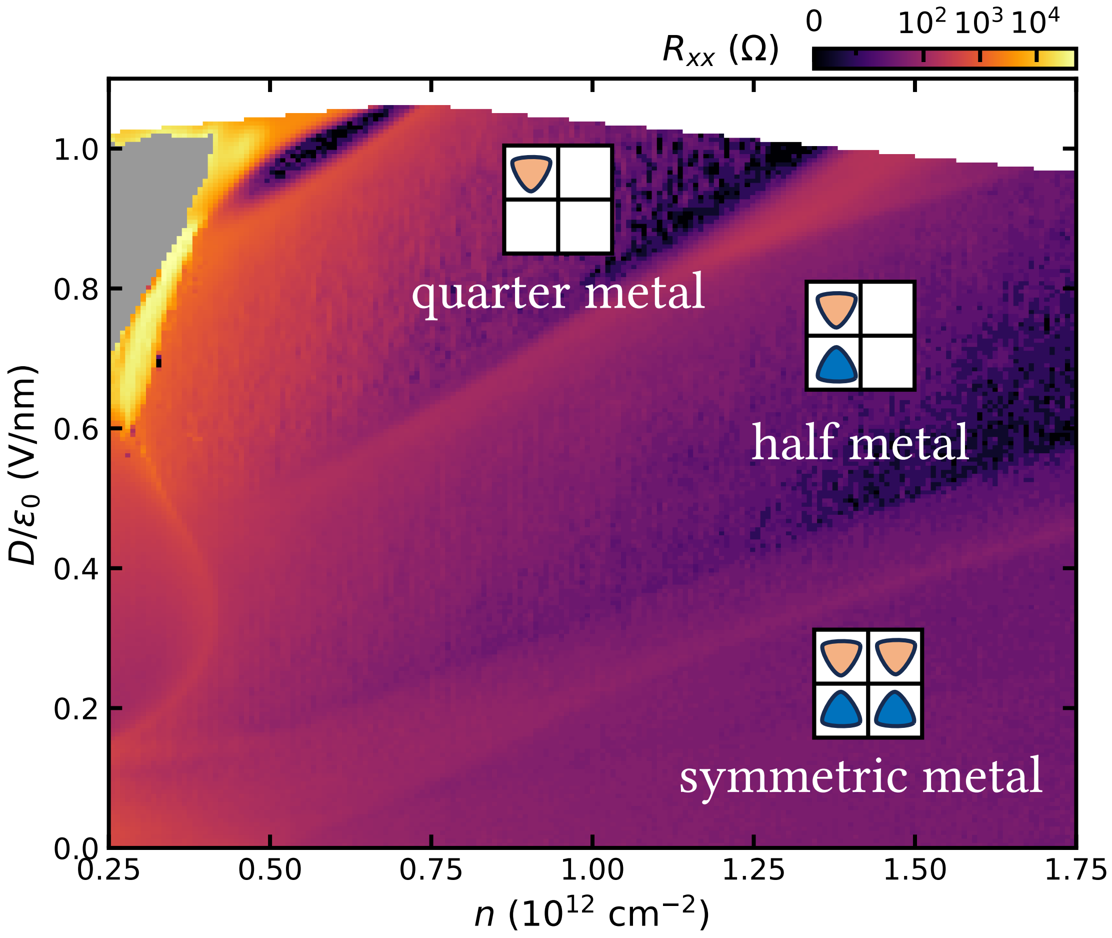

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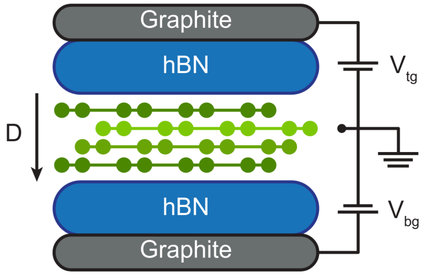

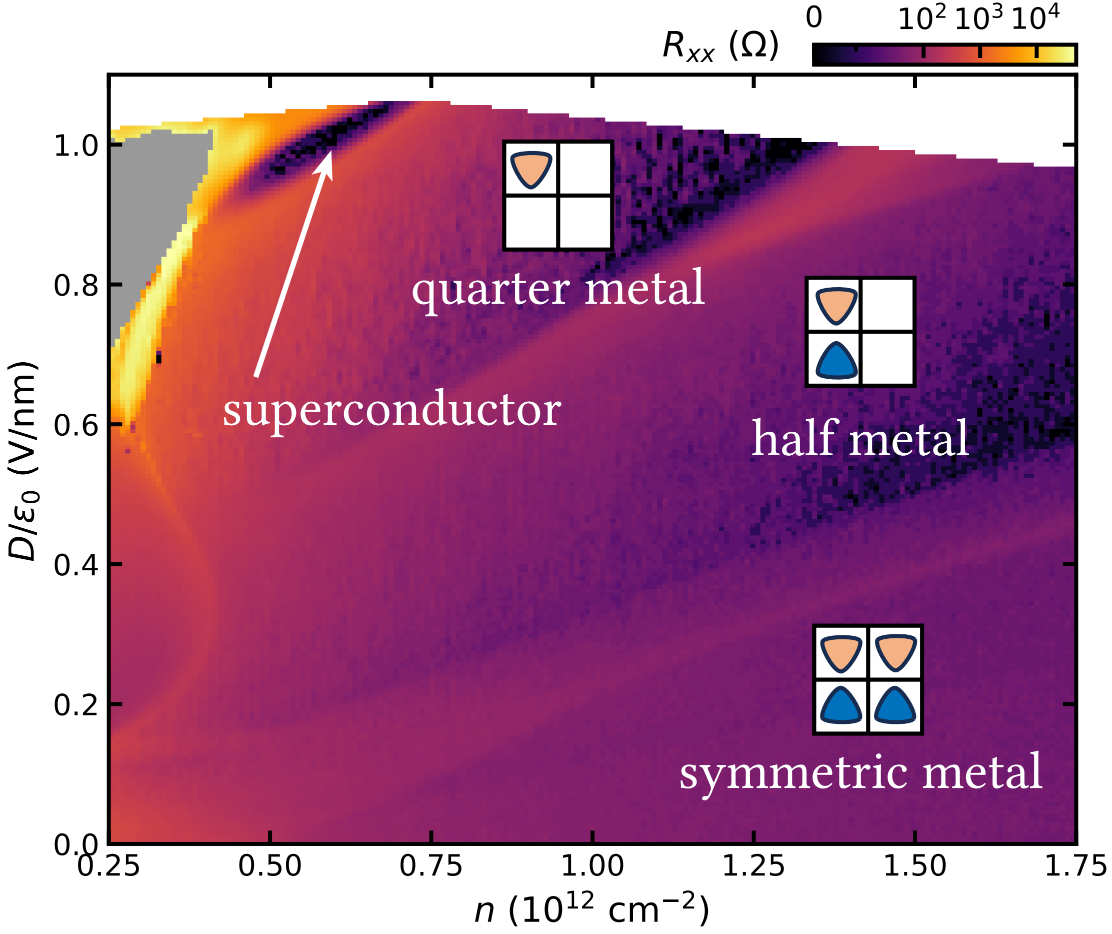

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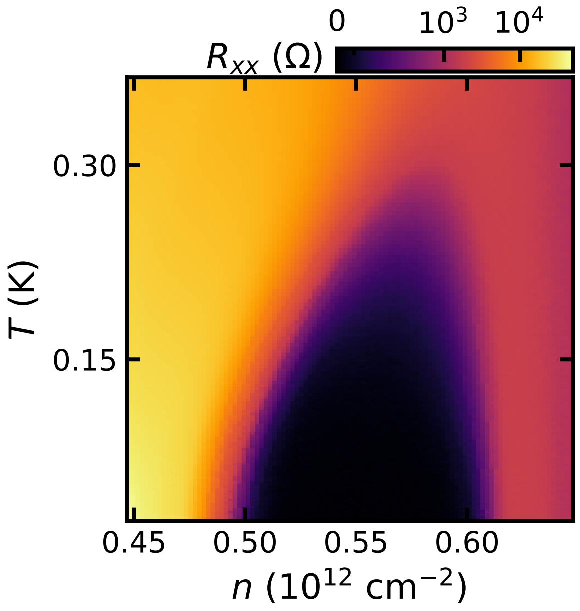
Basic Phenomenology of the Candidate CSC



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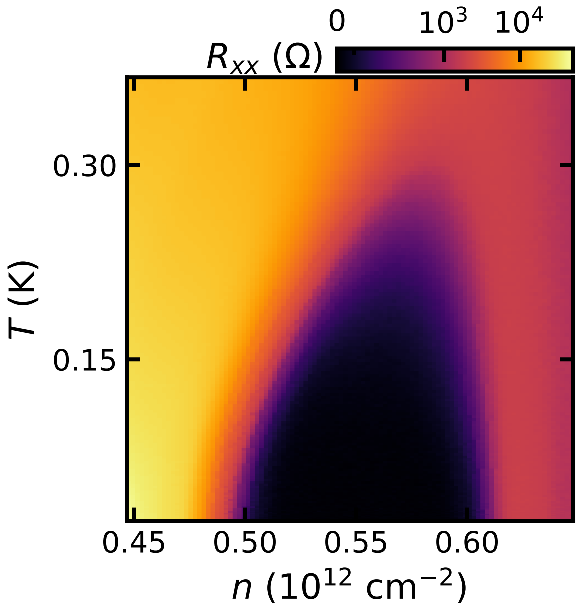

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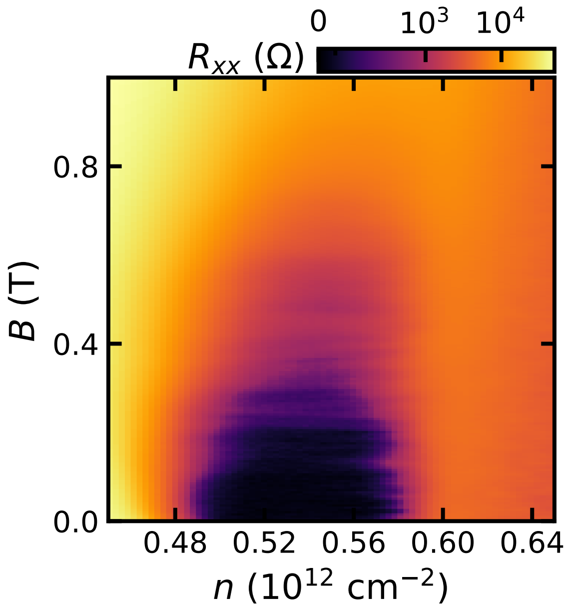

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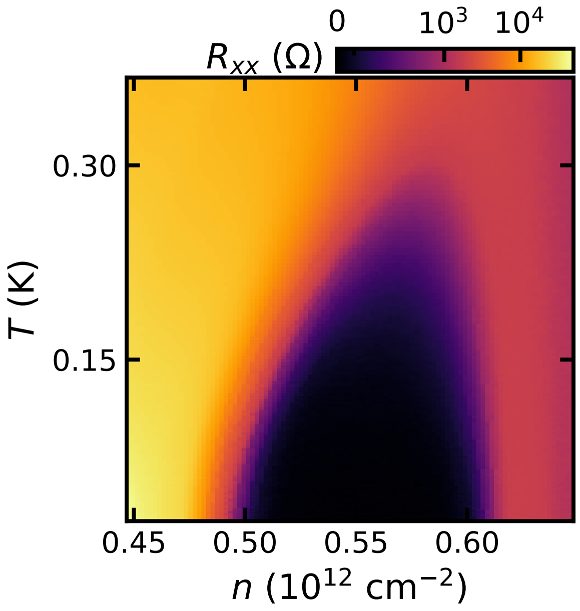

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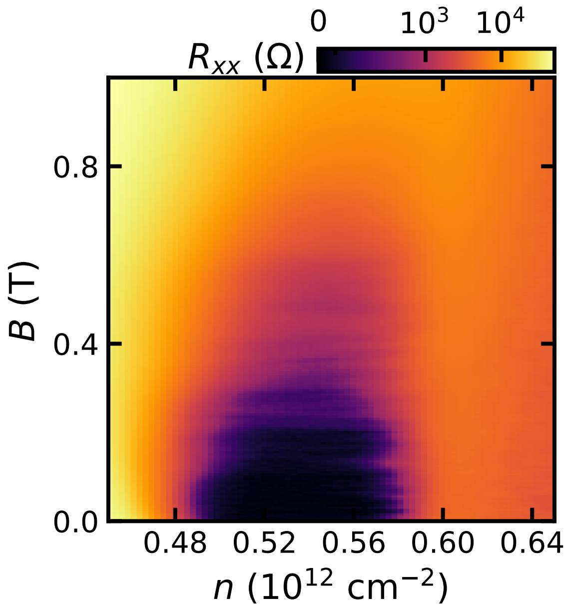

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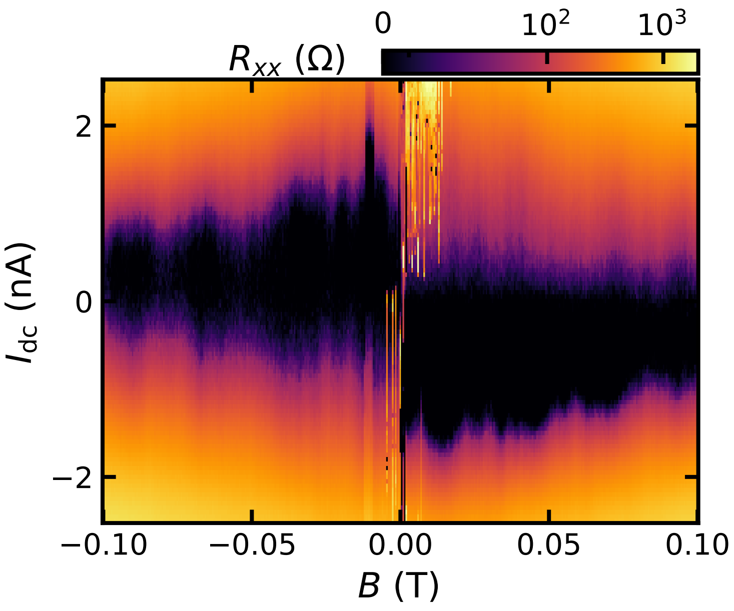

Phase coherence

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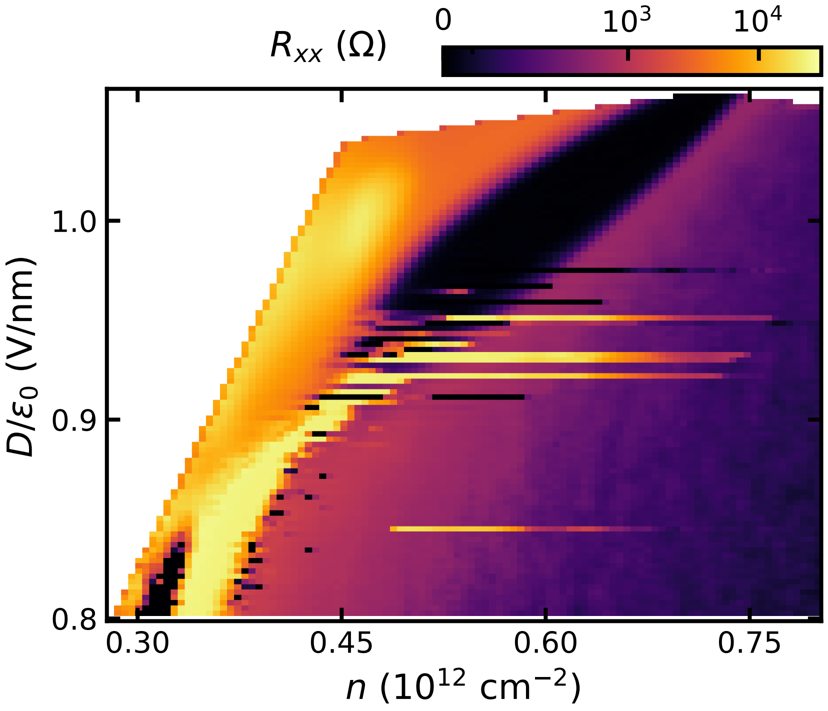
Time-Reversal Symmetry Breaking



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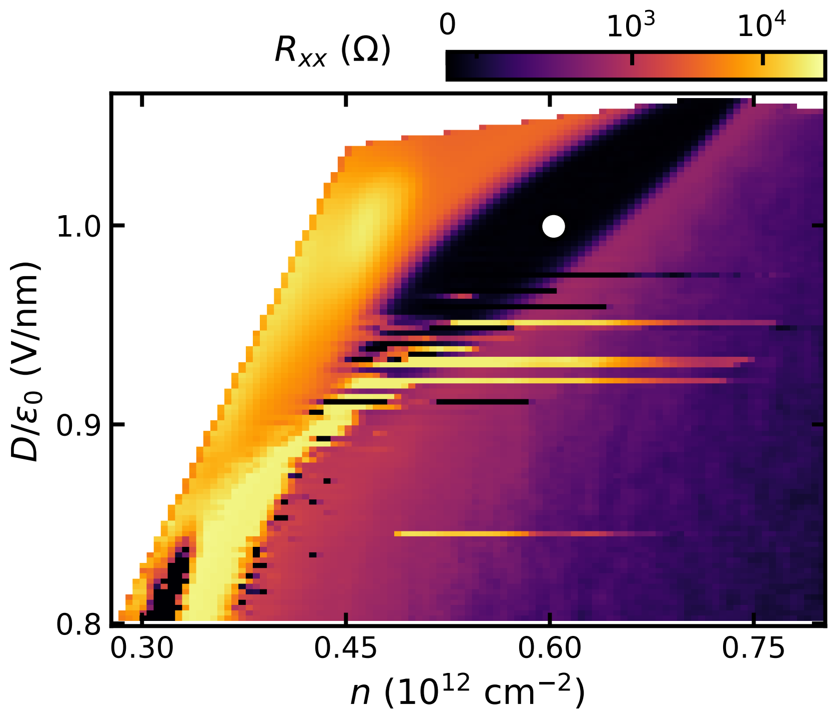

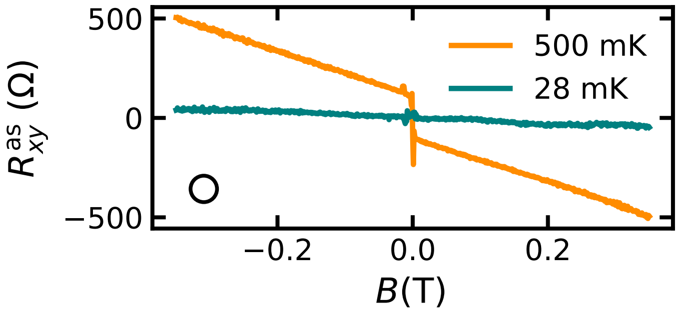

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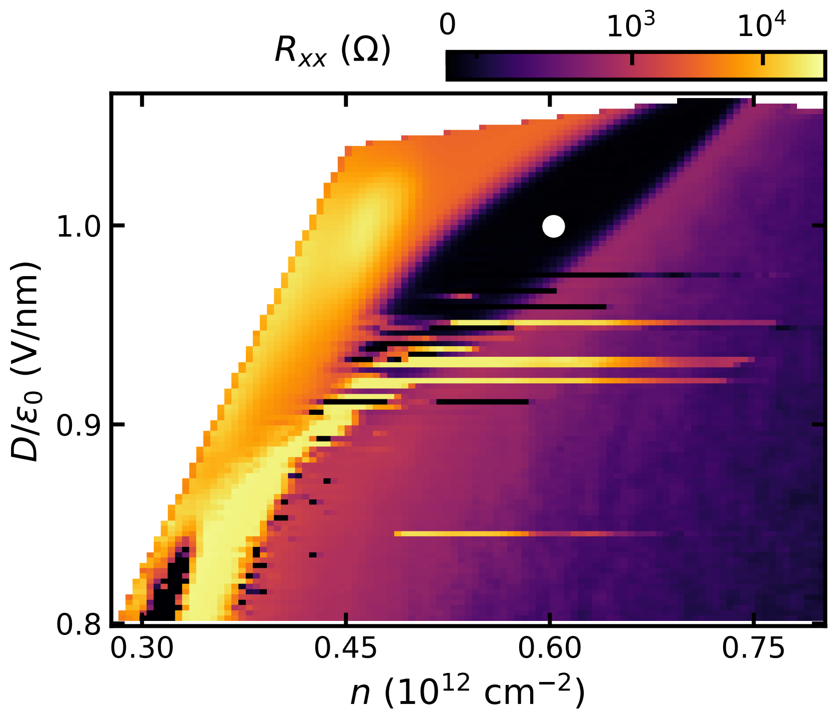

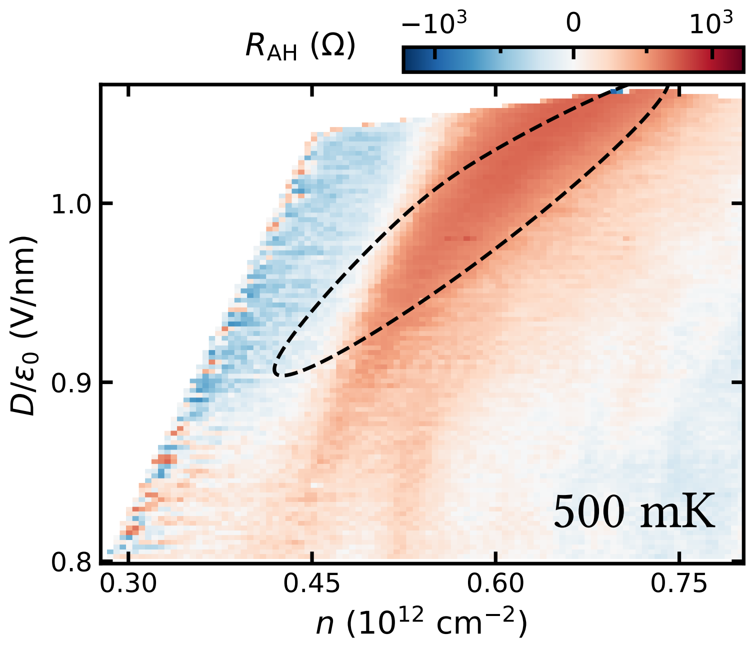

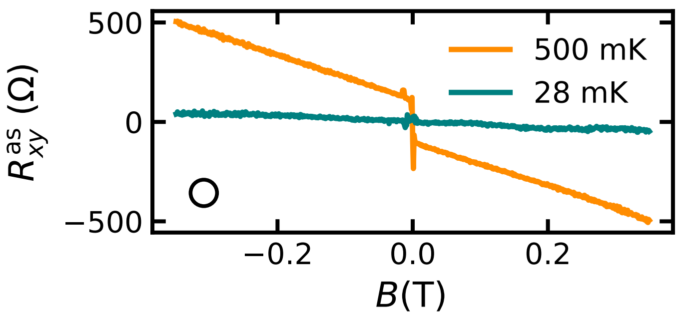

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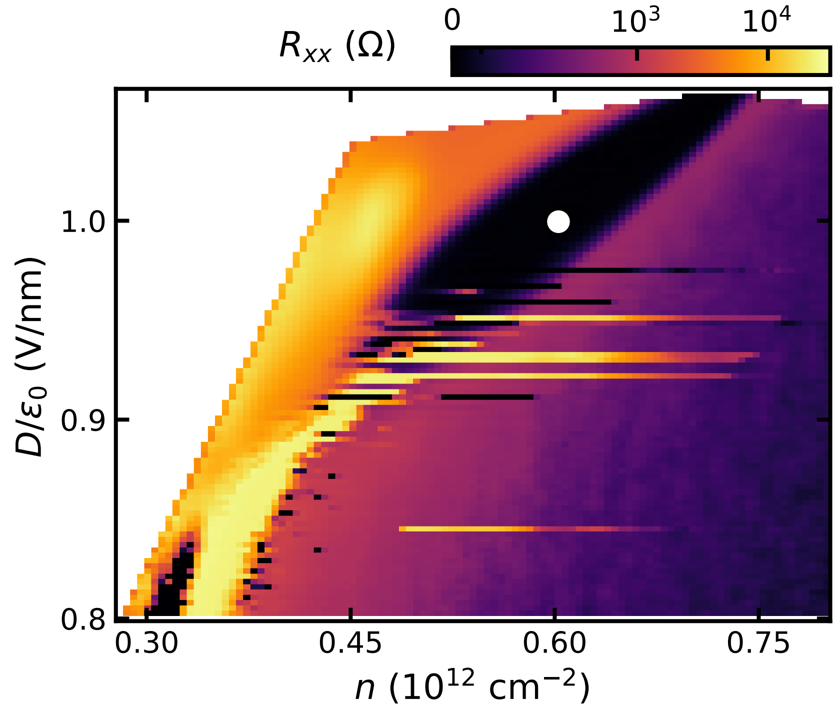

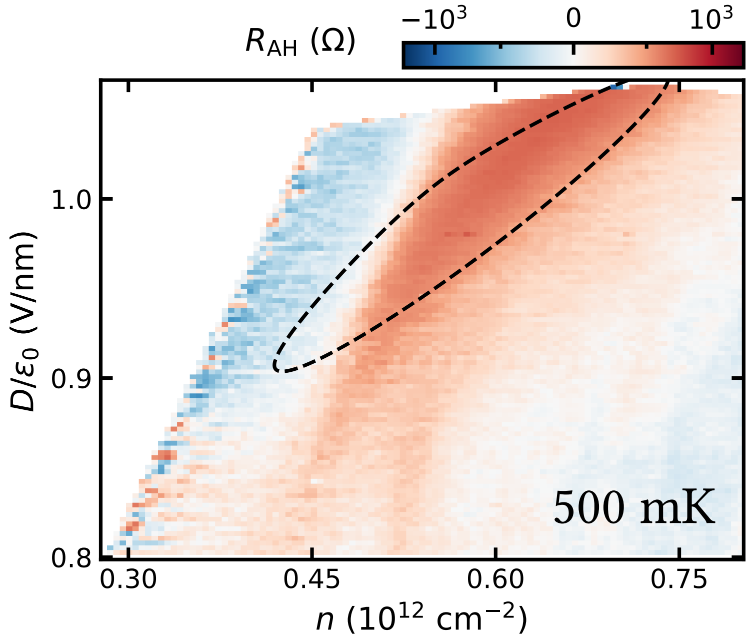

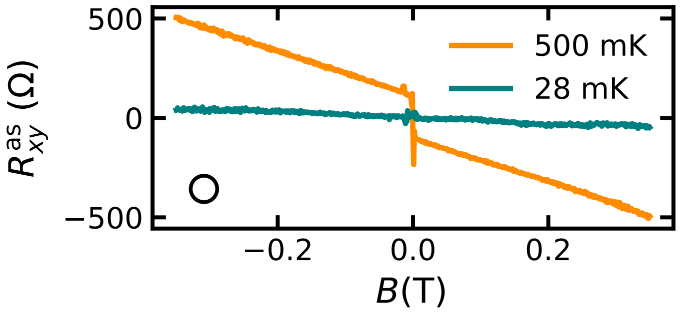

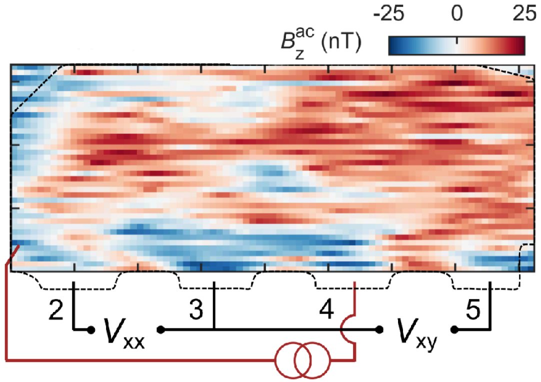

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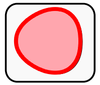
  - e.g. $p_x \pm ip_y$ pairing of 

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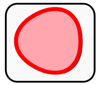
  - e.g. $p_x \pm ip_y$ pairing of 
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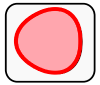
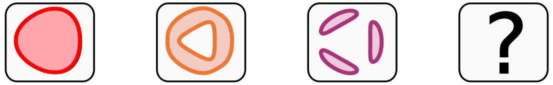
  - e.g. $p_x \pm ip_y$ pairing of 
- Mechanism driving superconductivity?
- Normal state? $\rightarrow$ quantum oscillations!



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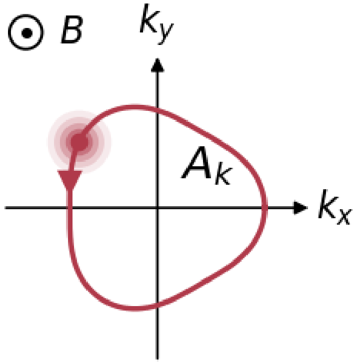

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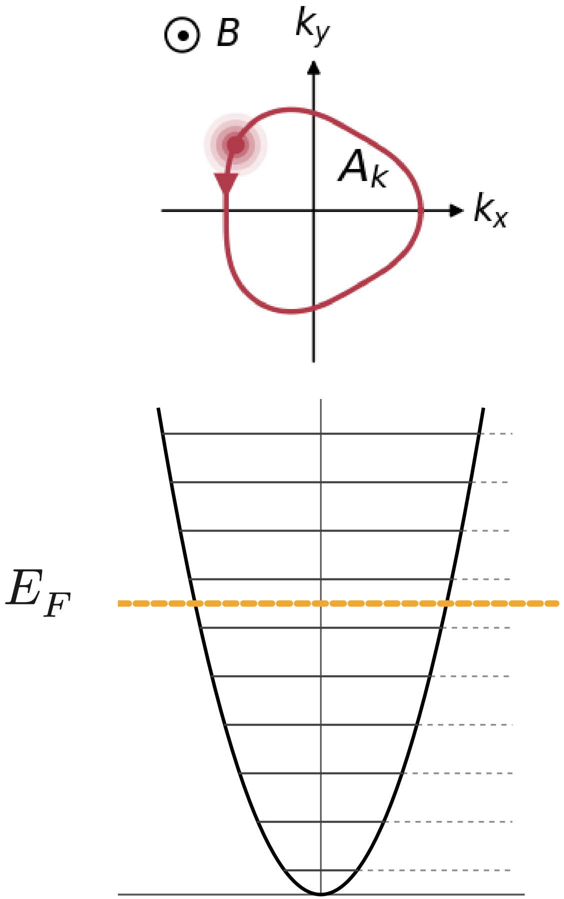

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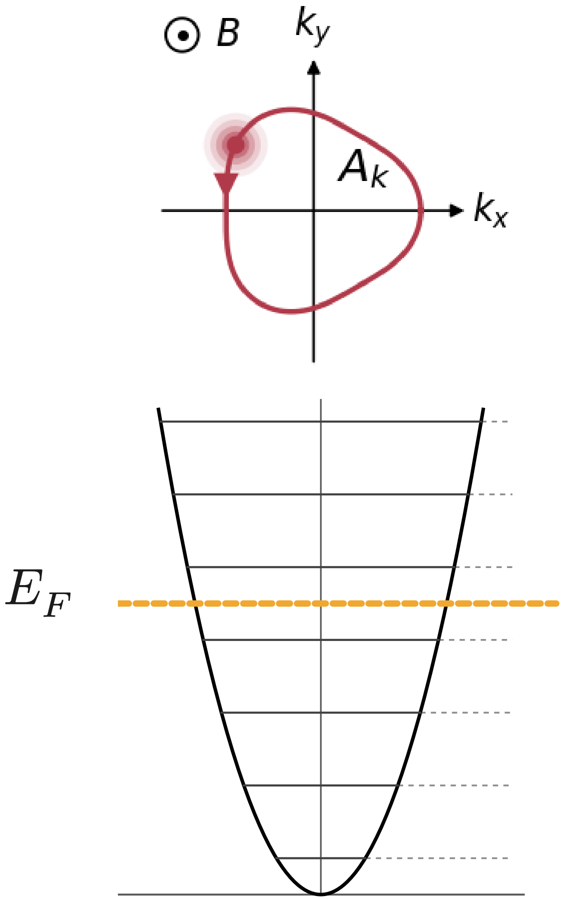

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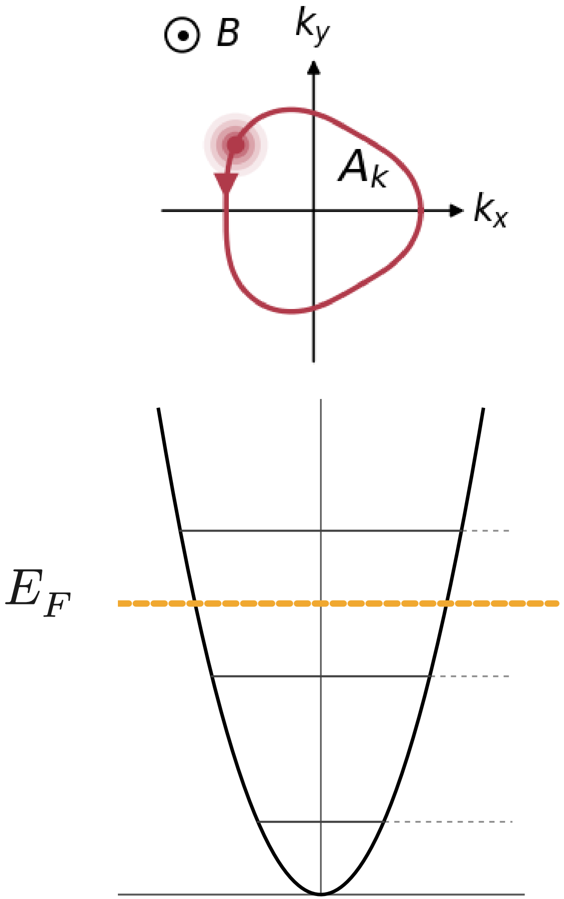

Landau levels evolve with magnetic field
$\rightarrow$ DOS at E_F periodically with $1/B$

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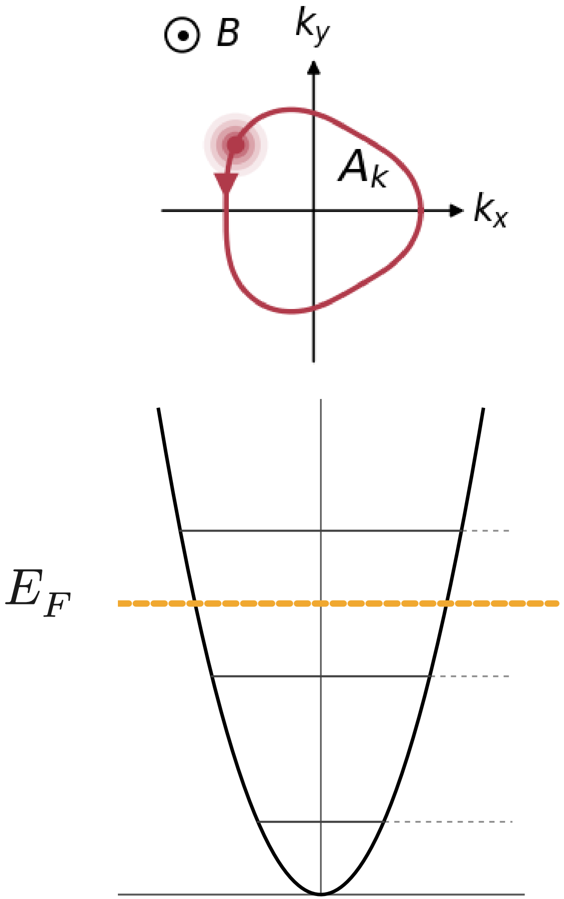

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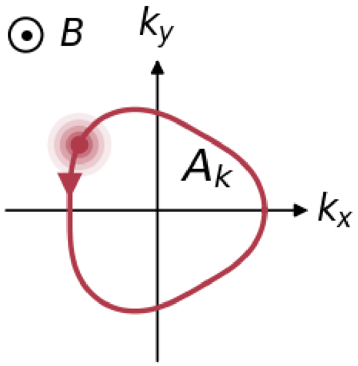

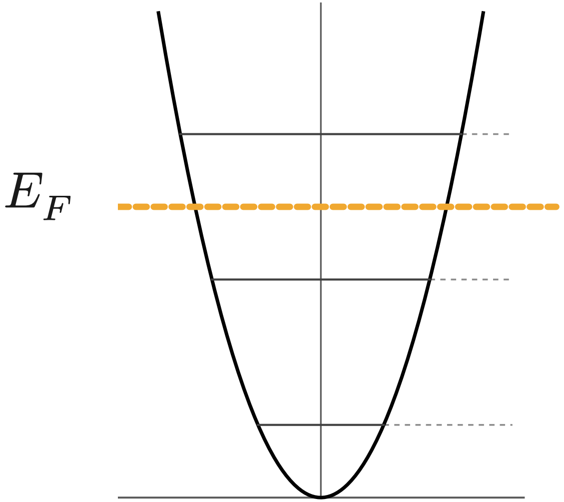

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- de Haas-van Alphen (dHvA): magnetic susceptibility
- **Shubnikov-de Haas (SdH):** resistivity

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Semiclassical (Onsager) Quantization

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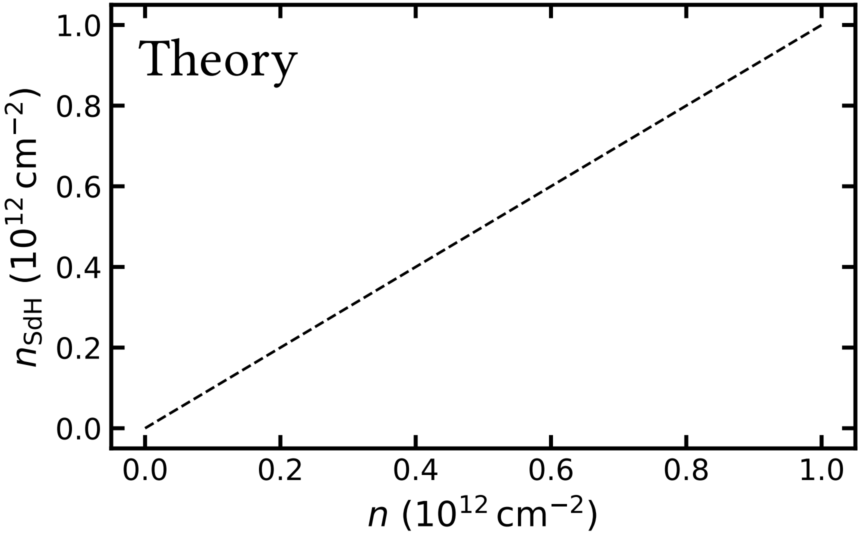

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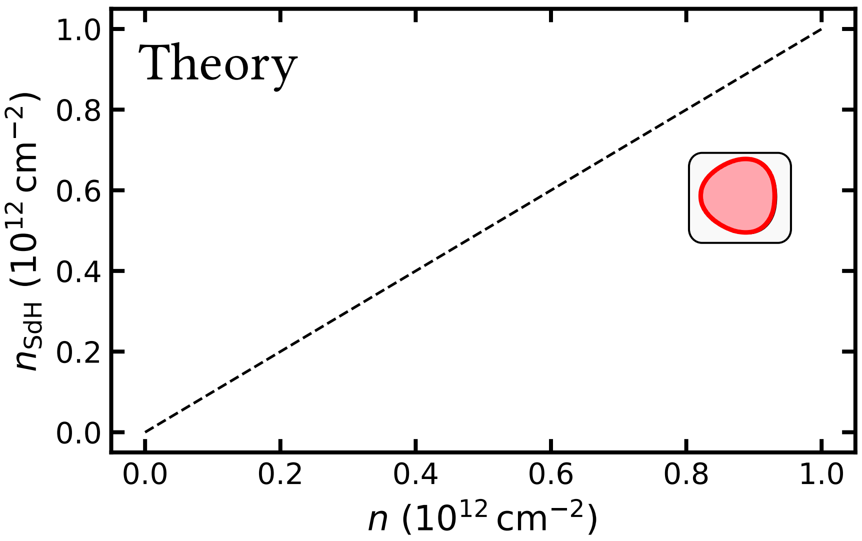

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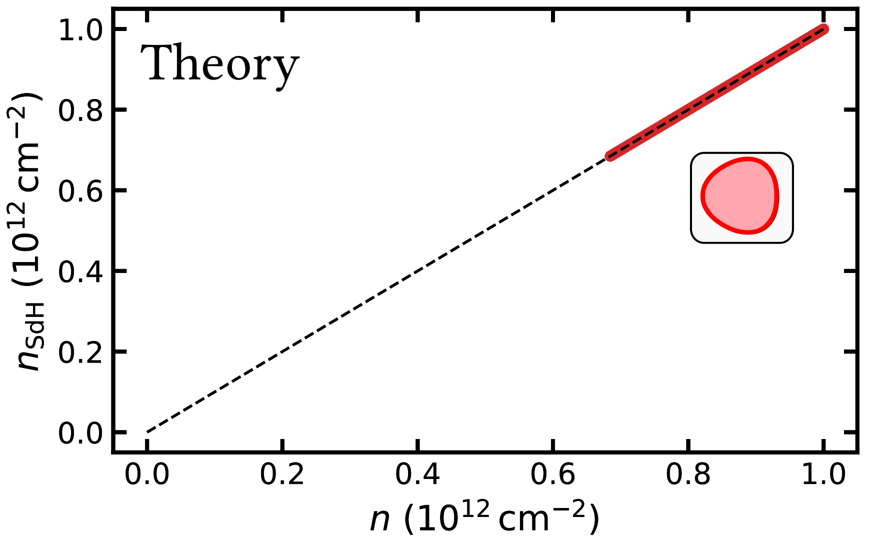

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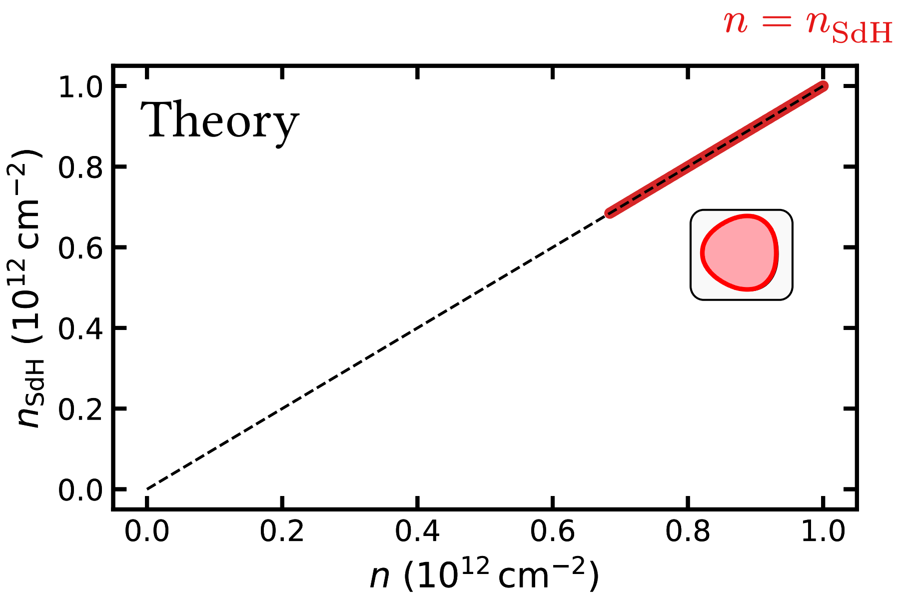

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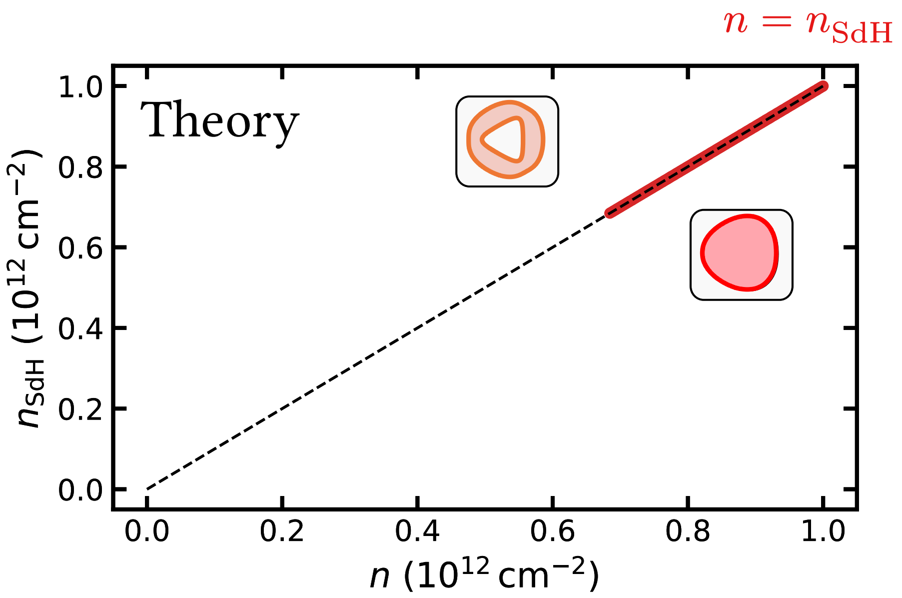

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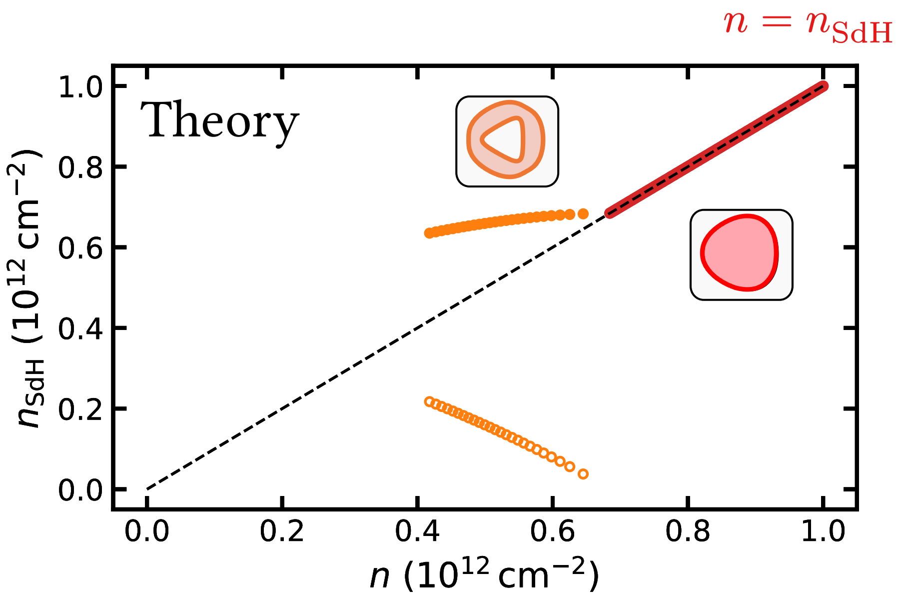

Onsager Phil. Mag. (1952)
Shoenberg Magnetic Oscillations in Metals (1984)

Semiclassical (Onsager) Quantization

- Effective density enclosed by orbit:

$$n_{\mathrm{SdH}} = \frac{A_k}{(2\pi)^2} = \frac{ef}{h}$$

- All carriers participating in quantum oscillations → **Luttinger sum rule**

$$n = \sum_j \theta_j g_j n_{\mathrm{SdH},j}$$

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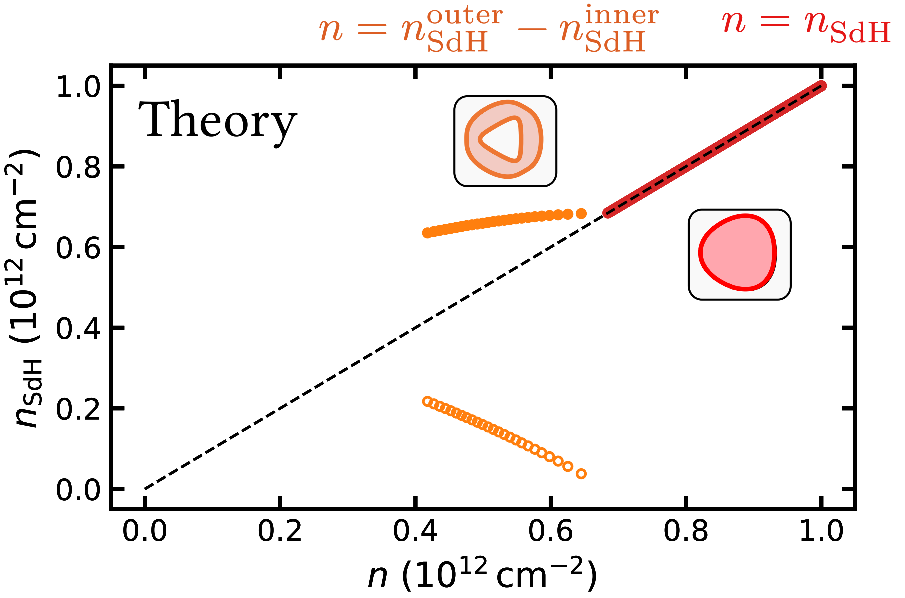

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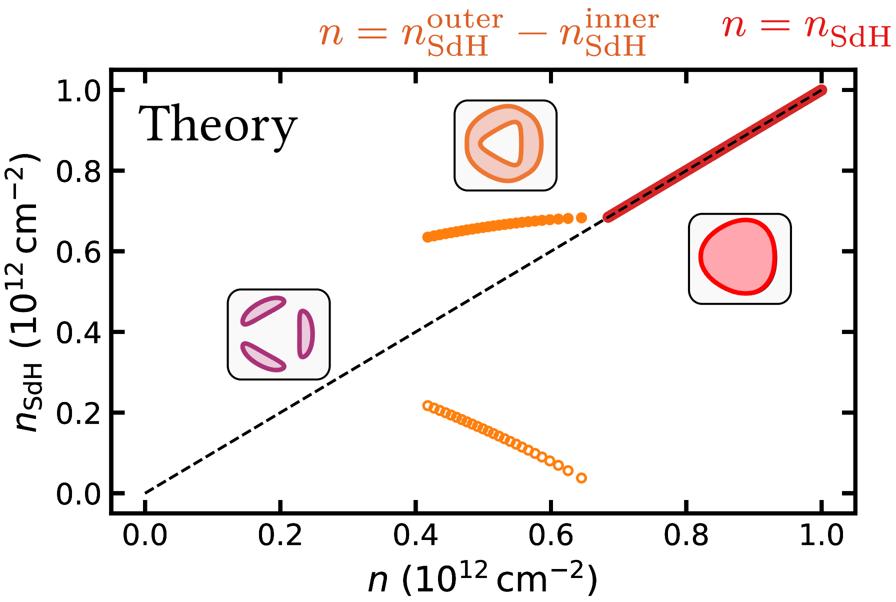

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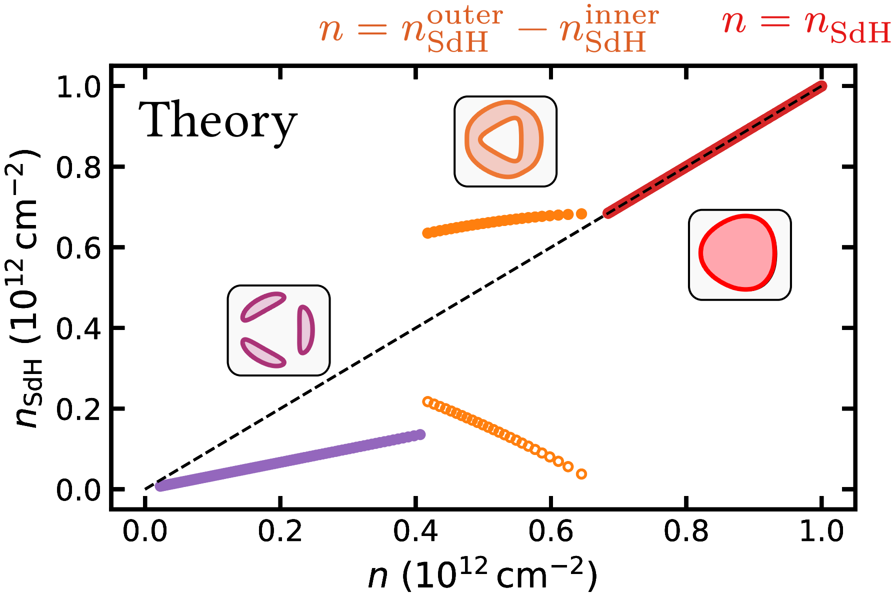

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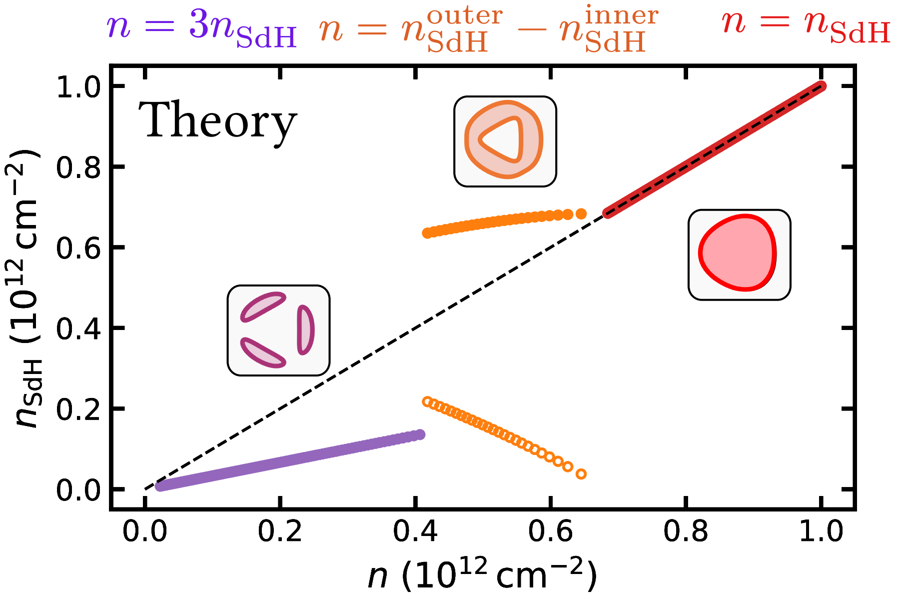

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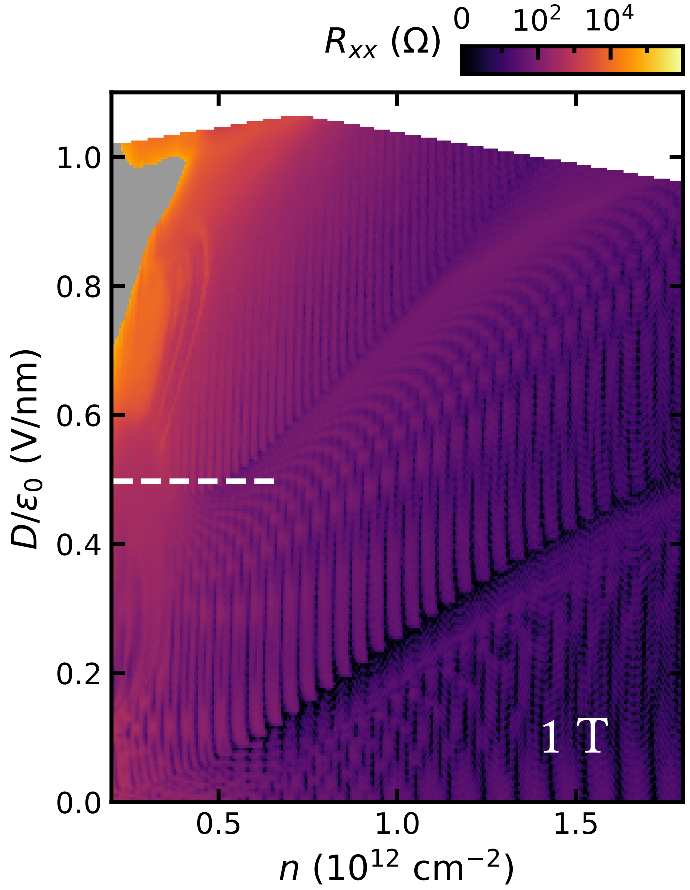
Flavor Transitions



Kalantre, ..., **Sharpe** arXiv:2606.05356

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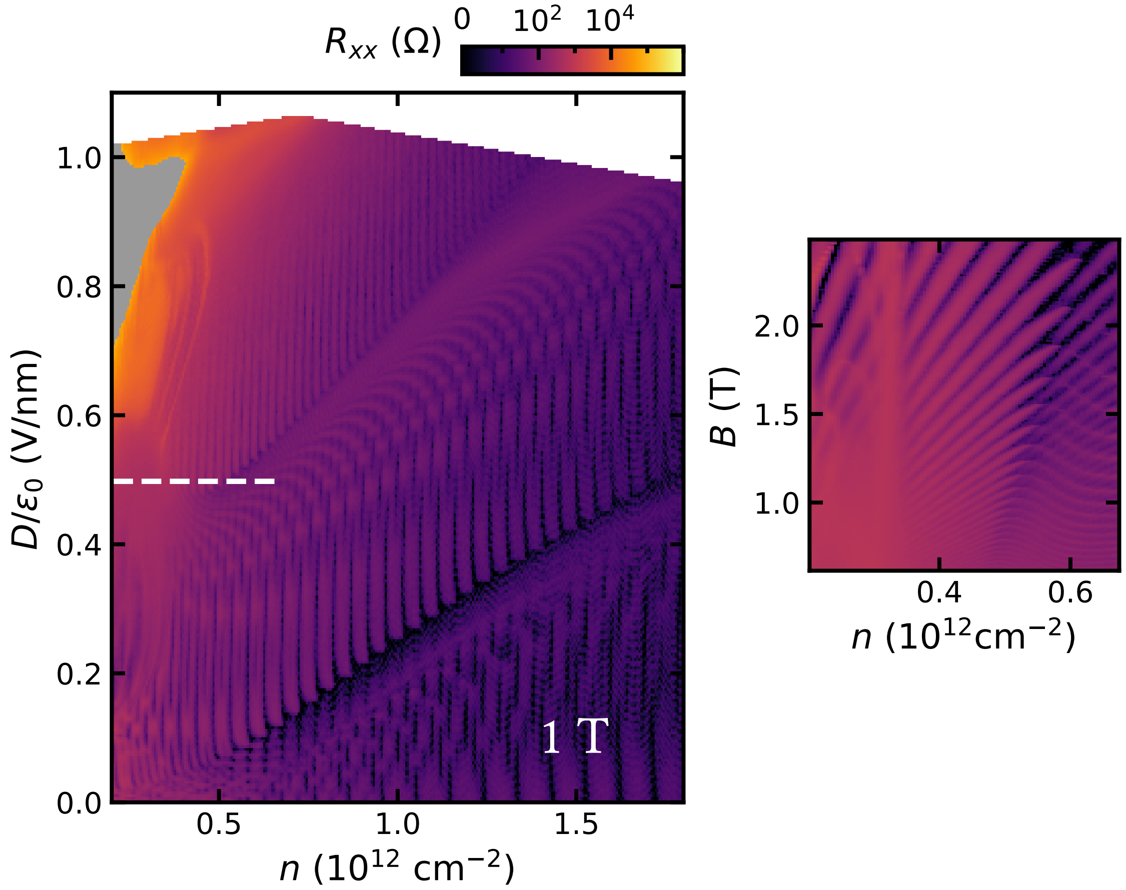

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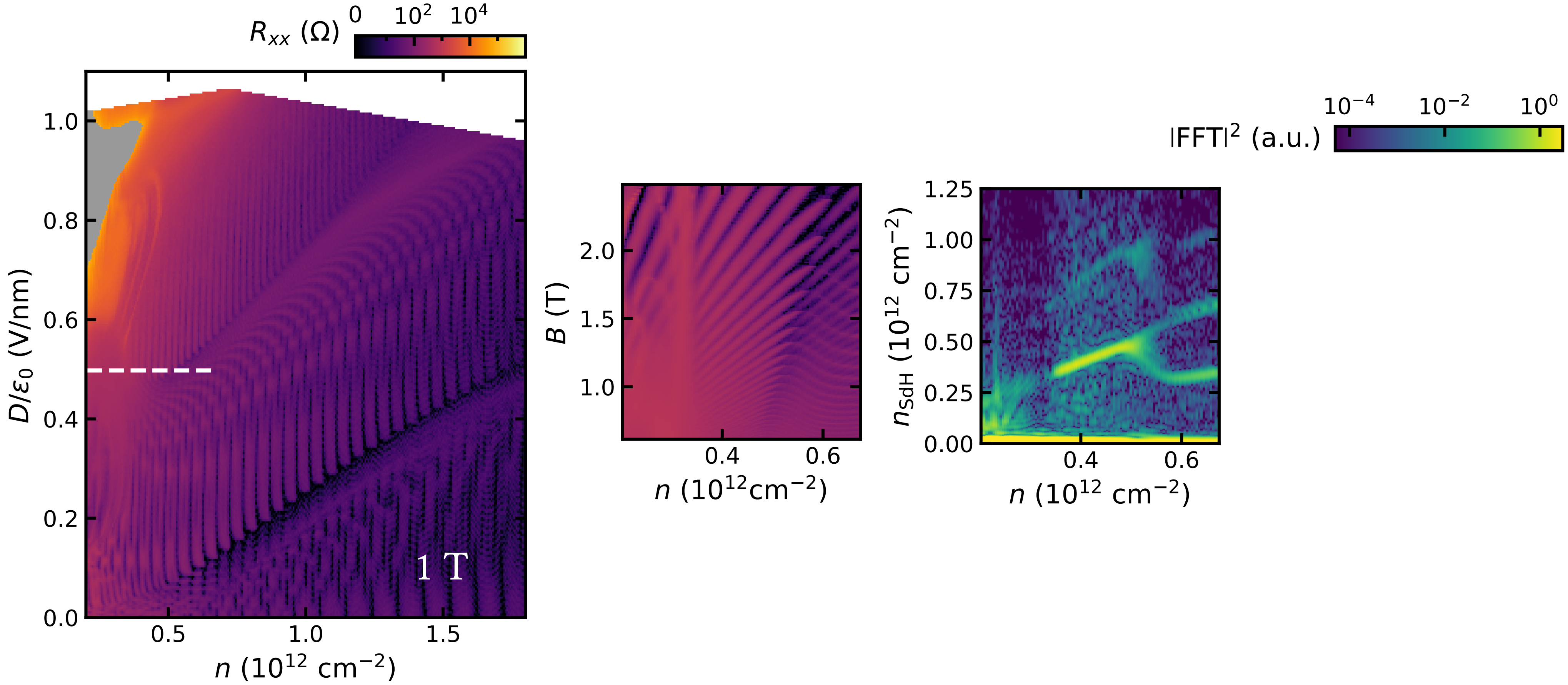

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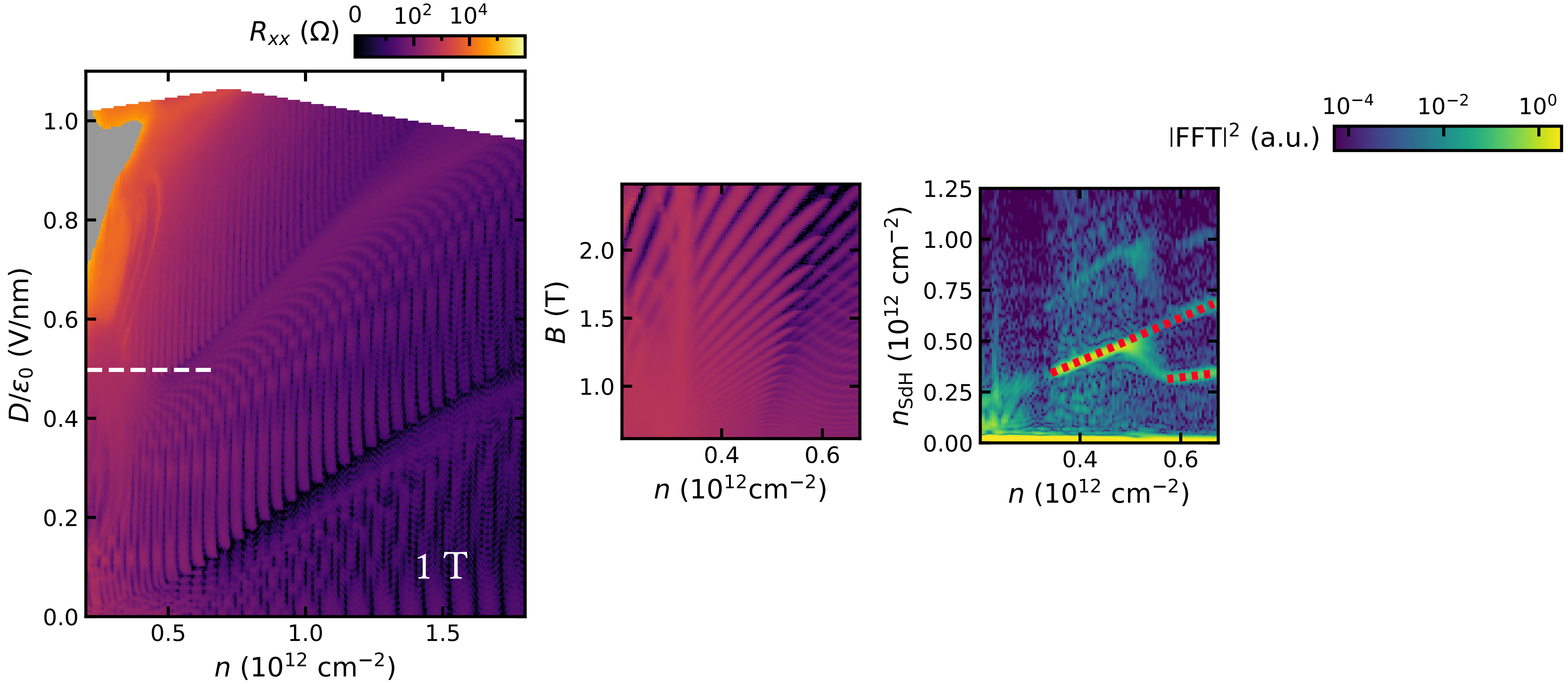

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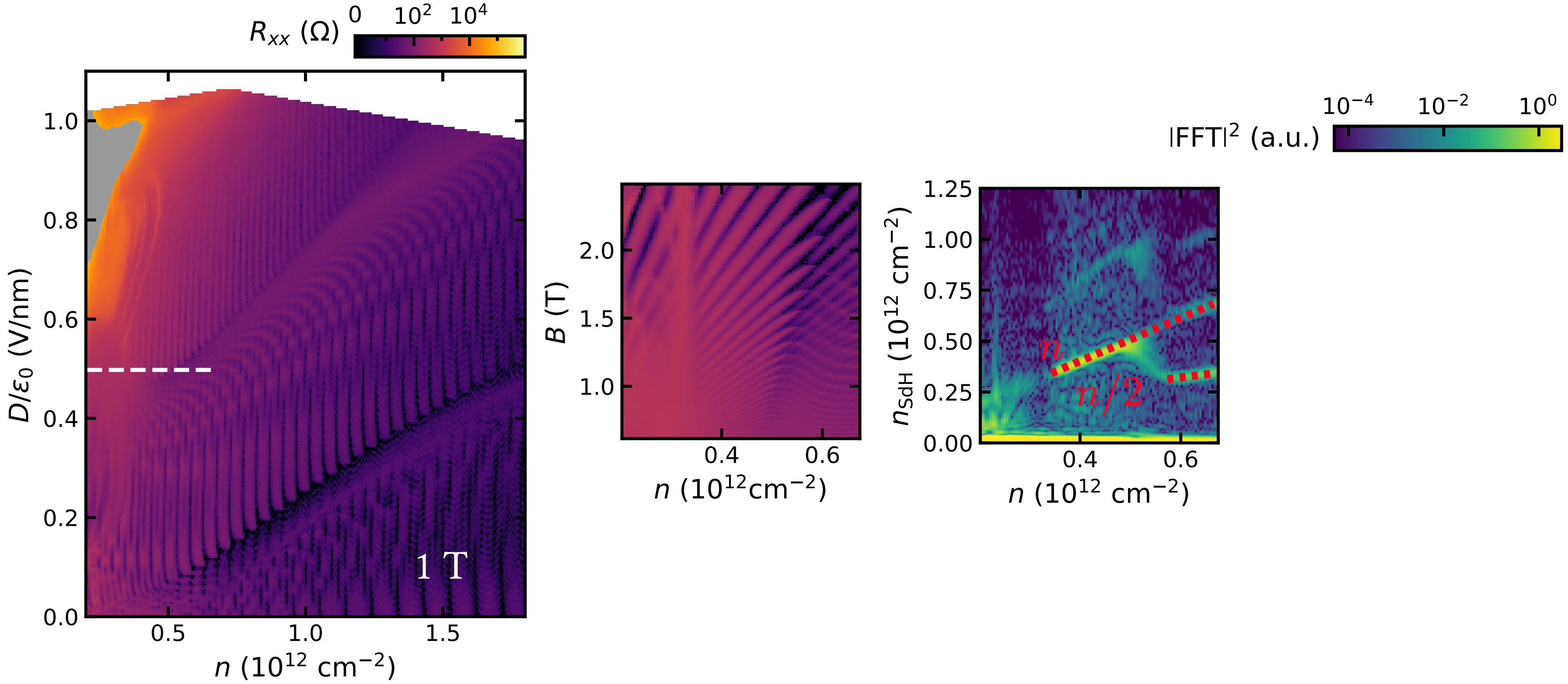

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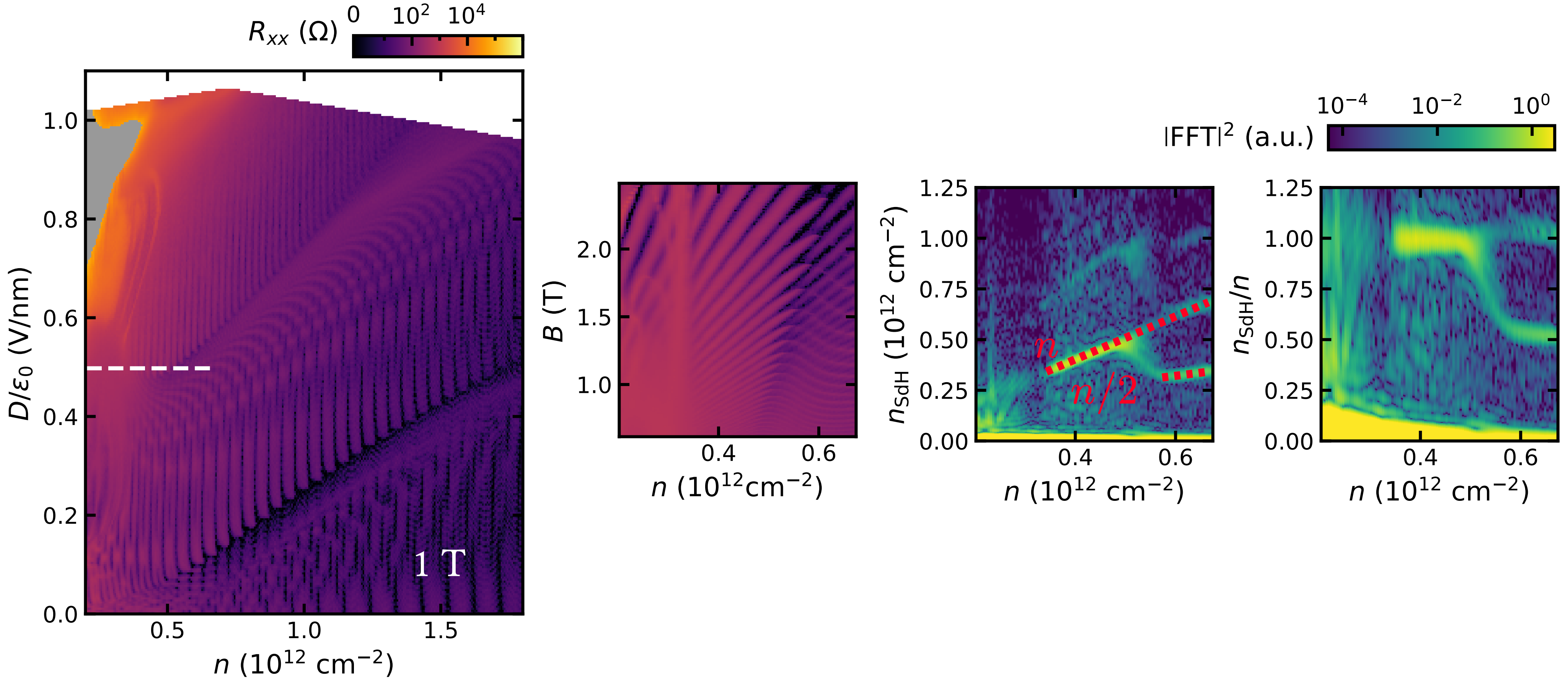

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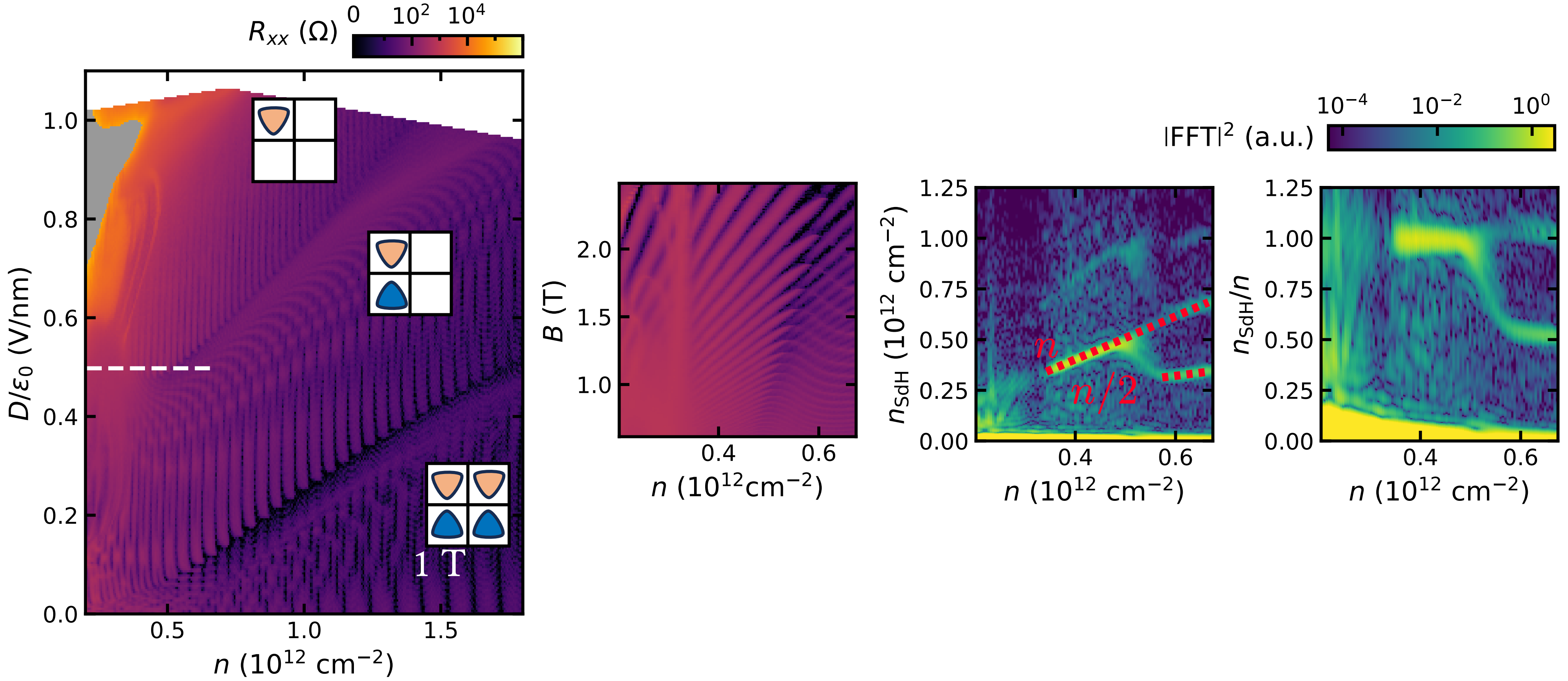
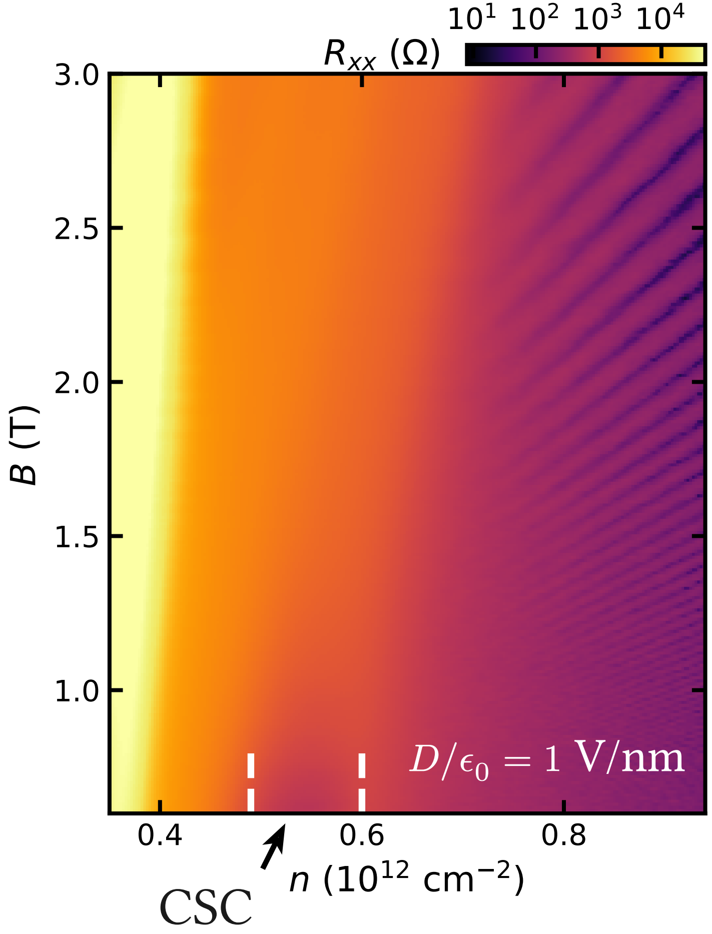


Oscillations near the CSC



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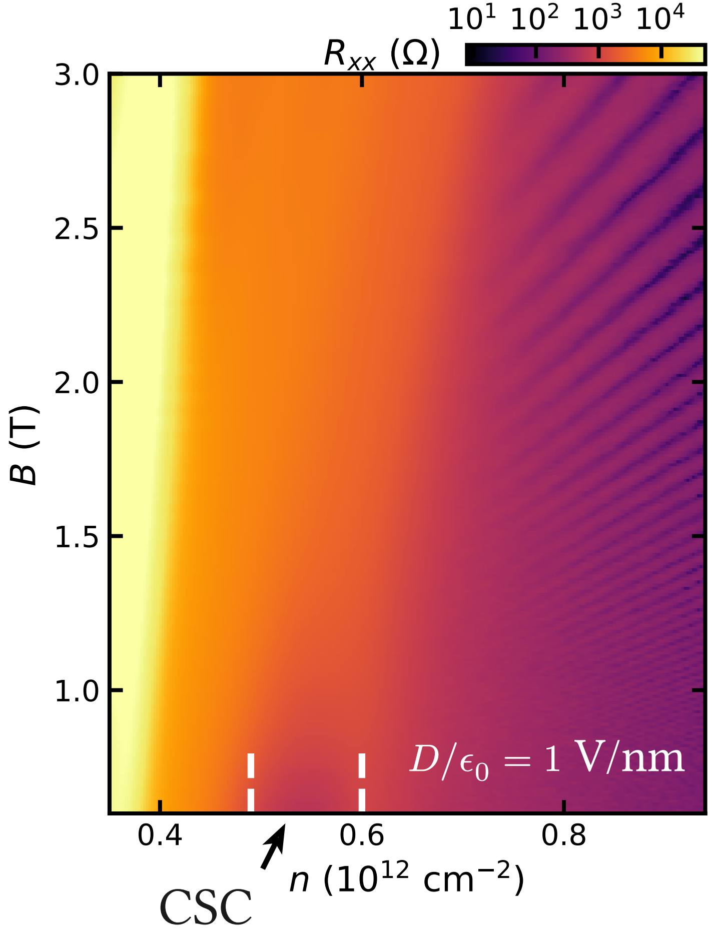

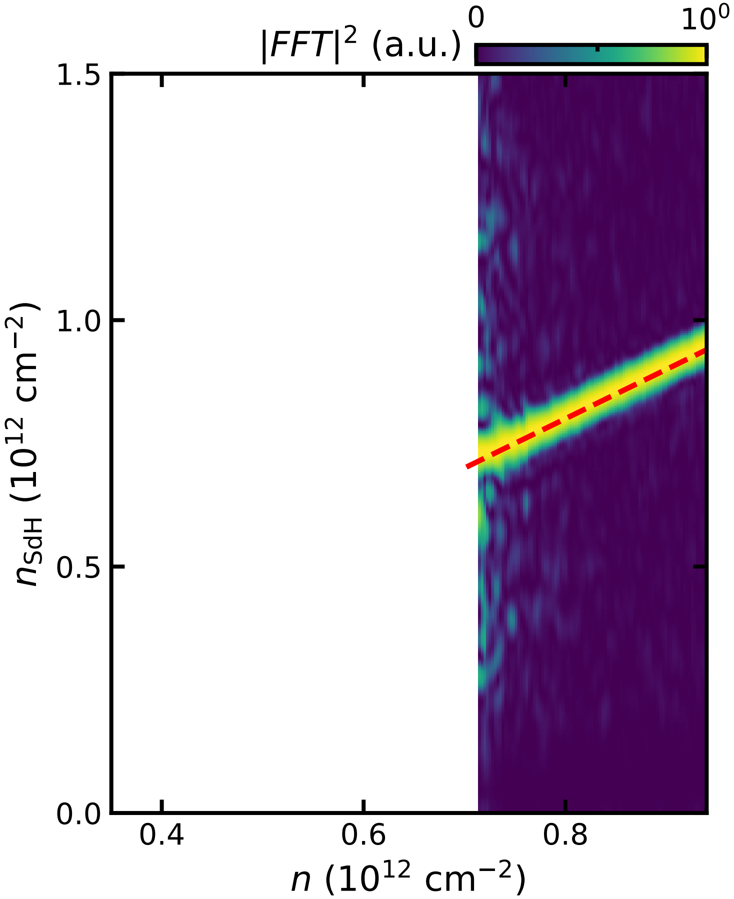

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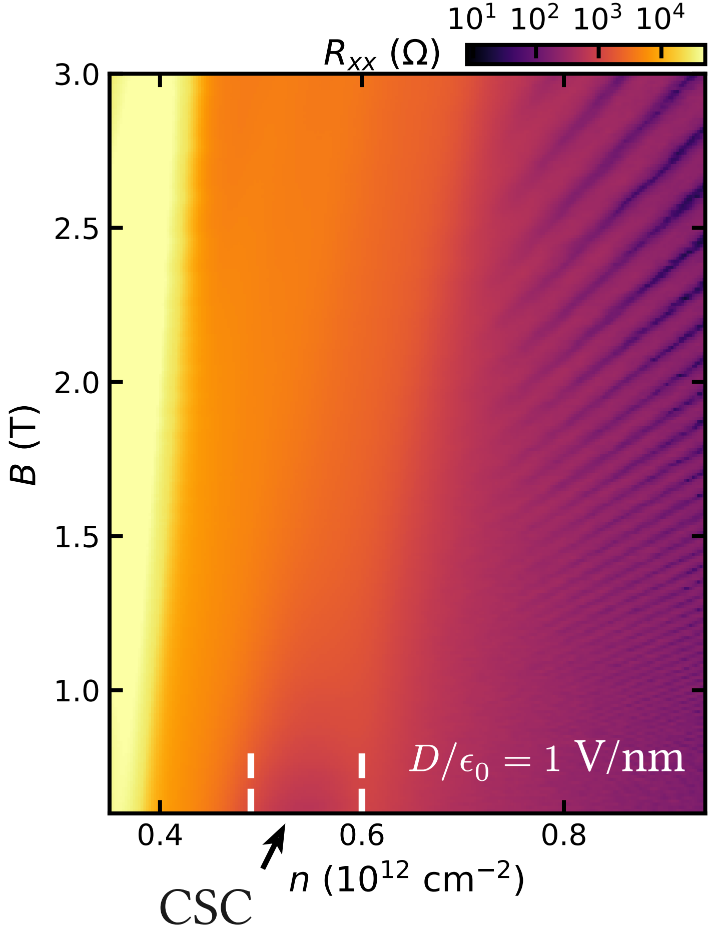

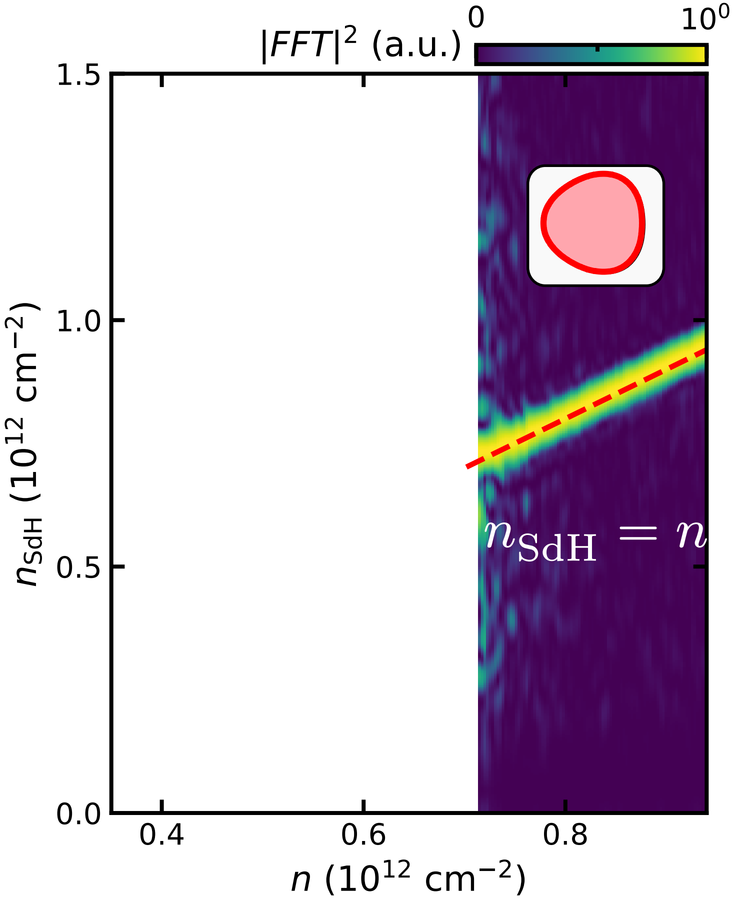

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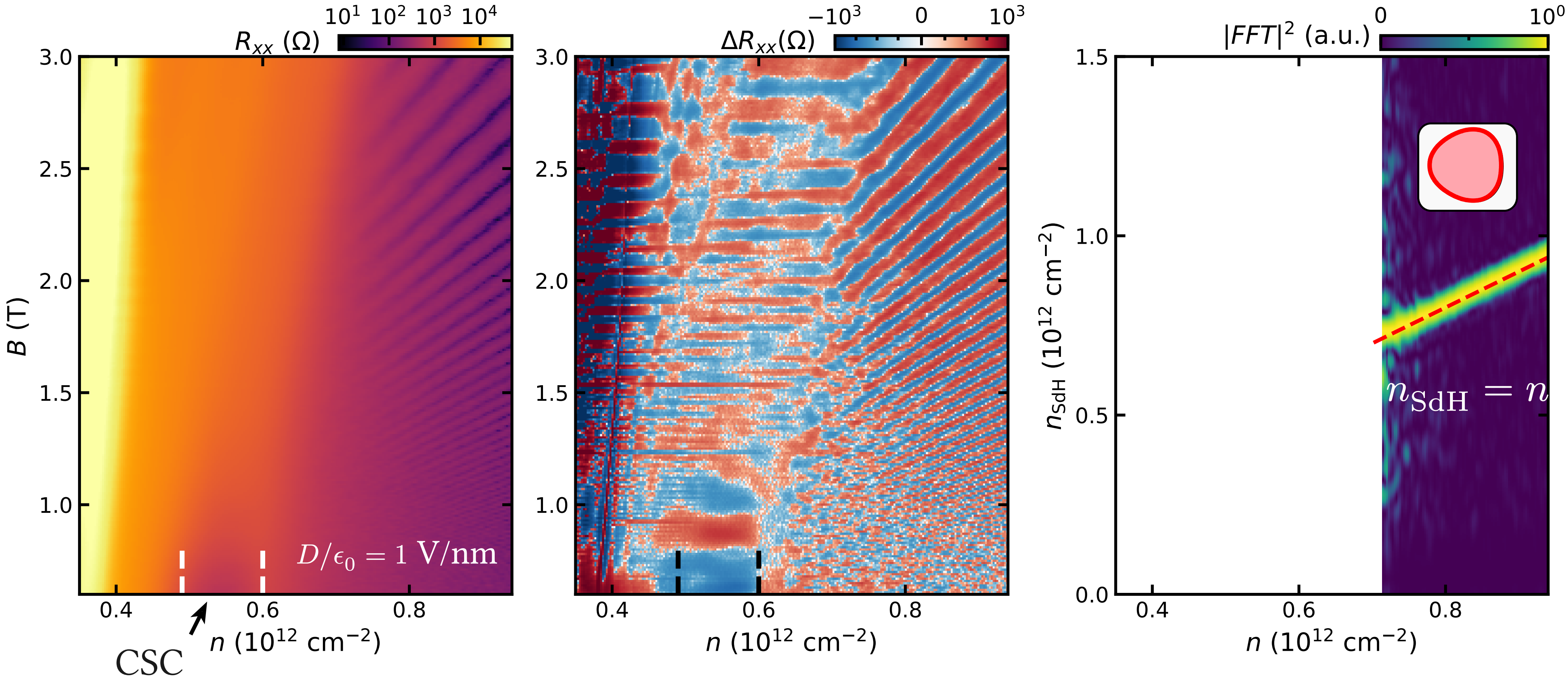

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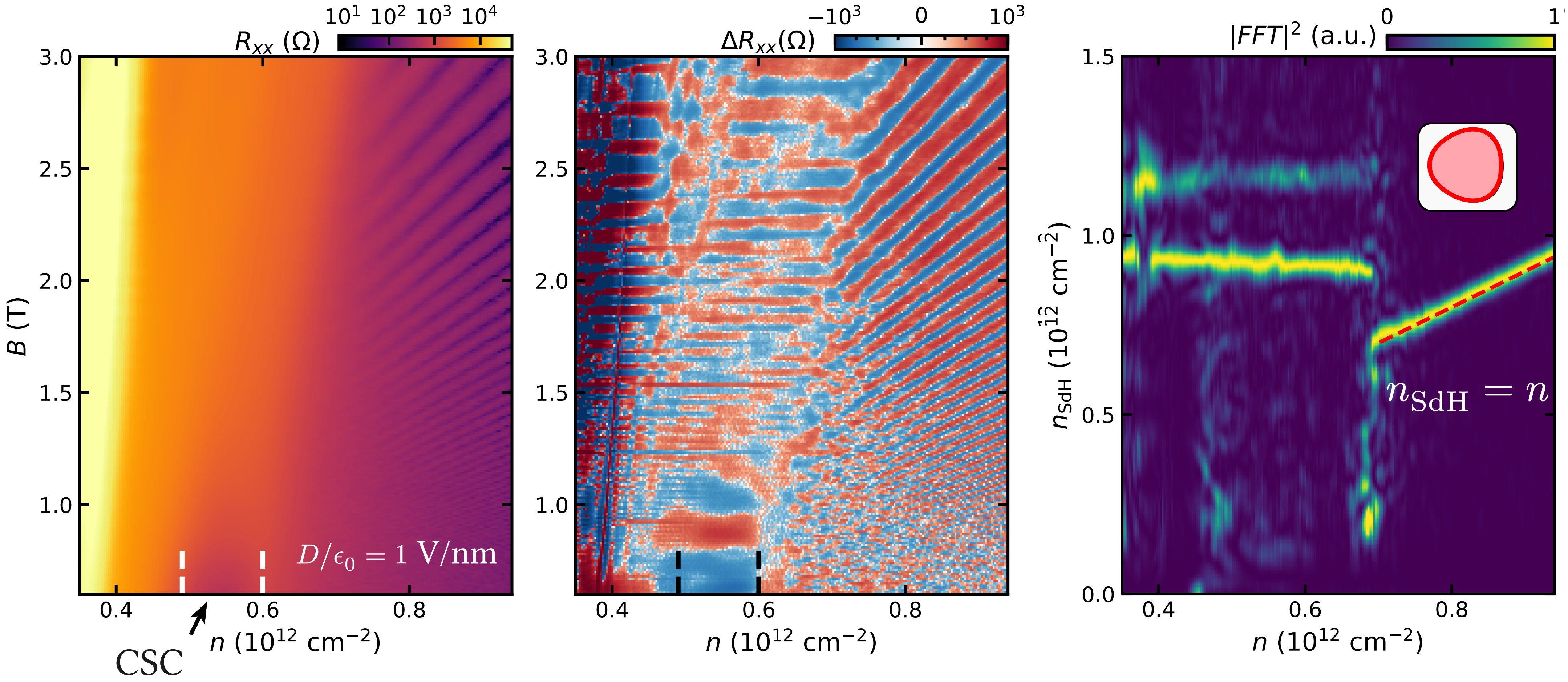

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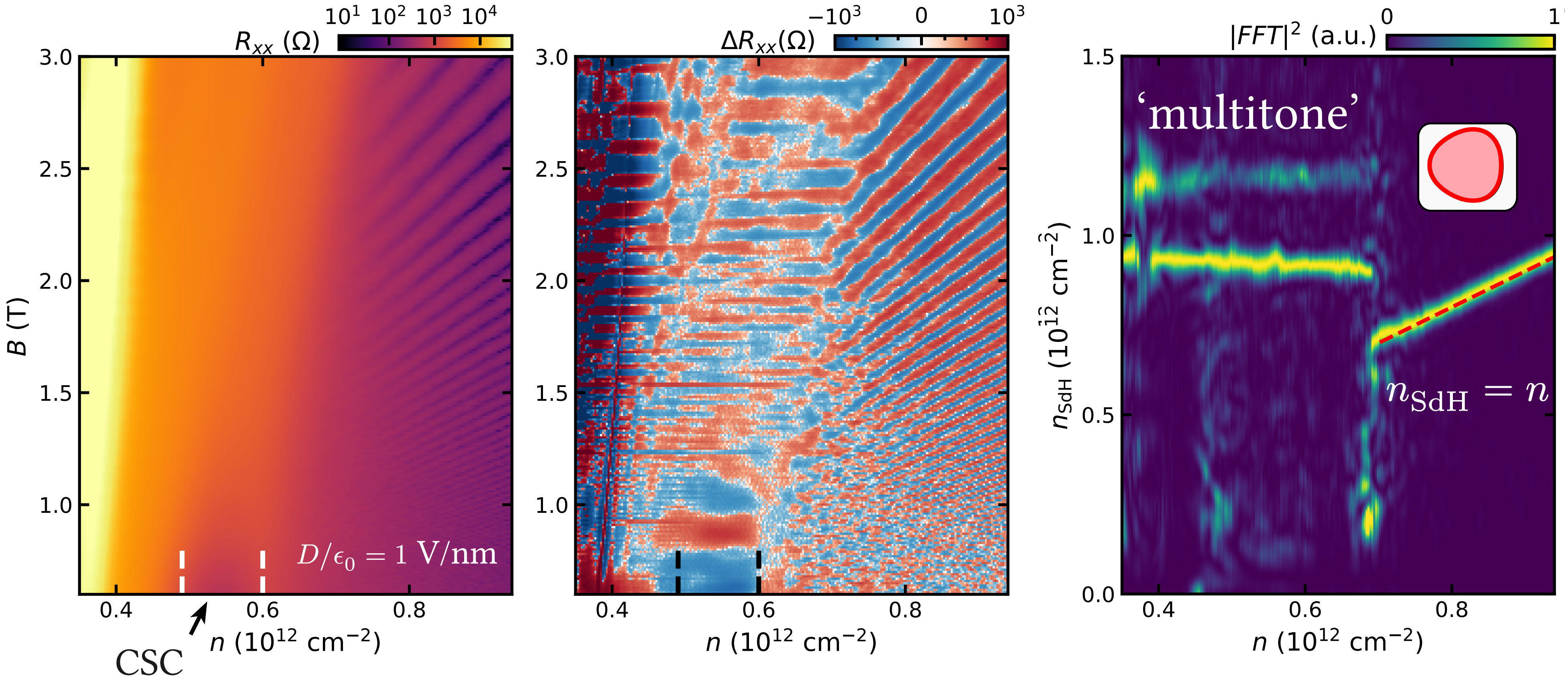
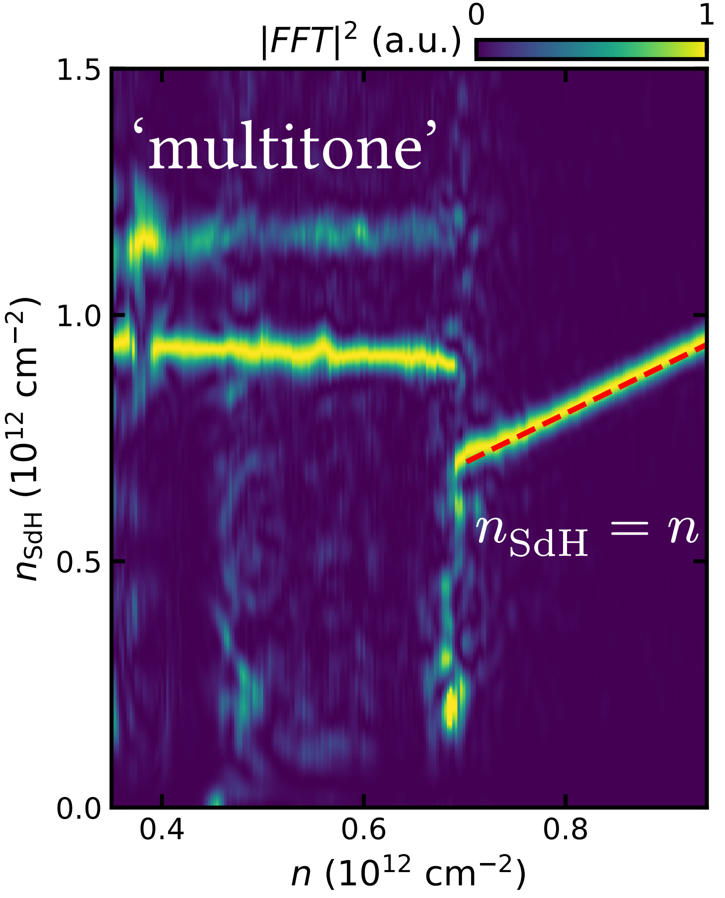


‘Multitone’ metal



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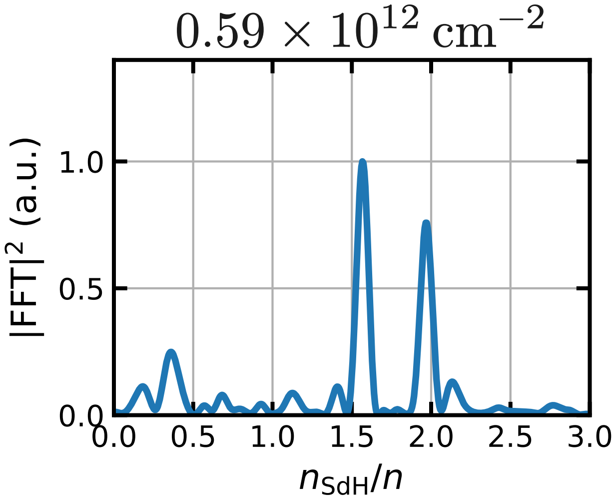

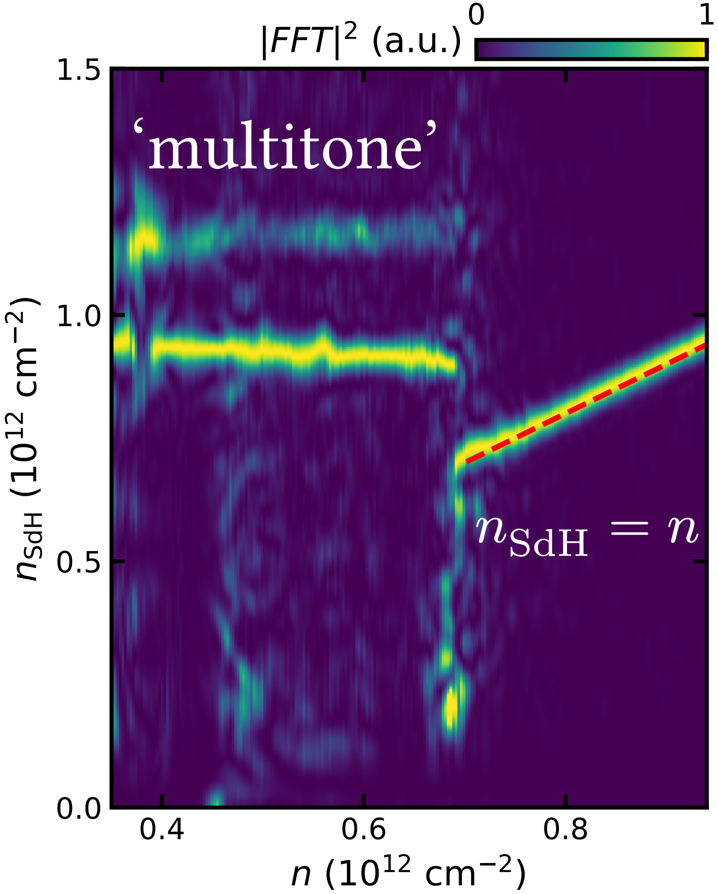

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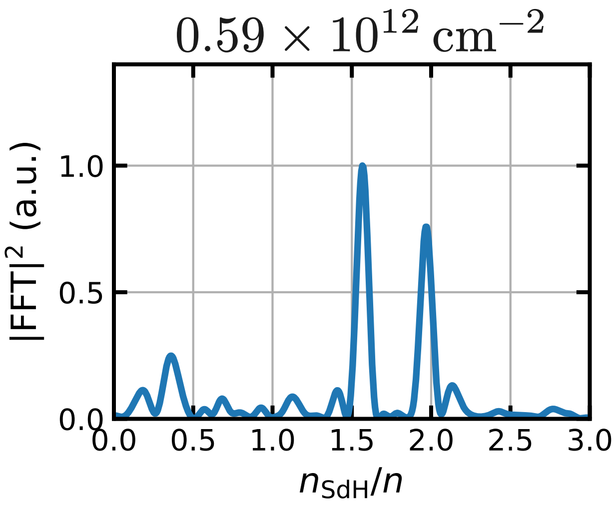
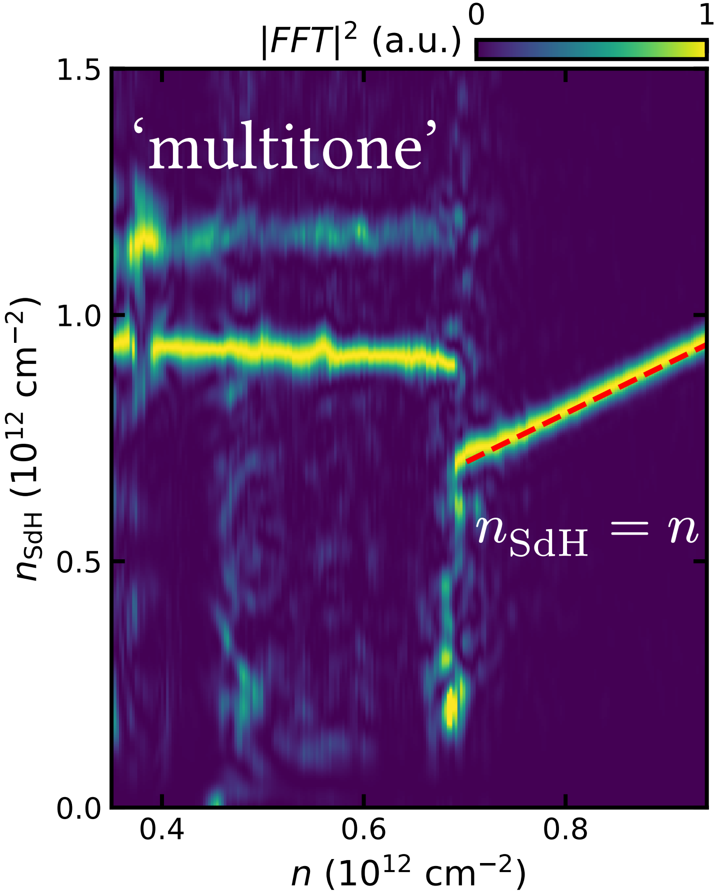


- jumps to two **high** and nearly **flat** frequencies



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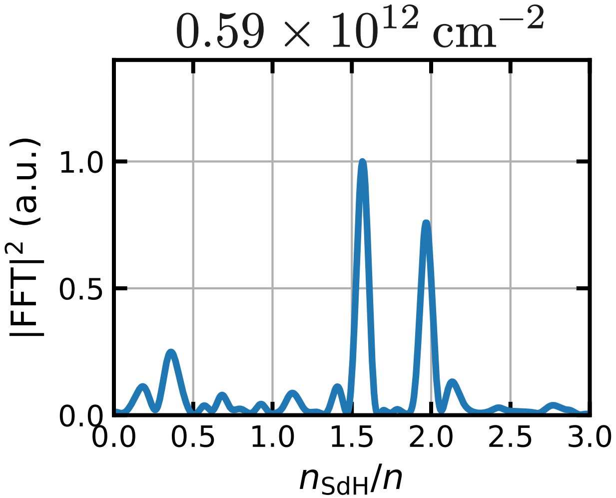

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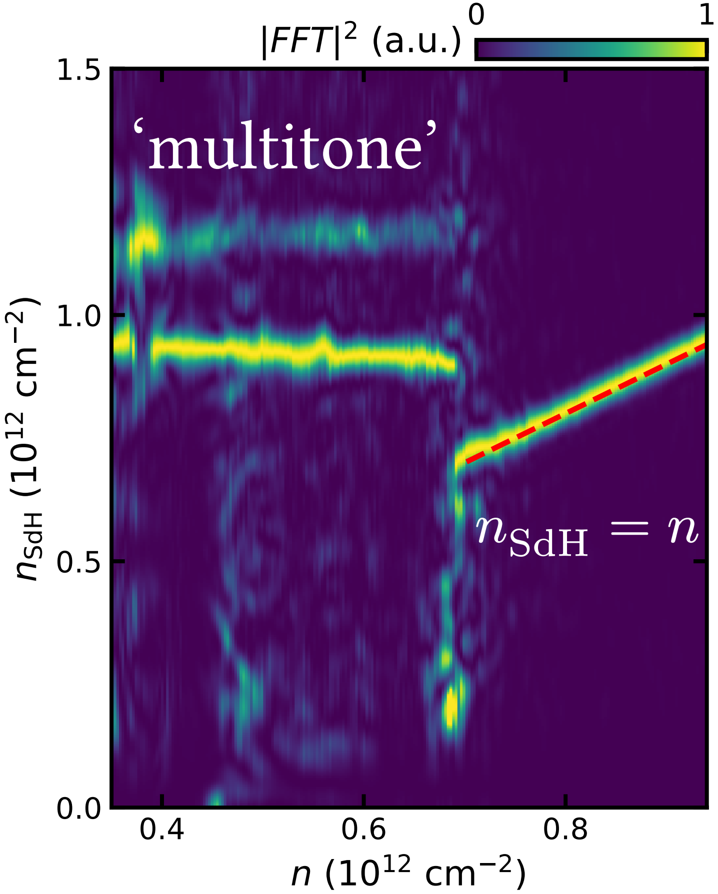
- Annulus?



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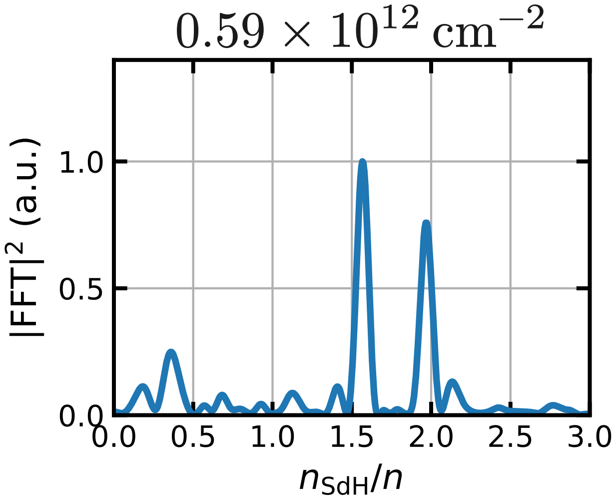
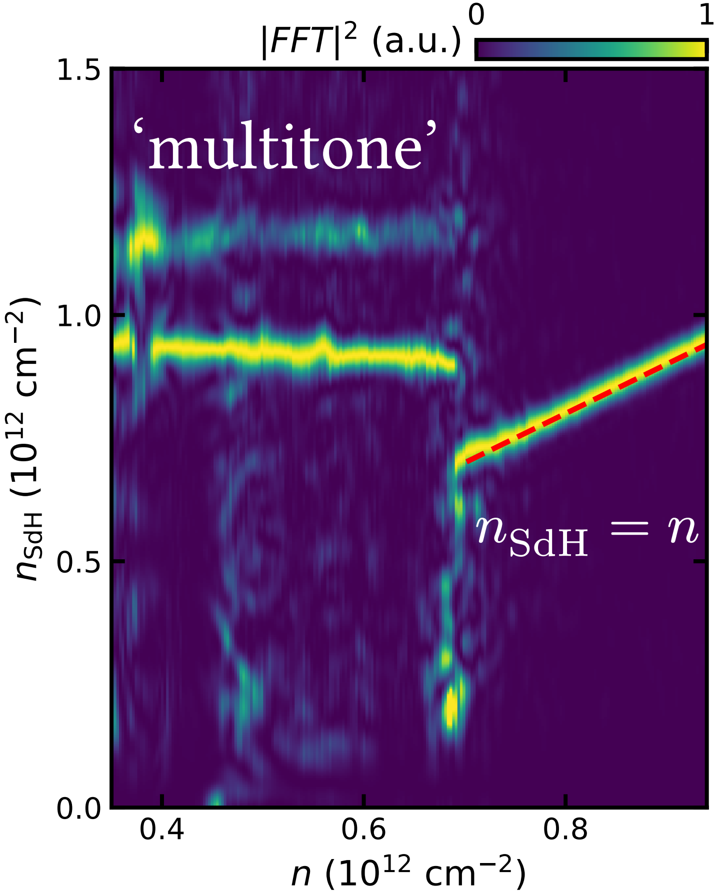
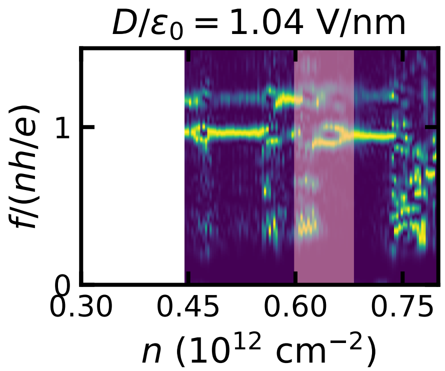


- jumps to two **high** and nearly **flat** frequencies
- Annulus?
  - $\Delta n_{\mathrm{SdH}}^{\mathrm{fast}} \approx 0.2 \times 10^{12}\ \mathrm{cm}^{-2}$

  $\rightarrow$ high tones inconsistent with a singly degenerate annulus*

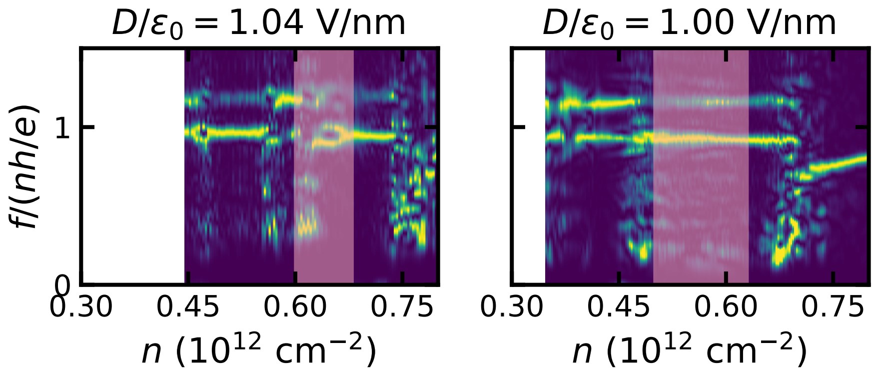


Mapping the Phase Diagram



Pink: extent of CSC

Mapping the Phase Diagram



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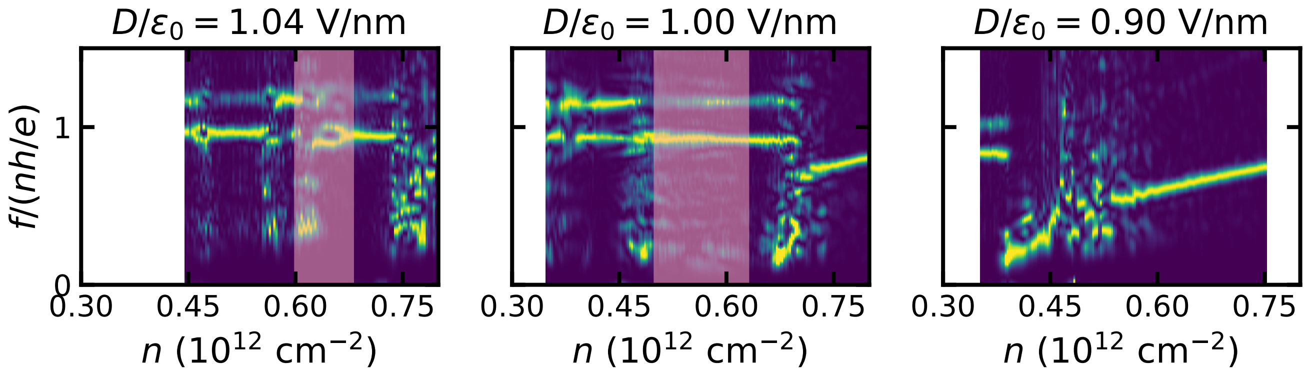

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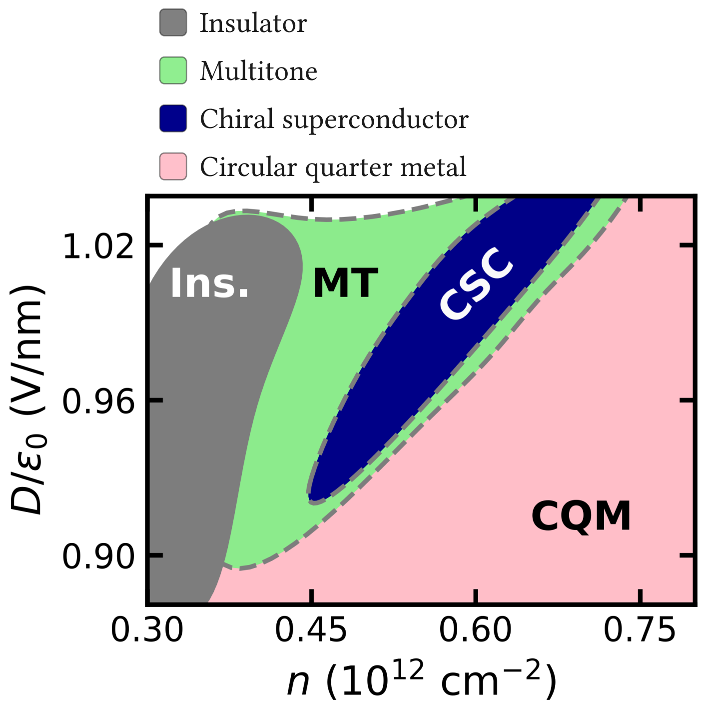

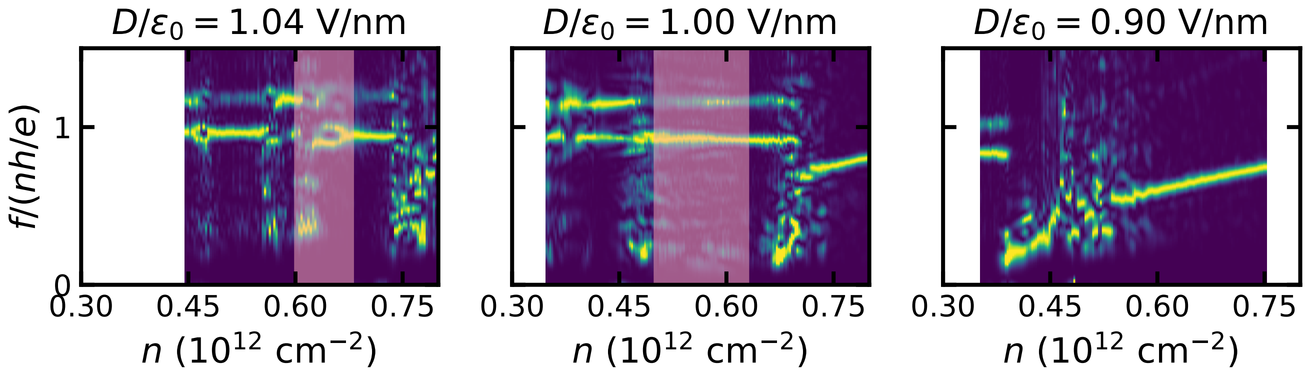

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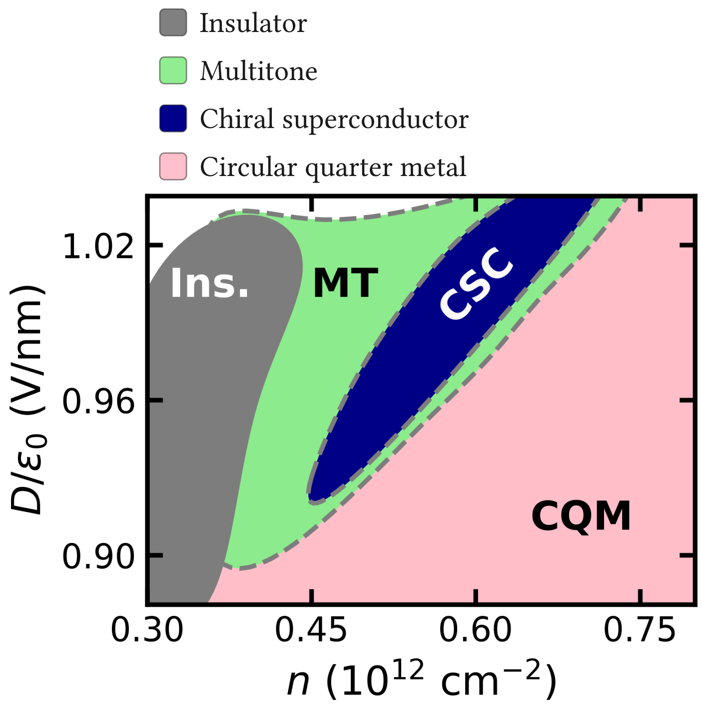
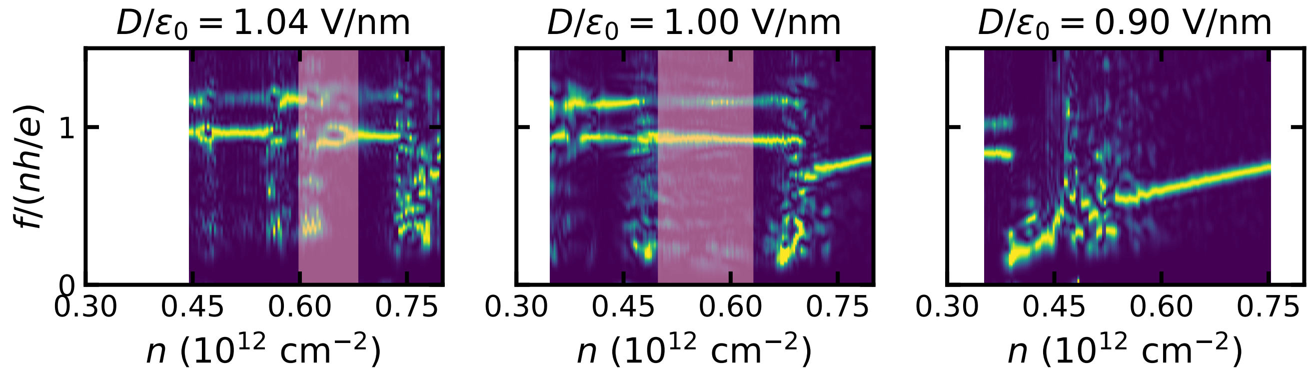
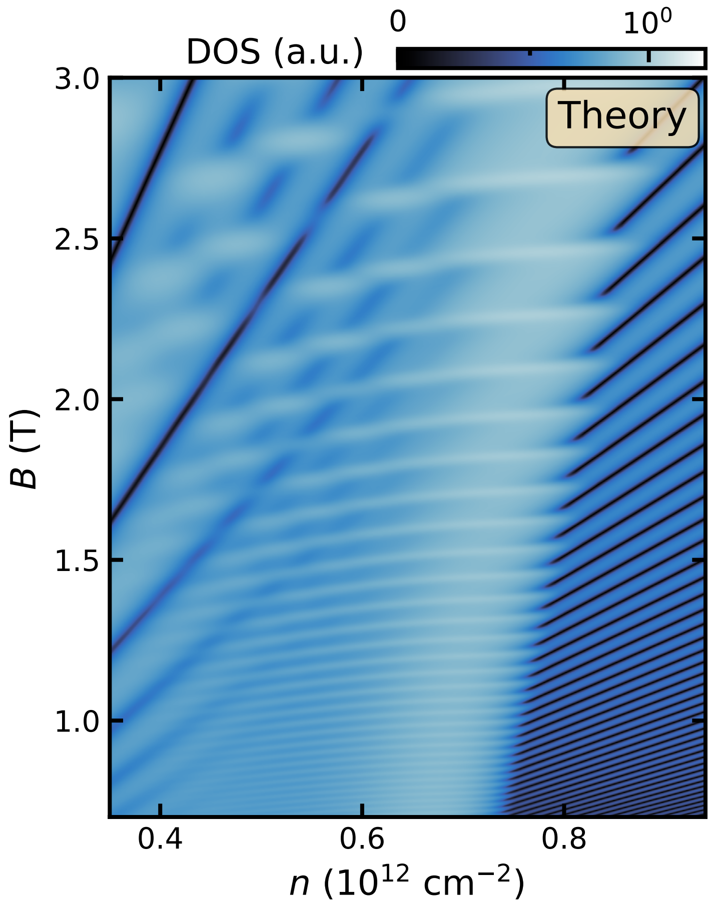




Pink: extent of CSC

- CSC resides primarily within multitone
  - → multitone = normal state (assuming no finite field transition)

Microscopic model



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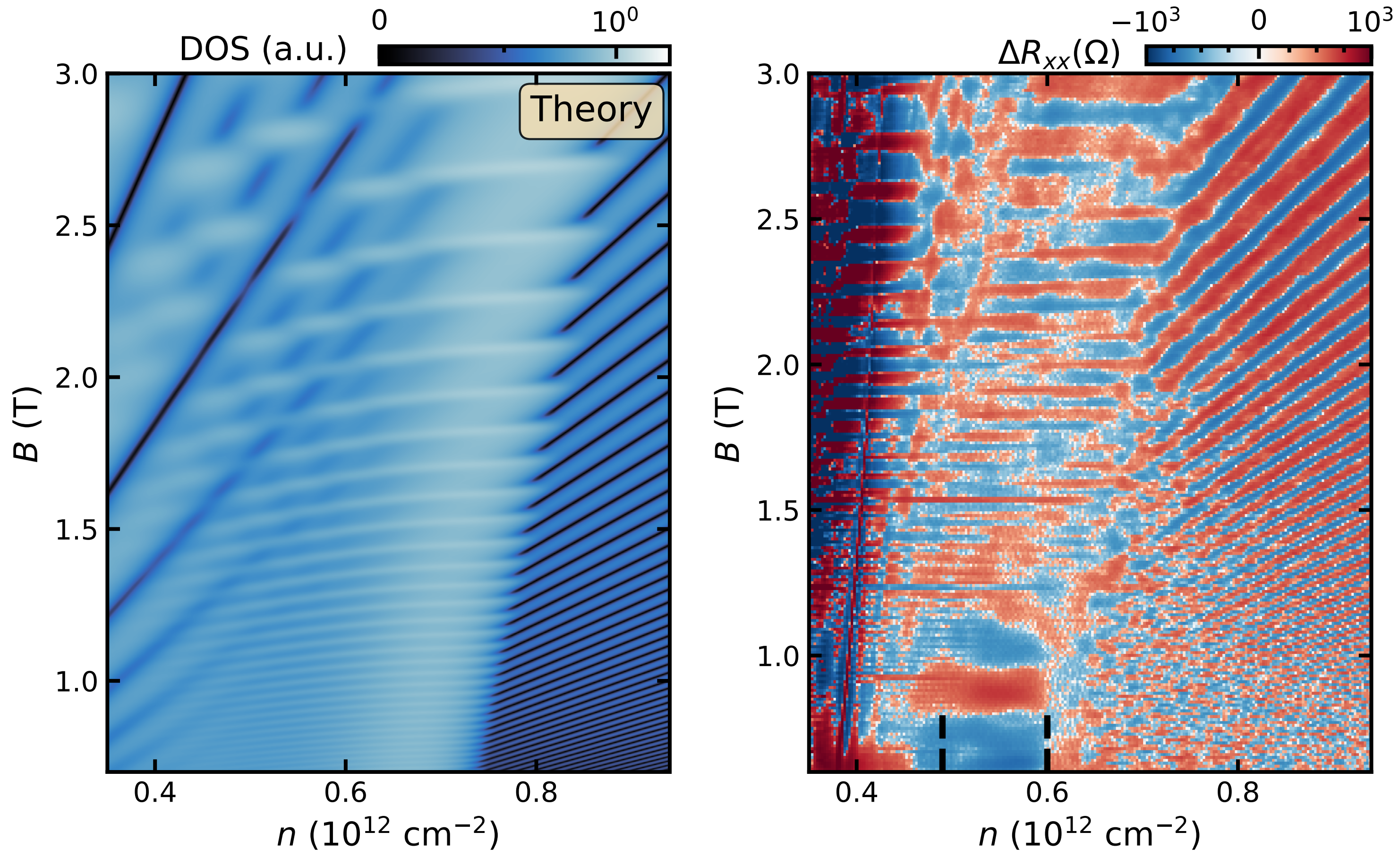

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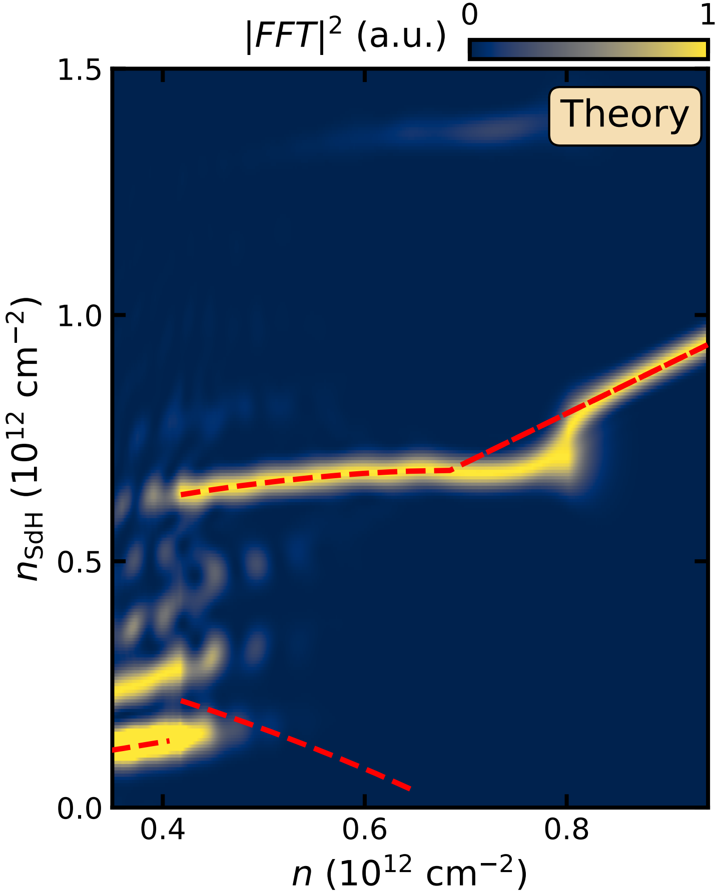

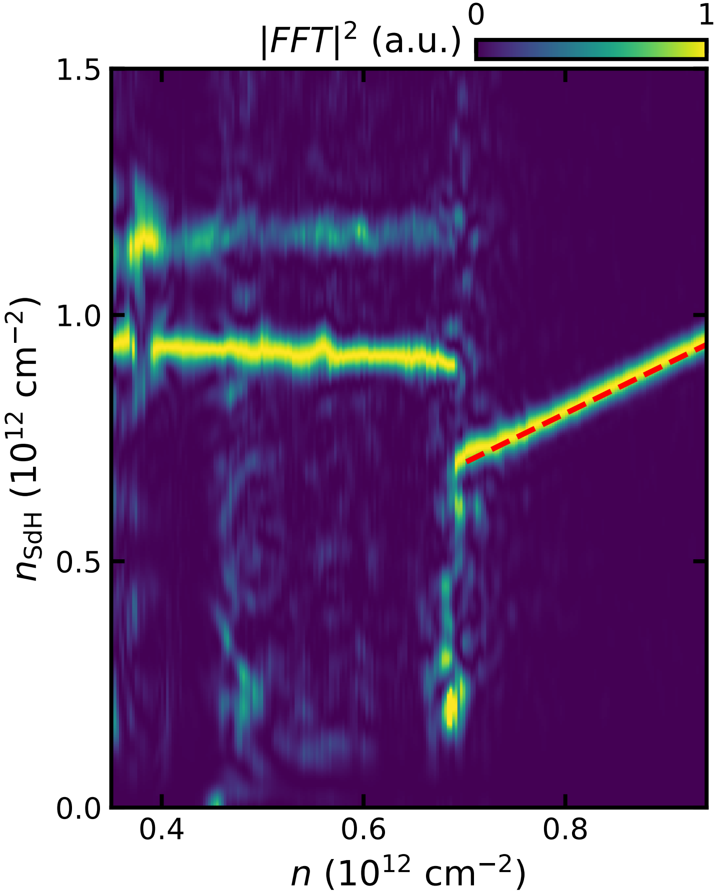

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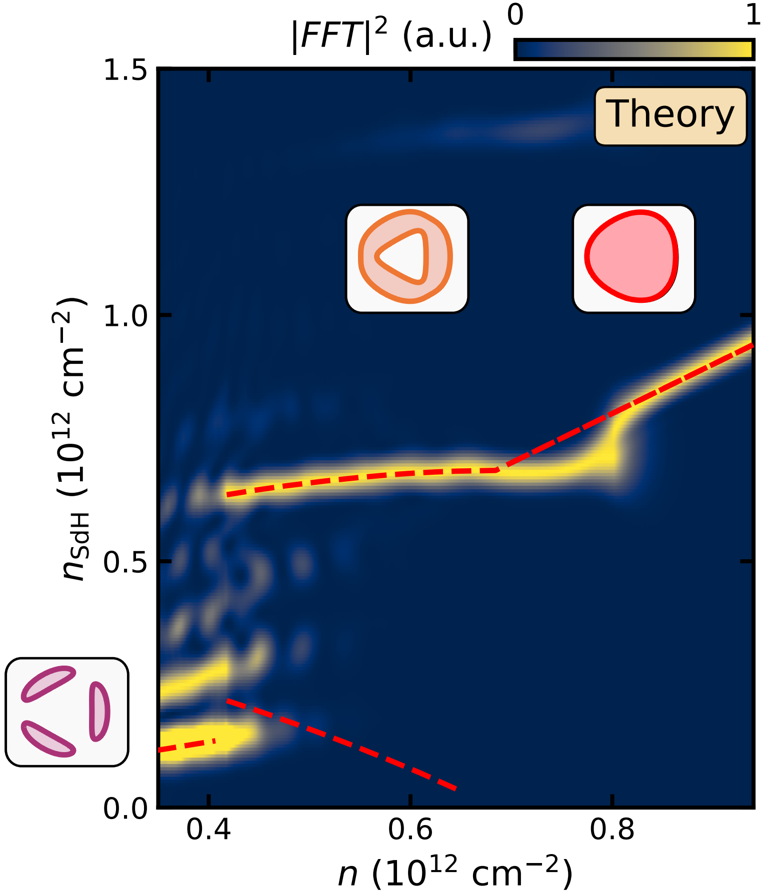
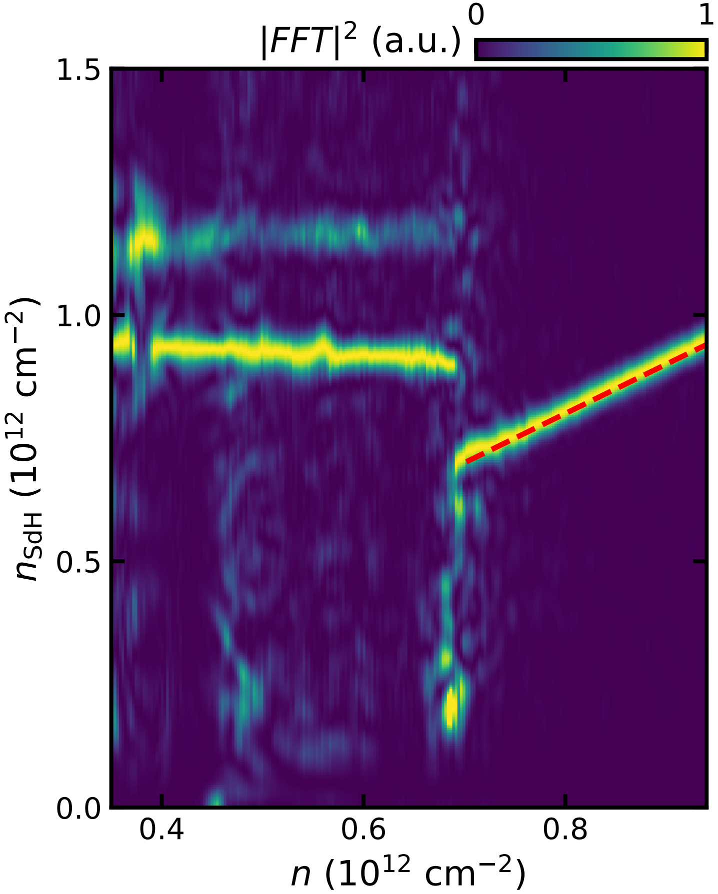
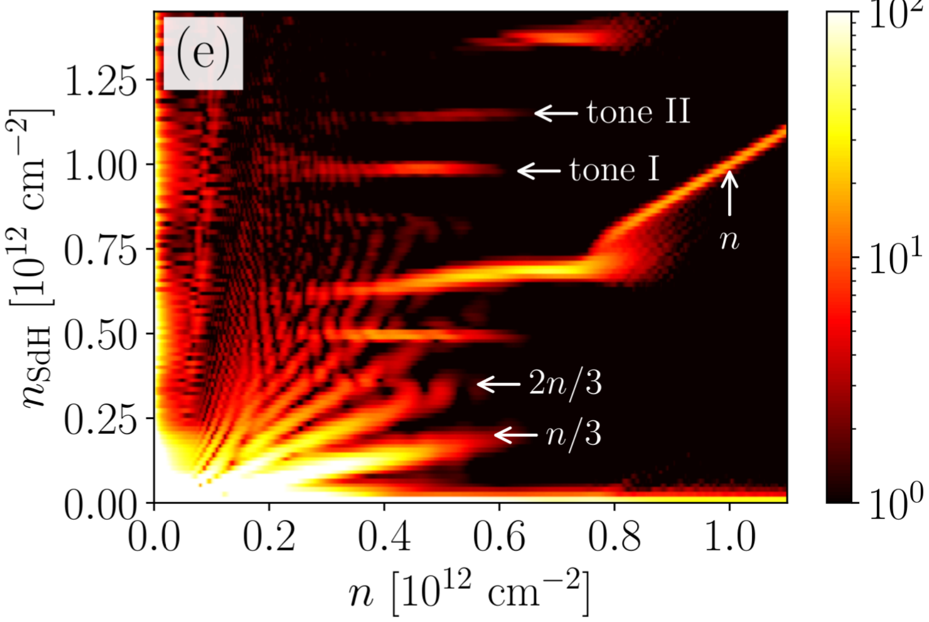


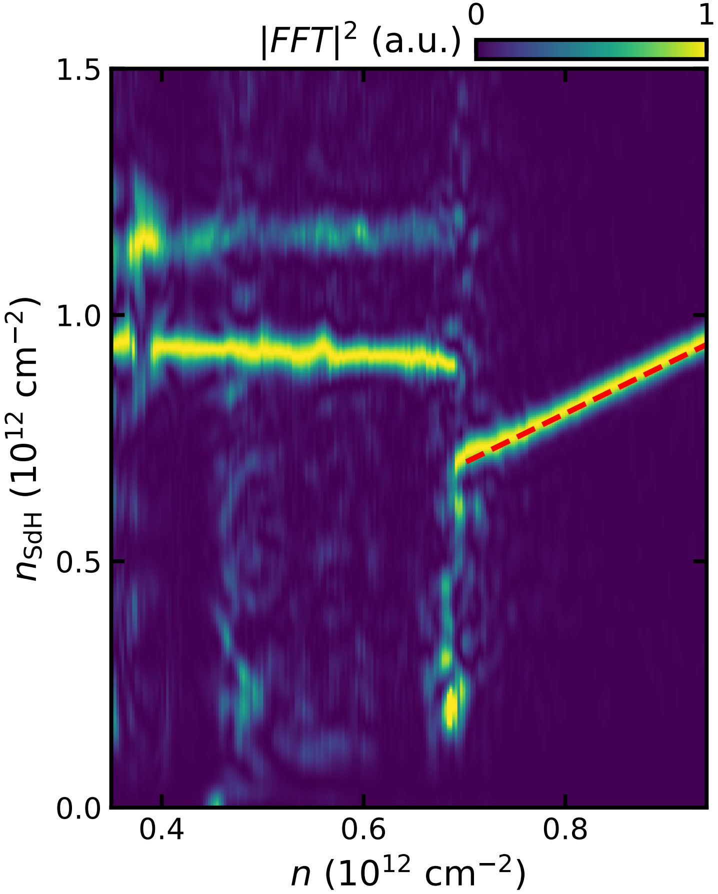


Kalantre, ..., **Sharpe** arXiv:2606.05356

Hot Off the Press Theory

Compute ρ_{xx} with Kubo and disorder





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Zhao et al. arXiv:2607.27207

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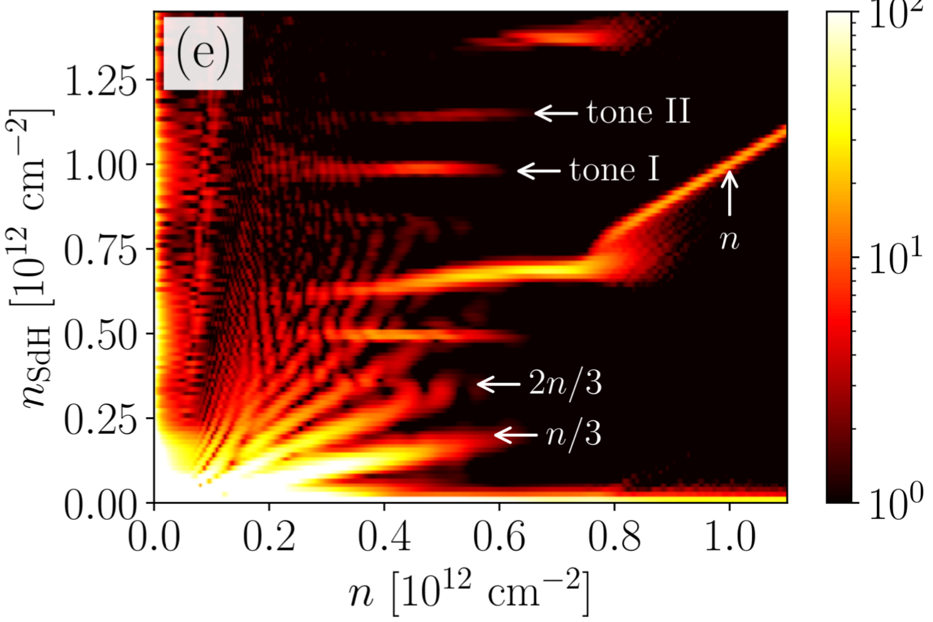

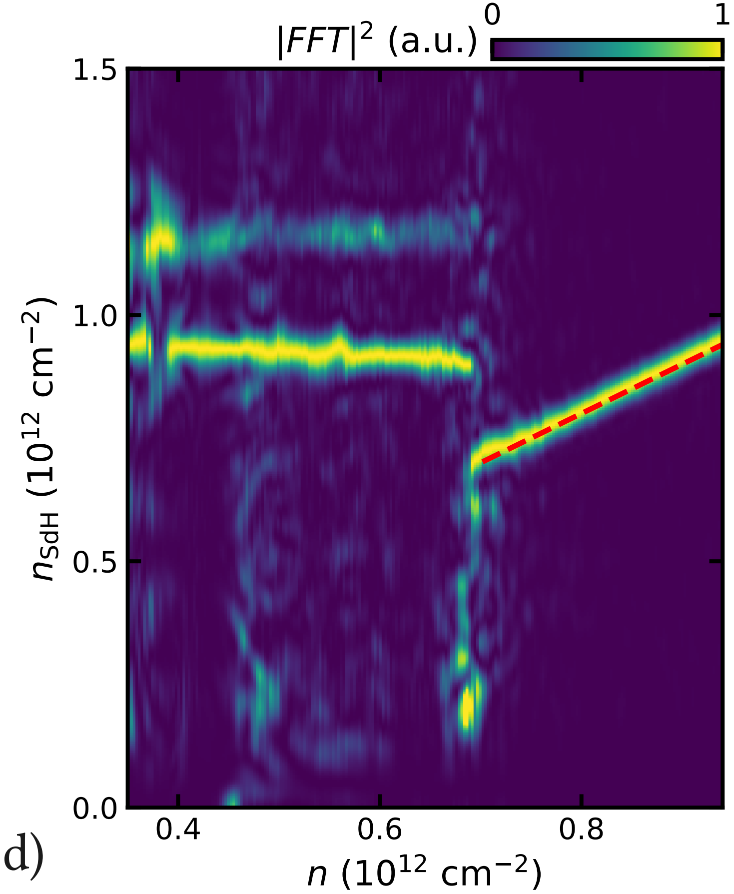

- Predicts multiple flat tones!
- Predicts strong annular tone (not observed)

Kalantre, ..., **Sharpe** arXiv:2606.05356
Zhao et al. arXiv:2607.27207

Crystallization and Onsager?

- Not all carriers are contributing to quantum oscillations?
  - $\Delta n_{\mathrm{SdH}}^{\mathrm{fast}} \approx 0.2 \times 10^{12}\,\mathrm{cm}^{-2}$

Kalantre, …, **Sharpe** arXiv:2606.05356
Dong et al. arXiv:2604.00114
Han et al. arXiv:2604.00113

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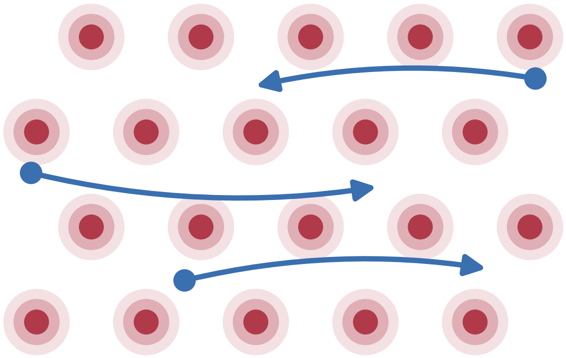
  - Itinerant fraction ~constant



localized carriers   itinerant carriers

Kalantre, ..., **Sharpe** arXiv:2606.05356
Dong et al. arXiv:2604.00114
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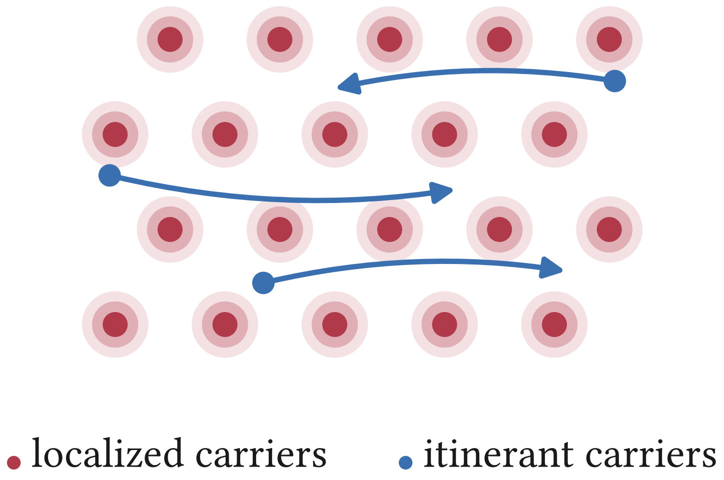
- Need microscopic calculations!



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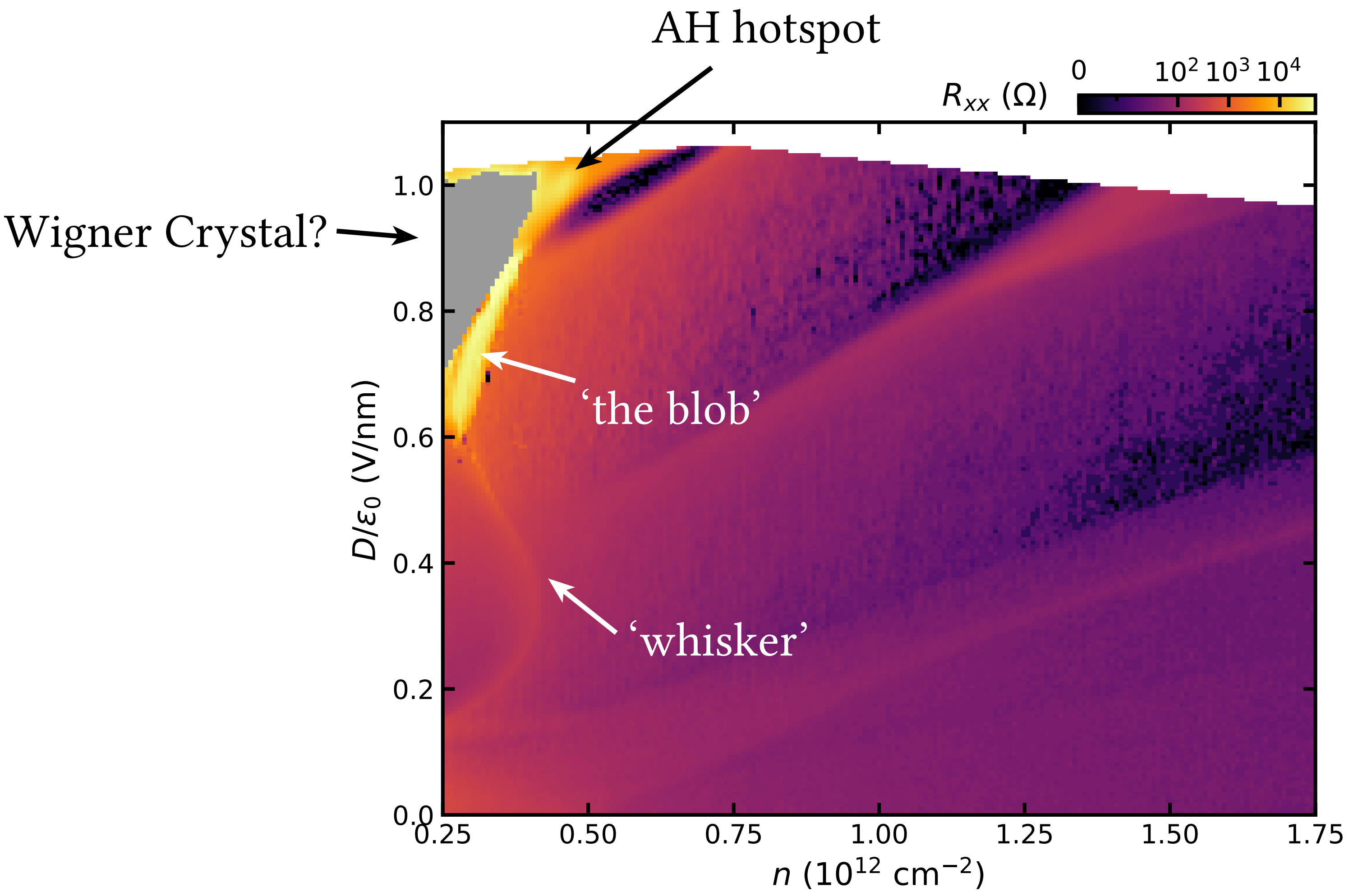
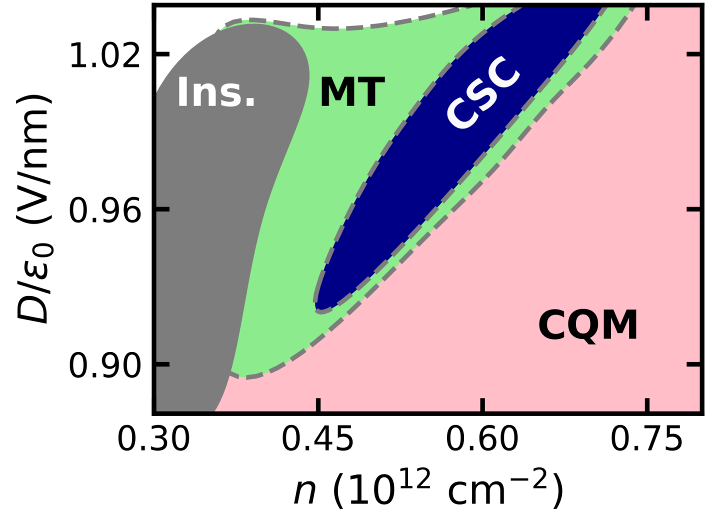
Nearby Notable Features

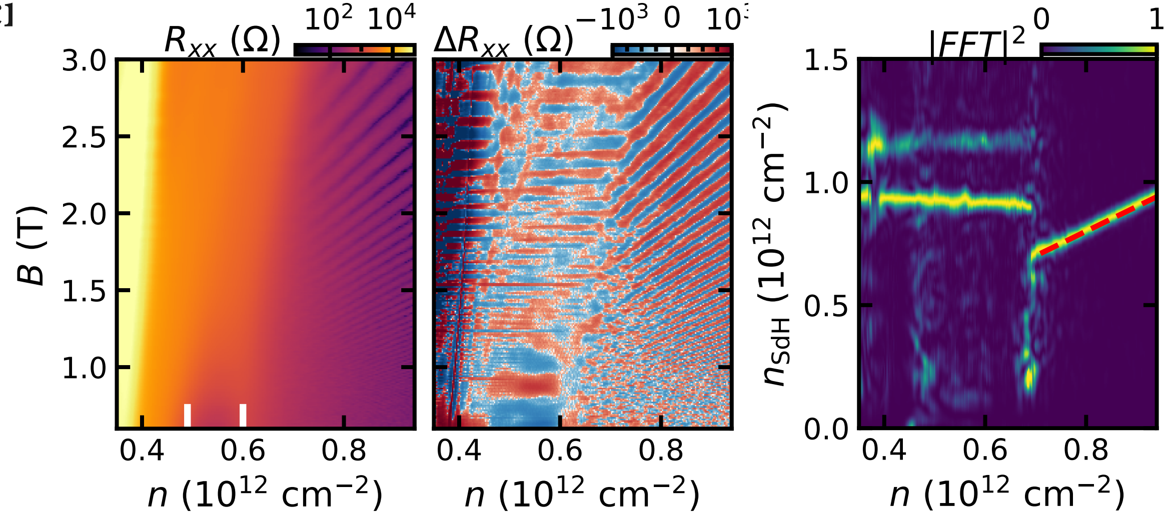


Summary

- CSC resides primarily within multitone
  - → multitone = normal state
  - **Not** circular quarter metal

- Multitone:
  1. Likely spin- and valley-polarized
  2. Two **high** and nearly **flat** frequenci
  3. Difference is ~ constant with D
  4. No obvious Luttinger sum rule
  5. **Inconsistent with simplest candidates**





Kalantre, ..., **Sharpe** arXiv:2606.05356

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David Goldhaber-Gordon, Andy Mannix